



**Dimensions and  
Subdimensions of the EQUI-T  
IA-OER Quality Framework**

**Deliverable 2.2**



---

This work is **the Deliverable 2.2– Drafting a Multi-Perspective IA-OER Quality Framework** of the Erasmus+ project [EQui-T](#) (ERASMUS-EDU-2022-PEX-TEACH-ACA 101104449)

Delivery date: April 2025

Authors: Catalina Jiménez Hurtado, Cristina Álvarez de Morales Mercado, Laura Carlucci, Emilia Iglesias Fernández, María Visitación Hurtado Torres, María José Rodríguez Fórtiz, María Luisa Rodríguez Almendros, Rosario Arroyo González, Silvia Martínez Martínez, Antonio Javier Chica Núñez, María Olalla Luque Colmenero, María Asunción Pérez de Zafra Arrufat, Celia Barnés Castaño, Silvia Toribio Camuñas, Vicente Bru García, Carlos Navas Vallejo (University of Granada)

We thank all EQui-T consortium partners for their valuable and helpful feedback in developing this framework.

---

The project EQui-T has been funded with support from the European Commission under the Erasmus+ Teacher Academy Programme.



**Co-funded by  
the European Union**

Funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or EACEA. Neither the European Union nor the granting authority can be held responsible for them.

---

This work is licensed under the Creative Commons Attribution-Non Commercial - Share Alike 4.0 licence.



## Partners



## Keywords

Quality Framework, Open Educational Resources, Inclusive OER, Accessible OER

## Abstract

**The EQUI-T OER Quality Framework (EQUIT-FW)** includes a set of recommendations or guidelines associated with five dimensions:

1. **Policies and guidelines**
2. **Didactic**
3. **Technology (ICT)**
4. **Diversity**
5. **Accessibility**

These recommendations are intended to guide the teacher in the process of developing or updating an OER specifically in each particular dimension, thus ensuring its quality. Each recommendation has a set of questions associated as a checklist for assessing the product. They are based on the previous bibliography, citing the source.

Dimensions and sub-dimensions of our framework follow the pattern of the most of the previous frameworks and guidelines in the literature, but complete them, specifically the dimensions Technology Diversity and Accessibility.



Co-funded by  
the European Union

Funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or EACEA. Neither the European Union nor the granting authority can be held responsible for them.

## Content

|          |                                                                                 |           |
|----------|---------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Dimension: Polices and Guidelines .....</b>                                  | <b>6</b>  |
| 1.1      | Policies .....                                                                  | 6         |
| 1.2      | Guidelines .....                                                                | 7         |
| <b>2</b> | <b>Didactics .....</b>                                                          | <b>13</b> |
| 2.1      | Competences and objectives .....                                                | 13        |
| 2.2      | Methodology .....                                                               | 17        |
| 2.3      | Contents .....                                                                  | 20        |
| 2.4      | Pedagogical principles .....                                                    | 28        |
| 2.5      | Student Assessment .....                                                        | 30        |
| <b>3</b> | <b>Dimension: ICT .....</b>                                                     | <b>35</b> |
| 3.1      | Technological base .....                                                        | 35        |
| 3.2      | Usability .....                                                                 | 41        |
| 3.3      | Platforms .....                                                                 | 45        |
| 3.4      | Metadata .....                                                                  | 46        |
| 3.5      | Licensing .....                                                                 | 50        |
| <b>4</b> | <b>Dimension: Diversity .....</b>                                               | <b>52</b> |
| 4.1      | General .....                                                                   | 52        |
| 4.2      | Gender, Language, Cultura, Race/Ethnicity, Age, Socio-economic .....            | 55        |
| 4.3      | Disability and specific educational needs .....                                 | 59        |
| 4.4      | Guidelines based on UDL .....                                                   | 61        |
| <b>5</b> | <b>Dimension: Accessibility .....</b>                                           | <b>77</b> |
| 5.1      | Accessibility for blind and partially sighted learners. Audio description ..... | 77        |
|          | Audio Description of videos .....                                               | 77        |
|          | Audio Description of images .....                                               | 91        |
| 5.2      | Accessibility for deaf and hard of hearing learners .....                       | 94        |
|          | Subtitles for the deaf and hard of hearing .....                                | 94        |
|          | Sign Language Interpreting .....                                                | 103       |
| 5.3      | Cognitive accessibility .....                                                   | 106       |
|          | Simplified Language .....                                                       | 106       |



|                                                                              |            |
|------------------------------------------------------------------------------|------------|
| Visual Aids and Pictograms .....                                             | 124        |
| 5.4 Technological accessibility .....                                        | 129        |
| Multimedia .....                                                             | 129        |
| Guidelines based on WCAG2.2 .....                                            | 134        |
| <b>6 Bibliographic references used to create the quality framework .....</b> | <b>147</b> |

## 1 Dimension: Policies and Guidelines

This dimension refers to educational laws and recommendations at the national and international levels that regulate the creation, implementation and evaluation of IA-OER and the legal conditions of use related to their intellectual property licence.

### 1.1 Policies

**Policies:** refer to the overarching law, rules and regulations that govern the design, implementation and dissemination of IA-OERs. They provide a framework for decision-making and ensure compliance with legal and ethical standards.

| RECOMMENDATION                                                                                                                                                                                            | QUESTION                                                                                                                                                                                        | SOURCE                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| It is desirable to design OER with accessibility from the start.<br>Accessibility is therefore embedded in the course design rather than a reactive accommodation for those with registered disabilities. | Has the OER been designed with accessibility from the start?<br>Is the accessibility embedded in the course design rather than a reactive accommodation for those with registered disabilities? | UNESCO, 2019;<br>QLT 2022                                |
| Strengthen the use of accessible OER in educational contexts, disseminating the guidelines widely to governments, education institutions, practitioners and students and gathering their feedback.        | Is the use of accessible OER in educational contexts strengthened and gather the feedback of faculty and students?                                                                              | UNESCO, 2019                                             |
| Ensure that faculty are supported in promoting inclusion and equity, and have the necessary tools to develop and evaluate inclusive and accessible OER.                                                   | Do you have the necessary tools to develop and evaluate inclusive and accessible OER?                                                                                                           | UNESCO, 2019                                             |
| Ensure that there is compliance with the main import/export and e-learning standards (e.g., IMS CP, IMS CC, SCORM, learning management system (LMS) and content management system (CMS).                  | Does the OER comply with any of the main import/export and e-learning standards (e.g., IMS CP, IMS CC, SCORM, learning management system (LMS) and Content management System (CMS)?             | UNESCO, 2019;<br>Tili et al., 2019;<br>Achieve Inc, 2011 |
| Involve communities in the development and implementation of policies that promote inclusion and equity in education.                                                                                     | Are communities involved in the development and implementation of policies that promote inclusion and equity in education?                                                                      | UNESCO, 2019                                             |



| RECOMMENDATION                                                                                                                                                                                                                                                                                        | QUESTION                                                                                                                                                                                                      | SOURCE                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Ensure that there is compliance with the WCAG guidelines for accessibility.                                                                                                                                                                                                                           | Does the OER comply with the WCAG guidelines for accessibility?                                                                                                                                               | Camilleri & Tannhäuser, 2012                                                      |
| Ensure that there is compliance with the UDL guidelines.                                                                                                                                                                                                                                              | Does the OER comply with the UDL guidelines?                                                                                                                                                                  |                                                                                   |
| Ensure that the institutions offer low-cost or free educational resources because they make a continuous effort to secure and allocate adequate financial resources (public or private) for OER operations, particularly for low-income, rural and urban communities, monitoring carefully the costs. | Does your institution offer low-cost or free educational resources, making a continuous effort to secure and allocate adequate financial resources?<br><br>Does your institution monitor carefully the costs? | Jung & Lee, 2014; Mhlanga, 2010; Latchem, 2012; Hasley et al., 2021; UNESCO, 2019 |

## 1.2 Guidelines

**Guidelines:** help institutions, teachers, or stakeholders to create IA-OER. They are specific recommendations and instructions that help shape the design of IA-OER. They also offer a more detailed guidance on creating IA-OERs that align with policies.

| ASPECT                             | RECOMMENDATION                                                                                                                                                     | QUESTION                                                                                                                                                     | SOURCE                                                                          |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>Accessibility and inclusion</b> | Ensure that every learner matters and matters equally. Use evidence to identify contextual barriers.                                                               | Do you ensure that every learner matters and matters equally?<br>Do you use evidence to identify contextual barriers?                                        | Save the Children, 2019<br>Latchem, 2012<br>Hasley et al., 2021<br>UNESCO, 2019 |
| <b>Accessibility and inclusion</b> | Ensure that the OER is appropriately localised and customised.                                                                                                     | Is the OER appropriately localised?<br>Can be the OER customised?                                                                                            | Latchem, 2012                                                                   |
| <b>Accessibility and inclusion</b> | Ensure that the institution demonstrates strong leadership in initiating and supporting educationally sound and ethical operations in inclusive and accessible OER | Does your institution demonstrate strong leadership in initiating and supporting educationally sound and ethical operations in inclusive and accessible OER? | Jung & Lee, 2014<br>Save the children 2019                                      |



| ASPECT              | RECOMMENDATION                                                                                                                                                                                                                                                                             | QUESTION                                                                                                                                                                                                                                                                                     | SOURCE                                                                   |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>Stakeholders</b> | Ensure that the OER is based on the community's needs.                                                                                                                                                                                                                                     | Is the OER based on national requirements and priorities, and community needs?                                                                                                                                                                                                               | Jung & Lee, 2014<br>Mhlanga, 2010<br>Latchem, 2012                       |
| <b>Stakeholders</b> | Ensure that the institution involves all stakeholders in the planning, conceptualisation and design processes, to ensure that the processes and outcomes accord with their needs and expectations.                                                                                         | Does your institution involve all stakeholders in the planning, conceptualisation and design processes? Are the processes and outcomes accord with the needs and expectations of the stakeholders?                                                                                           | Mhlanga, 2010<br>Latchem, 2012<br>Ehlers, 2012                           |
| <b>Stakeholders</b> | Ensure that the managers, teachers/trainers, facilitators/mentors, technical and other support personnel are recruited with care and appropriately trained in the systems, methods and technologies to be used in the programme.<br>Ensure that their performance is examined and reviewed | Are the managers, teachers/trainers, facilitators/mentors, technical, and other support personnel of your institution recruited with care and are appropriately trained in the systems, methods and technologies to be used in the programme?<br>Is their performance examined and reviewed? | Latchem, 2012<br>Binns & Otto, 2006<br>Ehlers, 2012<br>Mhlanga, 2010     |
| <b>Stakeholders</b> | Ensure that there are efficient and effective systems for communication between the stakeholders, especially between faculty and students to supervise and stimulate students.                                                                                                             | Are there efficient and effective systems for communication between the stakeholders, especially between faculty and students to supervise and stimulate students?                                                                                                                           | Latchem, 2012<br>Bakken & Bridges, 2011                                  |
| <b>Stakeholders</b> | State and make visible institutional policies, aligned with national policy, with which the students and faculty are expected to comply.                                                                                                                                                   | Are the institutional policies aligned with national policy?<br>Are the institutional policies visible so students and faculty can comply them?                                                                                                                                              | Framework of Quality Matters Program, 2011<br>QLT, 2022<br>Mhlanga, 2010 |





| ASPECT              | RECOMMENDATION                                                                                                                                                                                                                                                           | QUESTION                                                                                                                                                                                                                                                    | SOURCE                                  |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| <b>Stakeholders</b> | Ensure that the students receive (or have access to) information about programs, including admission requirements, tuition and fees, books and supplies, technical and proctoring requirements, and student support services prior to admission and course registration. | Do students receive (or have access to) information about programs, including admission requirements, tuition and fees, books and supplies, technical and proctoring requirements, and student support services prior to admission and course registration? | Sloan, 2013                             |
| <b>Stakeholders</b> | Ensure that the institution develops competences, encourages and rewards (e.g., with course load reductions) its faculty and staff regarding OER development, use and reuse.                                                                                             | Does your institution develop competences, encourage and reward (e.g., with course load reductions) its faculty and staff regarding OER development, use and reuse?                                                                                         | Jung & Lee, 2014<br>Hasley et al., 2021 |
| <b>Stakeholders</b> | Ensure that the institution promotes and supports research in OER by its faculty/staff and applies these research findings in improving its OER.                                                                                                                         | Does your institution promote and support research in OER by its faculty/staff?<br>Does your institution apply these research findings in improving its OER?                                                                                                | Jung & Lee, 2014                        |
| <b>Stakeholders</b> | Ensure that people responsible for the analysis, design, development, implementation, and evaluation of the programme (internal staff/external experts) are adequately qualified.                                                                                        | Are the people responsible for the analysis, design, development, implementation, and evaluation of the programme (internal staff/external experts) adequately qualified?                                                                                   | Ehlers, 2012                            |



| ASPECT              | RECOMMENDATION                                                                                                                                                                                                                                                                                                                      | QUESTION                                                                                                                                                                                                                                                                                                     | SOURCE                                                                  |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| <b>Stakeholders</b> | <p>Organise learning activities and create learning materials with the stakeholders, especially with the most marginalised, working as a team, and with a democratic organisation (sharing leadership, open meetings, community decisions).</p> <p>Everyone must feel committed, supporting each other as a cohesive community.</p> | Do you organise learning activities and create learning materials with the stakeholders, especially with the most marginalised, working as a team, and with a democratic organisation, supporting each other as a cohesive community?                                                                        | Latchem, 2012<br>Murillo, 2020                                          |
| <b>Stakeholders</b> | Establish partnerships with who can support the OER.                                                                                                                                                                                                                                                                                | Do you or your organisation establish partnerships with who can support the OER?                                                                                                                                                                                                                             | Latchem, 2012<br>Kear & Rosewell 2012                                   |
| <b>Stakeholders</b> | Ensure that your institution establishes collaborative relationships with staff, administration, learners, and other key stakeholders like parents, volunteers, associations, local NGOs, businesses, funding bodies, media channels, community carers, and governmental and non-governmental education institutions.               | Does your institution establish collaborative relationships with staff, administration, learners, and other key stakeholders like parents, volunteers, associations, local NGOs, businesses, funding bodies, media channels, community carers, and governmental and non-governmental education institutions? | Mhlanga, 2010<br>Latchem, 2012<br>Jung & Lee, 2014<br>Jung et al., 2016 |
| <b>Evaluation</b>   | Ensure that the materials are developed by people with expertise in UDL and are subjected to rigorous quality reviews, as peer review, before used by learners.                                                                                                                                                                     | Are the materials developed by people with expertise in UDL and are subjected to rigorous quality reviews, as peer review, before use by learners?                                                                                                                                                           | Mhlanga, 2010<br>OER Africa, 2015                                       |
| <b>Evaluation</b>   | There is an effective, trans-                                                                                                                                                                                                                                                                                                       | Is there an effective, transpar-                                                                                                                                                                                                                                                                             | Latchem, 2012                                                           |



| ASPECT            | RECOMMENDATION                                                                                                                                                                                                                                                                               | QUESTION                                                                                                                                                                                                            | SOURCE                                                                                                               |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
|                   | parent and democratic management of the programme, human resources and communications.                                                                                                                                                                                                       | ent and democratic management of the programme, human resources and communications?                                                                                                                                 |                                                                                                                      |
| <b>Evaluation</b> | Ensure that the institution has a standing policy on periodical reviewing the effectiveness of systems and procedures for the planning, designing, approval, development, documentation, and accreditation of courses.                                                                       | Does your institution have a standing policy on periodically reviewing the effectiveness of systems and procedures for the planning, designing, approval, development, documentation, and accreditation of courses? | Mhlanga, 2010                                                                                                        |
| <b>Evaluation</b> | Ensure that the institution establishes guidelines related to all the quality criteria such as: minimum standards used for course development and design, and delivery of online instructions (such as course syllabus elements, course materials, assessment strategies, faculty feedback). | Does your institution establish guidelines related to all the quality criteria such as: minimum standards used for course development and design, and delivery of online instructions?                              | Mhlanga, 2010<br>Williams et al., 2012                                                                               |
| <b>Evaluation</b> | Ensure that the institution collects basic ongoing quantitative and qualitative data on who, when, where and why OERs are being used for setting priorities and planning changes for quality improvement                                                                                     | Does your institution collect basic ongoing quantitative and qualitative data on who, when, where and why OERs are being used for setting priorities and planning changes for quality improvement?                  | Leicester University, 2010<br>Mhlanga, 2010<br>Sloan, 2013<br>Ehlers, 2012<br>Jung & Lee, 2014                       |
| <b>Evaluation</b> | Ensure to receive formal and informal feedback from students and faculty informing ongoing continuous im-                                                                                                                                                                                    | Does your institution receive formal and informal feedback from students and faculty, informing ongoing continuous improvement of the programme,                                                                    | Mhlanga, 2010<br>Ehlers, 2012<br>Binns & Otto, 2006<br>Bakken & Bridges, 2011<br>Sloan, 2013<br>Leicester University |



| ASPECT            | RECOMMENDATION                                                                                                                                                                                                                                                                               | QUESTION                                                                                                                                                                                                                                       | SOURCE                                                           |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
|                   | provement of the programme, design, materials, methods, tasks, schedule, and assessment tools for students.                                                                                                                                                                                  | design, materials, methods, tasks, schedule, and assessment tools for students?                                                                                                                                                                | 2010                                                             |
| <b>Evaluation</b> | Ensure that auditing systems are in place regularly to compare goals, budgets, outcomes and impacts, to ensure effectiveness, clarity, utility, and appropriateness of the OER.                                                                                                              | Are there auditing systems in place that regularly compare goals, budgets, outcomes and impacts, to ensure effectiveness, clarity, utility, and appropriateness of the OER?                                                                    | Sloan, 2013<br>Latchem, 2012<br>Mhlana, 2010<br>Jung & Lee, 2014 |
| <b>Evaluation</b> | Generalise good practices as a reference or model for other organisations: innovative, effective, sustainable, replicable and that can be maintained over time and produce lasting effects. Disseminate them in forums and media so that there is a positive impact of awareness in society. | Does your institution generalise good practices as a reference or model for other organisations?<br>Do you or does your institution disseminate good practices in forums and media so that there is a positive impact of awareness in society? | Latchem, 2012<br>Murillo, 2020                                   |
| <b>Evaluation</b> | Ensure that the OER is sustainable from an economic and temporal point of view.                                                                                                                                                                                                              | Is the OER sustainable from an economic and temporal point of view?                                                                                                                                                                            | EDIA-CEDEC, 2020<br>Save the children, 2019                      |

## 2 Didactics

The didactic dimension refers to teaching, instructional, or educational models in which the IA-OER is integrated to promote specific user learning.

### 2.1 Competences and objectives

**Competences and objectives:** this subdimension refers to actual knowledge, skills and attitudes promoted in students by the IA-OER.

| ASPECT              | RECOMMENDATION                                                                                                                                          | QUESTION                                                                                                                        | SOURCE                                |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| Learning objectives | Ensure that the purpose, benefit and importance of the WBLE are presented clearly.                                                                      | Are the purpose, benefit and importance of the WBLE presented clearly?                                                          | Baya'a et al., 2009<br>Sreedher, 2009 |
| Learning objectives | Ensure that on the cover page of the resource, the objective of the resource and the final product or result are made explicit.                         | On the cover page of the resource, are the objective of the resource and the final product or result made explicit?             | EDIA-CEDEC, 2020                      |
| Learning objectives | Check that objectives and outcomes are up to date and relevant to course curriculum.                                                                    | Are the objectives and outcomes up to date and relevant to course curriculum?                                                   | Mhlanga, 2010<br>Binns & Otto, 2006   |
| Learning objectives | Ensure that all learning objectives are achievable, measurable, relevant, clearly stated, and concise for learners and teachers.                        | Are all learning objectives achievable, measurable, relevant, clearly stated, and concise for learners and teachers?            | Haughey, 2005<br>Ehlers, 2012         |
| Learning objectives | Check that the content includes all the learning objectives in a clear and precise manner and is appropriate to the level of knowledge of the learners. | Does the content include all the learning objectives in a clear and precise manner and is appropriate to my level of knowledge? | UNE 71362<br>Achieve, 2011            |
| Learning objectives | Ensure that the didactic objectives are achievable by                                                                                                   | Are the didactic objectives achievable by the target audience according to the profile required                                 | UNE 71362                             |



| ASPECT                     | RECOMMENDATION                                                                                                                                                                 | QUESTION                                                                                                                                                               | SOURCE                                                          |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
|                            | the target audience according to the profile required in the resource itself.                                                                                                  | in the resource itself?                                                                                                                                                |                                                                 |
| <b>Learning objectives</b> | Ensure that contents have a balance of knowledge and skills, and contribute to the achievement of the stated course and module/unit learning objectives.                       | Do contents have a balance of knowledge and skills, and contribute to the achievement of the stated course and module/unit learning objectives?                        | Quality Matters Program, 2011<br>REA-Andalucía<br>Mhlanga, 2010 |
| <b>Learning objectives</b> | Ensure that the resource includes a didactic justification.                                                                                                                    | Does the resource include a didactic justification?                                                                                                                    | REA-Andalucía                                                   |
| <b>Learning objectives</b> | Ensure that the achieved learning experience is relevant to professional and social practice.                                                                                  | Is the achieved learning experience relevant to professional and social practice?                                                                                      | Ehlers, 2012                                                    |
| <b>Learning objectives</b> | Check that there is a link between what has been learned and the target's living environment (professional and/or social).                                                     | Is there a link between what has been learned and the target's living environment (professional and/or social)?                                                        | UNE-71362                                                       |
| <b>Learning objectives</b> | Ensure that the OER enables the transfer of the new knowledge and skills to different tasks, problems or contexts to consolidate their learning or construct personal meaning. | Does the OER enable the transfer of the new knowledge and skills to different tasks, problems or contexts to consolidate their learning or construct personal meaning? | Jung et al., 2016                                               |
| <b>Competences</b>         | Ensure that the competences and/or skills to be developed are clearly specified; they are coherent with the objectives and the target group.                                   | Are the competences and/or skills to be developed clearly specified?<br>Are they coherent with the objectives and the target group?                                    | UNE 71362                                                       |



| ASPECT                              | RECOMMENDATION                                                                                                                                                                         | QUESTION                                                                                                                                                                             | SOURCE             |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Competences</b>                  | Provide a list of technical competencies necessary for course completion, identifying and delineating the role/extent the online environment plays in the total course.                | Does the OER provide a list of technical competencies necessary for course completion, identifying and delineating the role/extent the online environment plays in the total course? | QLT, 2022          |
| <b>Competences</b>                  | Ensure that the OER provides a sound structure for knowledge and skills development.                                                                                                   | Does the OER provide a sound structure for knowledge and skills development?                                                                                                         | Jung et al., 2016  |
| <b>Competences</b>                  | Ensure that the OER helps the students recall, relate or apply prior knowledge, skills, experience, etc.                                                                               | Does the OER help students recall, relate or apply prior knowledge, skills, experience, etc.?                                                                                        | Jung et al., 2016  |
| <b>Competences</b>                  | Ensure that students learn appropriate methods for effective research, including assessment of the validity of resources and the ability to master resources in an online environment. | Do students learn appropriate methods for effective research, including assessment of the validity of resources and the ability to master resources in an online environment?        | Sloan, 2013        |
| <b>Implementation of objectives</b> | Ensure that the course description includes the purpose and format of the course, as well as prerequisite knowledge and competencies, if applicable.                                   | Does the course description include the purpose and format of the course, as well as prerequisite knowledge and competencies, if applicable?                                         | QLT, 2022          |
| <b>Implementation of objectives</b> | Align the OER provisions with the institution's vision, mission and goals.                                                                                                             | Are the OER provisions aligned with the institution's vision, mission and goals?                                                                                                     | Jung & Lee, 2014   |
| <b>Implementation of objectives</b> | Ensure that the programme is amended to conform to                                                                                                                                     | Is the programme amended to conform to the recently developed national curriculum?                                                                                                   | Binns & Otto, 2006 |



| ASPECT                              | RECOMMENDATION                                                                                                                                                                | QUESTION                                                                                                                                                          | SOURCE                 |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
|                                     | the recently developed national curriculum.                                                                                                                                   |                                                                                                                                                                   |                        |
| <b>Implementation of objectives</b> | Align the course content and assignments with the state's content standards, common core curriculum, or other accepted content standards.                                     | Are the course content and assignments aligned with the state's content standards, common core curriculum, or other accepted content standards?                   | Bakken & Bridges, 2011 |
| <b>Implementation of objectives</b> | Ensure that knowledge cover the descriptors of the selected key competences and the assessment criteria of the specific competences                                           | Does the knowledge cover the descriptors of the selected key competences and the assessment criteria of the specific competences?                                 | REA-Andalucía          |
| <b>Implementation of objectives</b> | Ensure that there is an alignment among learning goals, activities, assessments, and learner characteristics.                                                                 | Is there an alignment with learning goals, activities, assessments, and learner characteristics?                                                                  | LORI, 2009             |
| <b>Implementation of objectives</b> | Ensure that there is coherence between theory and practice (between the content and proposed activities or tasks, objectives, competencies, and the purpose of the resource). | Is there coherence between theory and practice (between the content and proposed activities or tasks, objectives, competencies, and the purpose of the resource)? | EDIA-CEDEC, 2020       |



## 2.2 Methodology

**Methodology:** this subdimension alludes to a possible sequence of strategies, activities or tasks in which the IA-OER participates. These strategies facilitate the achievement of the established objectives.

| ASPECT                                | RECOMMENDATION                                                                                                                                       | QUESTION                                                                                                                                     | SOURCE                                |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>Learner control and management</b> | Ensure that students can control and manage their learning by choosing content and learning paths.                                                   | Can students control and manage their learning by choosing content and learning paths?                                                       | UNE-71362<br>Alberta University, 2014 |
| <b>Learner control and management</b> | Ensure that students can estimate the time needed for learning according to their capabilities.                                                      | Are students able to estimate the time needed for learning according to their capabilities?                                                  | UNE-71362                             |
| <b>Learner control and management</b> | Ensure that students are informed of the time required to complete all learning activities.                                                          | Are students informed of the time required to complete all learning activities?                                                              | UNE-71362<br>Alberta University, 2014 |
| <b>Learner control and management</b> | Make a distinction between compulsory and recommended study/reading materials.                                                                       | Is there a distinction between compulsory and recommended study/reading materials?                                                           | Ehlers, 2012                          |
| <b>Learner-centred design</b>         | Ensure that teaching and learning are learner-centred, and this is considered during the course development process.                                 | Are the teaching and learning learner-centred? Is this considered during the course development process?                                     | Latchem, 2012<br>Sloan, 2013          |
| <b>Learner control and management</b> | Ensure that the course design reflects a clear understanding of all students' needs and incorporates varied ways to learn and master the curriculum. | Does the course design reflect a clear understanding of all students' needs and incorporates varied ways to learn and master the curriculum? | Bakken & Bridges, 2011                |
| <b>Learner control and management</b> | Ensure that the location, timing, and pace of the program are appropriate for learners and their circumstances.                                      | Are the location, timing, and pace of the program appropriate for learners and their circumstances?                                          | Latchem, 2012                         |



| ASPECT                                | RECOMMENDATION                                                                                                                  | QUESTION                                                                                                             | SOURCE                                                    |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>Learner control and management</b> | Check that the program workload and schedule are consistent with the requirements of the target group.                          | Are the program workload and schedule consistent with the requirements of the target group?                          | Ehlers, 2012                                              |
| <b>Learner control and management</b> | Ensure that the program combines an appropriate mix of teaching, group study, independent study, and practical work.            | Does the program combine an appropriate mix of teaching, group study, independent study, and practical work?         | Latchem, 2012                                             |
| <b>Interaction and collaboration</b>  | Ensure that the instructor uses specific strategies to create a presence in the course and facilitate student participation.    | Does the instructor use specific strategies to create a presence in the course and facilitate student participation? | Sloan, 2013<br>Framework of Quality Matters Program, 2011 |
| <b>Learner control and management</b> | Ensure that the communication and collaboration with teachers and classmates are facilitated through chats, email, forums, etc. | Are communication and collaboration with teachers and classmates facilitated through chats, email, forums, etc?      | Baya'a et al., 2009                                       |
| <b>Learner control and management</b> | Provide students with a way to interact with other students in the school and in an online community.                           | Are students provided with a way to interact with other students in the school and in an online community?           | Sloan, 2013                                               |
| <b>Learner control and management</b> | Ensure that learning activities provide opportunities for interaction that support active learning.                             | Do learning activities provide opportunities for interaction that support active learning?                           | Quality Matters Program, 2011                             |
| <b>Learner control and management</b> | Ensure that activities are interactive and foster active learning, encouraging frequent and ongoing peer-to-peer engagement.    | Are activities interactive and foster active learning, encouraging frequent and ongoing peer-to-peer engagement.     | Bakken & Bridges, 2011                                    |
| <b>Learner control and management</b> | Include social and collaborative activities in the program methodologies to contribute to the                                   | Are there social and collaborative activities included in the program methodologies? Do                              | Ehlers, 2012                                              |



| ASPECT                                                | RECOMMENDATION                                                                                                                                                                                                                              | QUESTION                                                                                                                                                                                                                        | SOURCE                        |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
|                                                       | achievement of learning objectives.                                                                                                                                                                                                         | social and collaborative activities contribute to the achievement of learning objectives?                                                                                                                                       |                               |
| <b>Learning activities and types of methodologies</b> | Ensure that the course methodology is clearly explained, showing how selected methods lead to the achievement of learning objectives.                                                                                                       | Is the course methodology clearly explained, showing how selected methods lead to the achievement of learning objectives?                                                                                                       | Ehlers, 2012<br>QLT, 2022     |
| <b>Learning activities and types of methodologies</b> | Ensure that the purpose of instructional materials, the methodologies, and how they are to be used for learning activities are clearly explained.                                                                                           | Are the purpose of instructional materials, the methodologies, and how they are to be used for learning activities clearly explained?                                                                                           | Quality Matters Program, 2011 |
| <b>Learning activities and types of methodologies</b> | Ensure that each learning unit is presented with the necessary and organised elements to guide learners in achieving the learning objectives.                                                                                               | Each learning unit is presented with the necessary and organised elements to guide learners in achieving the learning objectives.                                                                                               | Ehlers, 2012                  |
| <b>Learning activities and types of methodologies</b> | Ensure that multiple learning resources and materials are available to increase student success before the course begins.                                                                                                                   | Are multiple learning resources and materials available to increase student success before the course begins?                                                                                                                   | Bakken & Bridges, 2011        |
| <b>Learning activities and types of methodologies</b> | Ensure that active methodologies such as flipped classroom, project-based learning, problem-based learning, inquiry-based learning, learning as a service, design thinking, gamification, etc., are applied to promote class participation. | Are active methodologies such as flipped classroom, project-based learning, problem-based learning, inquiry-based learning, learning as a service, design thinking, gamification, etc., applied to promote class participation? | OER Africa, 2017              |
| <b>Learning activities and types of methodologies</b> | Ensure that learning activities promote the achievement of the stated learning objectives                                                                                                                                                   | Do learning activities promote the achievement of the stated learning objectives?                                                                                                                                               | Quality Matters Program, 2011 |



| ASPECT                                                | RECOMMENDATION                                                                                                       | QUESTION                                                                                                  | SOURCE             |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------|
| <b>Learning activities and types of methodologies</b> | Check that exercises and activities are clearly written and supported by accurate answer keys or scoring guidelines. | Are exercises and activities clearly written and supported by accurate answer keys or scoring guidelines? | Achieve Inc., 2011 |

### 2.3 Contents

**Contents:** this subdimension shows how the materials and information are presented in the IA-OERs to reach the established objectives.

| ASPECT          | RECOMMENDATION                                                                                                                                                                                                                     | QUESTION                                                                                                                                                                                                               | SOURCE                                                                                                                                                                                                                                                                                                |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Accuracy</b> | Ensure that the content (information) of the materials is accurate, free of bias or advertising.                                                                                                                                   | Is content (information) of the materials accurate, free of bias or advertising?                                                                                                                                       | Jung & Lee, 2014<br>Mhlanga, 2010<br>Leicester University, 2010<br>JISC Tiger Project, 2011<br>MERLOT<br>Baya'a et al., 2009<br>Bakken & Bridges, 2011<br>Sreedher, 2009<br><a href="https://www.achieve.org/files/AchieveOERRubrics_1.pdf">https://www.achieve.org/files/AchieveOERRubrics_1.pdf</a> |
| <b>Accuracy</b> | Ensure that the content based on factual knowledge and professional thinking, reflecting the ways in which knowledge is conceptualised within the domain.                                                                          | Is the content based on factual knowledge and professional thinking, reflecting the ways in which knowledge is conceptualised within the domain?                                                                       | Haughey, 2005                                                                                                                                                                                                                                                                                         |
| <b>Accuracy</b> | Ensure that there is an accurate and balanced presentation of significant concepts grounded in the discipline, appropriate level of detail, and reusability in varied contexts, e.g., connecting concepts with the subject matter. | Is there an accurate and balanced presentation of significant concepts grounded in the discipline, appropriate level of detail, and reusability in varied contexts, e.g., connecting concepts with the subject matter? | Jung & Lee, 2014<br>Mhlanga, 2010<br>Leicester University, 2010<br>JISC Tiger Project, 2011<br>MERLOT<br>Baya'a et al., 2009<br>Bakken & Bridges, 2011<br>Sreedher, 2009<br>Achieve Inc., 2011                                                                                                        |



| ASPECT          | RECOMMENDATION                                                                                                                                                                                                 | QUESTION                                                                                                                                                                                               | SOURCE                                                                                                                                                  |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Accuracy</b> | Ensure that the OER provides sufficient comprehensive information so effectively that the target audience should be able to understand the subject matter without needing additional explanation or materials. | Does the OER provide sufficient comprehensive information so effectively that the target audience should be able to understand the subject matter without needing additional explanation or materials? | <a href="https://www.achieve.org/files/AchieveOERRubrics_1.pdf">https://www.achieve.org/files/AchieveOERRubrics_1.pdf</a><br>Baya'a et al., 2009        |
| <b>Accuracy</b> | Ensure that multimedia content is compliant with quality standards specified during design.                                                                                                                    | Is multimedia content compliant with quality standards specified during design?                                                                                                                        | Sreedher, 2009<br>UNE-71362<br>EDIA-CEDEC, 2020                                                                                                         |
| <b>Accuracy</b> | Ensure that all resources and materials used are appropriately cited and referenced, offering the original source of the resource.                                                                             | Are all the resources and materials used appropriately cited and referenced, offering the original source of the resource?                                                                             | Quality Matters Program, 2011<br>Alberta University, 2014<br>EDIA-CEDEC, 2020<br>QLT, 2018<br>Ehlers, 2012                                              |
| <b>Accuracy</b> | Ensure that contents are of sufficient rigor, depth and breadth, do not omit relevant data, and do not include duplicate or marginal data.                                                                     | Are the contents of sufficient rigor, depth and breadth, not omitting relevant data, and avoiding duplicate or marginal data?                                                                          | Bakken & Bridges, 2011<br>Leicester University, 2010<br>Baya'a et al., 2009<br>Checklist "CREA con DUA"<br>JISC Tiger Project, 2011<br>OER Africa, 2015 |
| <b>Accuracy</b> | Ensure that contents are based on sound learning theories and lead learners to develop their knowledge rather than simply memorise facts.                                                                      | Are contents based on sound learning theories that lead learners to develop their knowledge rather than simply memorise facts?                                                                         | Mhlanga, 2010                                                                                                                                           |
| <b>Accuracy</b> | Identify clearly the main concepts addressed in the OER.                                                                                                                                                       | Are the main concepts addressed in the OER clearly identified?                                                                                                                                         | Achieve Inc., 2011<br>Baya'a et al., 2009                                                                                                               |



| ASPECT                     | RECOMMENDATION                                                                                                                                                                                                             | QUESTION                                                                                                                                                                                                       | SOURCE                                                                                                                                                  |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Accuracy</b>            | Ensure that the language is clear, concise and readily comprehensible for non-specialists.                                                                                                                                 | Is the language clear, concise and readily comprehensible for non-specialists?                                                                                                                                 | Bakken & Bridges, 2011<br>Leicester University, 2010<br>Baya'a et al., 2009<br>Rubio Pulido, 2018<br>JISC Tiger Project, 2011<br>OER Africa, 2015       |
| <b>Accuracy</b>            | Check that the OER uses correct grammar, punctuation, and spelling.                                                                                                                                                        | Are grammar, punctuation, and spelling correct?                                                                                                                                                                | Bakken & Bridges, 2011<br>Leicester University, 2010<br>Baya'a et al., 2009<br>Checklist "CREA con DUA"<br>JISC Tiger Project, 2011<br>OER Africa, 2015 |
| <b>Updated</b>             | Ensure that the OER is updated periodically to ensure that the content and links are current and relevant.                                                                                                                 | Is the OER updated periodically to ensure that the content and links are current and relevant?                                                                                                                 | Bakken & Bridges, 2011<br>Jung & Lee, 2014<br>MERLOT<br>Baya'a et al., 2009                                                                             |
| <b>Updated</b>             | Ensure that contents are current or timeless (not needing to be updated).                                                                                                                                                  | Are contents current or timeless (not needing to be updated)?                                                                                                                                                  | UNE-71362<br>Jung et al., 2016<br>Alberta University, 2014<br>Quality Matters Program, 2011                                                             |
| <b>Updated</b>             | Provide information about the degree of contents updated.                                                                                                                                                                  | Is information about the degree of contents updated provided?                                                                                                                                                  | UNE-71362                                                                                                                                               |
| <b>Adaptation to users</b> | Ensure that contents are adequate to the learners' prior knowledge, abilities and needs (age, diversity, educational level, motivation, preferences, personal situation, socioeconomic status, ethnicity, religion, etc.). | Are contents adequate to the learners' prior knowledge, abilities and needs (age, diversity, educational level, motivation, preferences, personal situation, socioeconomic status, ethnicity, religion, etc.)? | Jung & Lee, 2014<br>UNE-71362<br>Sreedher, 2009<br>Murillo, 2020<br>Achieve Inc., 2011<br>Alberta University, 2014<br>Quality Matters Program, 2011     |
| <b>Adaptation to users</b> | Ensure that contents are appropriate to the duration of the study.                                                                                                                                                         | Are contents appropriate for the duration of the study?                                                                                                                                                        | Jung & Lee, 2014<br>MERLOT                                                                                                                              |



| ASPECT                      | RECOMMENDATION                                                                                                                                                                                                                | QUESTION                                                                                                                                                                                                         | SOURCE                                                                                                                                            |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Adaptation to users</b>  | Ensure that contents are flexible and transformable with respect to learners' needs and preferences, allowing for different learning paths.                                                                                   | Are contents flexible and transformable with respect to learners' needs and preferences, allowing for different learning paths?                                                                                  | Inclusive Learning Design Handbook<br>MERLOT<br>Ehlers, 2012<br>Save the children, 2019<br>UNE-71362<br>EDIA-CEDEC, 2020                          |
| <b>Content organisation</b> | Ensure that contents are organised in a logical sequence from simpler to more complex concepts.                                                                                                                               | Are contents organised in a logical sequence from simpler to more complex concepts?                                                                                                                              | Bakken & Bridges, 2011<br>Leicester University, 2010<br>Baya'a et al., 2009<br>Rubio Pulido, 2018<br>JISC Tiger Project, 2011<br>OER Africa, 2015 |
| <b>Content organisation</b> | Ensure that the course is structured to support independent learning and meets the requirements of the programme.                                                                                                             | Is the course structured to support independent learning? Does it meet the requirements of the programme?                                                                                                        | Mhlanga, 2010                                                                                                                                     |
| <b>Content organisation</b> | Ensure that the content of the course/ programme is coherently presented and subdivided in logical manageable sequences of modules and/or lessons/sections, organised in such a way that enables comprehension and retention. | Is the content of the course/programme coherently presented and subdivided in logical manageable sequences of modules and/or lessons/sections, organised in such a way that enables comprehension and retention? | Ehlers, 2012<br>Leicester University, 2010                                                                                                        |
| <b>Content organisation</b> | Ensure that instructions make clear how to get started and where to find various course components.                                                                                                                           | Do instructions make clear how to get started and where to find various course components?                                                                                                                       | Quality Matters Program, 2011                                                                                                                     |
| <b>Content organisation</b> | Ensure that instructional manuals are detailed and self-explanatory.                                                                                                                                                          | Are instructional manuals detailed and self-explanatory?                                                                                                                                                         | Sreedher, 2009                                                                                                                                    |



| ASPECT                      | RECOMMENDATION                                                                                                                                                                                                            | QUESTION                                                                                                                                                                                                          | SOURCE                                                                   |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>Content organisation</b> | Ensure that each unit and lesson include an overview describing objectives, activities, assignments, assessments, and resources to provide multiple learning opportunities for students to master the content.            | Does each unit and lesson include an overview describing objectives, activities, assignments, assessments, and resources to provide multiple learning opportunities for students to master the content?           | Bakken & Bridges, 2011                                                   |
| <b>Content organisation</b> | Ensure that students are introduced to the general description, purpose and structure of the course (table of contents) in the very beginning of the resource.                                                            | Are students introduced to the general description, purpose and structure of the course (table of contents) in the very beginning of the resource?                                                                | Quality Matters Program, 2011<br>EDIA-CEDEC, 2020<br>Baya'a et al., 2009 |
| <b>Content organisation</b> | Ensure that the OER clearly states expectations for academic integrity, use of copyrighted materials, plagiarism and netiquette (Internet etiquette) regarding lesson activities, discussions, and e-mail communications. | Does the OER clearly state expectations for academic integrity, use of copyrighted materials, plagiarism and netiquette (Internet etiquette) regarding lesson activities, discussions, and e-mail communications? | Bakken & Bridges, 2011<br>Quality Matters Program, 2011                  |
| <b>Content organisation</b> | Ensure that a distinction is made between compulsory and recommended study/reading materials.                                                                                                                             | Is there a distinction between compulsory and recommended study/reading materials?                                                                                                                                | Ehlers, 2012<br>Quality Matters Program, 2011                            |
| <b>Content organisation</b> | Ensure that privacy policies are clearly stated.                                                                                                                                                                          | Are privacy policies clearly stated?                                                                                                                                                                              | Bakken & Bridges, 2011                                                   |
| <b>Content organisation</b> | Ensure that the OER includes examples, formative self-assessment activities, and other means of support to enable learners to study independently or at a distance.                                                       | Does the OER include examples, formative self-assessment activities, and other means of support to enable learners to study independently or at a distance?                                                       | Latchem, 2012<br>Jung & Lee, 2014                                        |





| ASPECT                      | RECOMMENDATION                                                                                                                                                                 | QUESTION                                                                                                                                                                | SOURCE                        |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| <b>Content organisation</b> | Ensure that course instructions articulate or link to the institution's accessibility policies and services.                                                                   | Do course instructions articulate or link to the institution's accessibility policies and services?                                                                     | Quality Matters Program, 2011 |
| <b>Content organisation</b> | Ensure that terms are consistent to name the same material, activity or section.                                                                                               | Are terms consistent to name the same material, activity or section?                                                                                                    | Rubio Pulido, 2018            |
| <b>Content organisation</b> | Ensure that graphic style is uniform to facilitate reading.                                                                                                                    | Is graphic style uniform to facilitate reading?                                                                                                                         | EDIA-CEDEC, 2020              |
| <b>Content organisation</b> | Ensure that learning scenarios have a semantically structured design that ensures access, comprehension, order and learning progress.                                          | Do learning scenarios have a semantically structured design that ensures access, comprehension, order and learning progress?                                            | UNE-71362                     |
| <b>Content organisation</b> | Ensure that the OER include text, audio or video orientation and introductory components to familiarise the learners with the courses and their instructors/support personnel. | Does the OER include text, audio or video orientation and introductory components to familiarise the learners with the courses and their instructors/support personnel? | Jung & Lee, 2014              |
| <b>Content organisation</b> | Ensure that the grading information is presented to the learners at the very beginning of the course and is easily accessible throughout the course.                           | Is the grading information presented to the learners at the very beginning of the course, being also easily accessible throughout the course?                           | Alberta University, 2014      |
| <b>Content organisation</b> | Ensure that the self-introduction by the instructor is appropriate and available online.                                                                                       | Is the self-introduction by the instructor appropriate and available online?                                                                                            | Quality Matters Program, 2011 |
| <b>Content organisation</b> | Ensure that the minimum technical skills expected of the student are clearly stated.                                                                                           | Are minimum technical skills expected of the student clearly stated?                                                                                                    | Quality Matters Program, 2011 |



| ASPECT                           | RECOMMENDATION                                                                                                                                                   | QUESTION                                                                                                                                             | SOURCE                                    |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Content organisation</b>      | Provide a course outline/syllabus and course description.                                                                                                        | Is there a course outline/syllabus and course description?                                                                                           | Alberta University, 2014                  |
| <b>Content organisation</b>      | Ensure that modules/lessons/units provide a glossary of terms associated to the learning materials.                                                              | Do modules/lessons/units provide a glossary of terms associated to the learning materials?                                                           | Ehlers, 2012                              |
| <b>Content organisation</b>      | Ensure that the OER includes a syllabus list indicating which textbooks and materials are required or recommended.                                               | Does the OER include a syllabus list indicating which textbooks and materials are required or recommended?                                           | QLT, 2026                                 |
| <b>Content organisation</b>      | Ensure that students have access to resources that enrich the course content.                                                                                    | Do students have access to resources that enrich the course content?                                                                                 | Bakken & Bridges, 2011                    |
| <b>Help, contact and support</b> | Check that the OER provides an option for sending suggestions and feedback.                                                                                      | Does the OER provide an option for sending suggestions and feedback?                                                                                 | Rubio Pulido, 2018                        |
| <b>Help, contact and support</b> | Ensure that the OER provides help and documentation files and links for students and teachers including contextual assistance.                                   | Does the OER provide help and documentation files and links for students and teachers including contextual assistance?                               | Haughey, 2005<br>Alberta University, 2014 |
| <b>Help, contact and support</b> | Ensure that information on contact persons is available, mentioning name, area of responsibility, and contact details.                                           | Is information on contact persons available, mentioning name, area of responsibility, and contact details?                                           | Ehlers, 2012                              |
| <b>Help, contact and support</b> | Ensure that there is an explanation of how the academic support services and resources can help students succeed in the course and how students can access them. | Is there an explanation of how the academic support services and resources can help students succeed in the course and how students can access them? | Quality Matters Program, 2011             |



| ASPECT                           | RECOMMENDATION                                                                                                                                           | QUESTION                                                                                                                                          | SOURCE       |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>Help, contact and support</b> | Ensure that students are provided access to library professionals and resources that help them to deal with the overwhelming number of online resources. | Are students provided with access to library professionals and resources that help them to deal with the overwhelming number of online resources? | Sloan, 2013  |
| <b>Help, contact and support</b> | Ensure that search functions are available for forums, discussion boards or blogs, whenever such tools are utilised.                                     | Are search functions available for forums, discussion boards or blogs, whenever such tools are utilised?                                          | Ehlers, 2012 |



## 2.4 Pedagogical principles

**Pedagogical principles:** These principles refer to personal and social development models that support IA-OER, including values like equality, equity, and the integration and coherence of the didactic process.

| ASPECT                      | RECOMMENDATION                                                                                                                                                                                                                                                             | QUESTION                                                                                                                                                              | SOURCE                  |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Social participation</b> | Ensure that the OER promotes the creation of a community of practice for sharing learning resources and professional development resources.                                                                                                                                | Does the OER promote the creation of a community of practice for sharing learning resources and professional development resources?                                   | Halsey et al., 2021     |
| <b>Social participation</b> | Ensure that the OER empowers participants and fosters a sense of self-worth and initiative.                                                                                                                                                                                | Does the OER empower participants and foster a sense of self-worth and initiative?                                                                                    | Latchem, 2012           |
| <b>Social participation</b> | Ensure that the instructor allows students to introduce themselves at the start of the course, building a sense of community.                                                                                                                                              | Does the instructor provide an opportunity for students to introduce themselves at the start of the course to build a sense of community?                             | QLT, 2022               |
| <b>Social participation</b> | Promote collaborative work by creating diverse work groups that are inclusive of students from different backgrounds, without segregation based on academic performance or disability, potentially involving students from different courses, families, and professionals. | Is collaborative work promoted by forming diverse groups inclusive of all abilities, possibly involving students from different courses, families, and professionals? | Save the children, 2019 |
| <b>Social participation</b> | Ensure that everyone's participation is facilitated according to their abilities, creating learning opportunities for all.                                                                                                                                                 | Is everyone's participation facilitated, creating learning opportunities tailored to individual abilities?                                                            | Save the children, 2019 |



| ASPECT                 | RECOMMENDATION                                                                                                                                                                                                                               | QUESTION                                                                                                                                                               | SOURCE                                                                                                             |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>OER development</b> | Ensure that program development uses available resources (e.g., technology, human resources, infrastructure) effectively and efficiently to achieve learning goals and outcomes.                                                             | Does program development ensure that available resources (technology, human resources, infrastructure) are used efficiently to support learning goals and outcomes?    | Mhlanga, 2010                                                                                                      |
| <b>OER development</b> | Include didactic suggestions in the OER, such as appropriate methodology, mode of use (e.g., classroom or self-learning), and educational context for teachers and students, including estimated learning time and required prior knowledge. | Does the OER include didactic suggestions like instructions on methodology, mode of use, and educational context, including estimated learning time and prerequisites? | UNE-71362<br>Latchem, 2012<br>Sreedher, 2009                                                                       |
| <b>Motivation</b>      | Ensure that OER modules and resources are engaging, motivating, and attractive.                                                                                                                                                              | Are the OER modules and resources engaging, motivating, and attractive?                                                                                                | Camilleri & Tannhäuser, 2012<br>Mhlanga, 2010<br>UNE-71362<br>EDIA-CEDEC, 2020<br>Nesbit et al, 2007<br>LORI, 2009 |
| <b>Motivation</b>      | Ensure that content (titles, tasks, images, multimedia) facilitates motivation, encouraging students to attend, be motivated, and see the relevance of their learning.                                                                       | Do content features like titles, tasks, images, and multimedia motivate students to attend, engage, and understand the relevance of what they learn?                   | UNESCO, 2015<br>UNE-71362<br>EDIA-CEDEC, 2020<br>Latchem, 2012                                                     |
| <b>Motivation</b>      | Ensure that the OER promotes autonomy and supports students' self-esteem and well-being, fostering respect and responsibility in their educational journey.                                                                                  | Does the OER promote autonomy, self-esteem, and well-being, encouraging students to take responsibility and actively engage in their education?                        | Murillo, 2020                                                                                                      |



| ASPECT            | RECOMMENDATION                                                                                                                       | QUESTION                                                                                                                                   | SOURCE                   |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <b>Motivation</b> | Ensure that the OER uses a positive tone, contributing to a supportive learning environment.                                         | Does the OER have a positive tone that contributes to a supportive learning environment?                                                   | Alberta University, 2014 |
| <b>Motivation</b> | Ensure that the OER maintains students' attention and interest.                                                                      | Does the OER maintain students' attention and interest?                                                                                    | Jung et al., 2016        |
| <b>Motivation</b> | Ensure that multimedia and humour (if present) are relevant, appropriate to the subject, and do not slow down web page presentation. | Are multimedia and humour (if present) relevant, clear, and appropriate to the subject matter, without slowing down web page presentation? | Baya'a et al., 2009      |
| <b>Motivation</b> | Ensure that the media mix (e.g., combination of audio, graphics) is engaging.                                                        | Is the media mix (combination of audio, graphics, etc.) engaging?                                                                          | Sreedher, 2009           |

## 2.5 Student Assessment

**Student Assessment:** This section refers to the criteria and tools used to evaluate the learning achieved by students before, during, and after using the IA-OER relative to the established objectives.

| ASPECT                  | RECOMMENDATION                                                                                                                                                   | QUESTION                                                                                                                                      | SOURCE                                                                                                  |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Assessment tools</b> | Ensure that expectations for assignment completion, assessment instruments, grading policy, and instructor response are clearly outlined in the course syllabus. | Does the course syllabus clearly outline assignment completion expectations, assessment instruments, grading policy, and instructor response? | Sloan, 2013<br>EDIA-CEDEC, 2020<br>Quality Matters Program, 2011<br>Bakken & Bridges, 2011<br>QLT, 2021 |
| <b>Assessment tools</b> | Ensure that formative, interim, and summative assessments, along with scoring rubrics, are included to support the core curriculum.                              | Does the OER include formative, interim, and summative assessments with scoring rubrics?                                                      | Hasley et al., 2021<br>Williams et al., 2012                                                            |



| ASPECT                  | RECOMMENDATION                                                                                                                                                     | QUESTION                                                                                                                               | SOURCE                                                                                    |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Assessment tools</b> | Ensure that knowledge, skills, attendance, attitudes, motivation, achievement, and participation of learners are evaluated throughout the program.                 | Are knowledge, skills, attendance, attitudes, motivation, achievement, and participation of learners evaluated throughout the program? | Latchem, 2012<br>Sreedher, 2009<br>Binns & Otto, 2006<br>Jung et al., 2016                |
| <b>Assessment tools</b> | Ensure that the learner evaluation scheme includes a variety of assessment techniques, such as self-assessment, peer review, and tracking of platform interaction. | Does learner evaluation schemes include diverse assessment techniques (self-assessment, peer review, platform interaction tracking)?   | Sreedher, 2009<br>Ehlers, 2012<br>Quality Matters Program, 2011<br>Bakken & Bridges, 2011 |
| <b>Assessment tools</b> | Ensure that assessment tools are objective, valid, and reliable.                                                                                                   | Are the assessment tools objective, valid, and reliable?                                                                               | Sreedher, 2009                                                                            |
| <b>Assessment tools</b> | Provide opportunities in the course design for appropriate instructor-student interaction, including frequent and timely feedback on student progress.             | Does the OER design provide opportunities for instructor-student interaction with timely and frequent feedback on student progress?    | Bakken & Bridges, 2011                                                                    |
| <b>Assessment tools</b> | Ensure records are maintained for all faculty and their performance within the program.                                                                            | Are there records of all faculty and their performance within the program?                                                             | Latchem, 2012                                                                             |
| <b>Assessment tools</b> | Scaffold a continuous formative assessment process with timely, constructive feedback to support a positive learning environment throughout the course.            | Is there a continuous formative assessment process with timely, constructive feedback supporting a positive learning environment?      | Mhlanga, 2010                                                                             |
| <b>Assessment tools</b> | Make reporting tools (e.g., e-Portfolio) available to document and track learning outcomes and the learning process.                                               | Are reporting tools available to document and track learning outcomes and processes?                                                   | Camilleri & Tannhäuser, 2012<br>UNE-71362                                                 |



| ASPECT                            | RECOMMENDATION                                                                                                                                                                | QUESTION                                                                                                               | SOURCE                                                                                                  |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Assessment tools</b>           | Ensure that assessment methods, such as selected responses, constructed responses, or group work, require students to demonstrate proficiency in intended concepts or skills. | Do the assessment methods require students to demonstrate proficiency in intended concepts or skills?                  | Achieve, 2011                                                                                           |
| <b>Assessment tools</b>           | Maintain a record of students' interactions (e.g., results, completion time, attempts) for review by both teacher and student.                                                | Is there a record of students' interactions (e.g., results, completion time, attempts) for teacher and student review? | UNE-71362                                                                                               |
| <b>Assessment tools</b>           | Ensure that assessment types are suitable for an online learning environment.                                                                                                 | Are assessment types suitable for an online learning environment?                                                      | Sloan, 2013                                                                                             |
| <b>Learner-Centred Assessment</b> | Ensure there are processes for recognising prior learning and identifying at-risk learners.                                                                                   | Are there processes for recognising prior learning and identifying at-risk learners?                                   | Mhlanga, 2010<br>Nesbit, et al., 2007<br>MERLOT                                                         |
| <b>Learner-Centred Assessment</b> | Ensure alignment among learning goals, activities, assessments, and learner characteristics.                                                                                  | Is there alignment among learning goals, activities, assessments, and learner characteristics?                         | LORI, 2009                                                                                              |
| <b>Learner-Centred Assessment</b> | Ensure that learners are linked to an appropriate tutor during OER implementation for tracking progress, advising, mentoring, and support.                                    | Are learners linked to a tutor for tracking progress, advising, mentoring, and support during OER implementation?      | Mhlanga, 2010<br>Binns & Otto, 2006<br>Sloan, 2013<br>Ehlers, 2012                                      |
| <b>Learner-Centred Assessment</b> | Ensure that assessment design and communication systems consider learners' special needs.                                                                                     | Do assessment design and communication systems consider learners' special needs?                                       | Sloan, 2013<br>EDIA-CEDEC, 2020<br>Quality Matters Program, 2011<br>Bakken & Bridges, 2011<br>QLT, 2022 |





| ASPECT                            | RECOMMENDATION                                                                                                                              | QUESTION                                                                                                                                        | SOURCE                                                                                                  |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Learner-Centred Assessment</b> | Ensure that all students experience optimal learning and positive outcomes by setting individualised objectives, so everyone can succeed.   | Do all students achieve optimal learning and positive outcomes?<br>Are individualised objectives and evaluation tools in place to support this? | Save the children, 2019                                                                                 |
| <b>Learner-Centred Assessment</b> | Ensure that learners' progress and achievements are monitored and evaluated.                                                                | Are learners' progress and achievements monitored and evaluated?                                                                                | Ehlers, 2012                                                                                            |
| <b>Learner-Centred Assessment</b> | Ensure that selected assessment tools are sequenced, varied, and appropriate for student work.                                              | Are assessment tools sequenced, varied, and appropriate for the work being assessed?                                                            | Quality Matters Program, 2011<br>OER Africa, 2015                                                       |
| <b>Learner-Centred Assessment</b> | Ensure that learner evaluations align with the learning objectives.                                                                         | Are learner evaluations aligned with the learning objectives?                                                                                   | Quality Matters Program, 2011<br>OER Africa, 2015                                                       |
| <b>Learner-Centred Assessment</b> | Encourage instructors to ask students to share or reflect on their learning goals.                                                          | Does the instructor ask students to share or reflect on their learning goals?                                                                   | Sloan, 2013<br>EDIA-CEDEC, 2020<br>Quality Matters Program, 2011<br>Bakken & Bridges, 2011<br>QLT, 2022 |
| <b>Learner-Centred Assessment</b> | Ensure that learner records (e.g., contact details, assessment results) are detailed, updated, and accessible to tutors and academic staff. | Are learner records detailed, up to date, and accessible to tutors, academic, and administrative staff?                                         | Mhlanga, 2010<br>Latchem, 2012                                                                          |
| <b>Learner-Centred Assessment</b> | Ensure student information remains confidential.                                                                                            | Is student information kept confidential?                                                                                                       | Bakken & Bridges, 2011                                                                                  |
| <b>Learner-Centred Assessment</b> | Ensure that learner inquiries, complaints, and issues are handled promptly, efficiently and effectively.                                    | Are learner inquiries, complaints, and issues handled promptly, efficiently, and effectively?                                                   | storage platforms                                                                                       |

| ASPECT                            | RECOMMENDATION                                                                       | QUESTION                                                                    | SOURCE                                        |
|-----------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------|
| <b>Learner-Centred Assessment</b> | Provide individual feedback to learners on time.                                     | Is individual feedback provided to learners in a timely manner?             | Ehlers, 2012<br>Quality Matters Program, 2011 |
| <b>Learner-Centred Assessment</b> | Ensure that feedback has an analytical approach and addresses the solution approach. | Does feedback take an analytical approach, addressing the solution process? | Ehlers, 2012                                  |



### 3 Dimension: ICT

In this context, Information and Communication Technology (ICT) refers to the software that enables open access to AI-OERs, facilitating inclusivity and intercultural interactions, as well as the platforms for storage.

#### 3.1 Technological base

**Technological base** refers to the human and technological resources necessary for creating, using, and reusing Open Educational Resources (OER).

| ASPECT                 | RECOMMENDATION                                                                                                                                                           | QUESTION                                                                                                                                                                                                                     | SOURCE                                                                           |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>OER Development</b> | Ensure OERs are compatible across multiple IT systems and can be accessed in content management and e-learning systems, with or without internet connection if feasible. | Can the OER be used in multiple environments and IT systems (interoperability)?<br>Can the OER be used in any content management system or e-learning system?<br>Can the OER be used with or without an Internet connection? | UNE-71362<br>Kurilovas, 2011<br>eQNet<br>EDIA-CEDEC, 2020<br>Hasley et al., 2021 |
| <b>OER Development</b> | Verify that the technologies used in OER are up to date.                                                                                                                 | Are the technologies used in OER current?                                                                                                                                                                                    | Quality Matters Program, 2011                                                    |
| <b>OER Development</b> | Provide technology that accommodates diverse users inclusively.                                                                                                          | Is the technology provided inclusive?                                                                                                                                                                                        | Latchem, 2012<br>Mhlanga, 2010                                                   |
| <b>OER Development</b> | Ensure stable and reliable OER performance, unaffected by technical or device changes.                                                                                   | Is technical robustness and stability assured, with resilience against errors and device changes?                                                                                                                            | UNE-71362<br>EDIA-CEDEC, 2020                                                    |
| <b>OER Development</b> | The OER platform should respond quickly and visibly to user actions, with technical support as needed.                                                                   | Does the technology respond quickly to user actions and provide technical support where necessary?                                                                                                                           | UNE-71362                                                                        |



| ASPECT                 | RECOMMENDATION                                                                                                                                                                                                                                                                                | QUESTION                                                                                                                                                                                                                                                                    | SOURCE                                                    |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>OER Development</b> | Use a documented technology plan that includes e-security measures (e.g., password protection, encryption, secure online or proctored exams, etc.) to ensure quality standards, FERPA compliance, and information integrity and validity.                                                     | Is there a documented technology plan that includes electronic security measures (e.g., password protection, encryption, secure online or proctored examinations, etc.) and ensures quality standards, FERPA compliance, and the integrity and validity of the information? | Sloan, 2013                                               |
| <b>OER Development</b> | Use tools to support users' communication, interaction and content creation.                                                                                                                                                                                                                  | Can tools be used to support users' communication, interaction and content creation?                                                                                                                                                                                        | Camilleri & Tannhäuser, 2012                              |
| <b>OER Development</b> | Provide information on etiquette in digital communication and collaboration tasks.                                                                                                                                                                                                            | Is information on etiquette in digital communication and collaboration tasks provided?                                                                                                                                                                                      | EDIA-CEDEC, 2020                                          |
| <b>OER Development</b> | Ensure that all file types composing the OER (i.e., text, image, audio, video or software application) are available in widely used formats (e.g., txt, MS Word, PDF, HTML, XML, WAV, mp3, mp4, jpeg, gif, etc.). If this is not the case, the necessary software to use it must be provided. | Are all file types that make up the OER (i.e., text, image, audio, video or software application) available in widely used formats (e.g., txt, MS Word, PDF, HTML, XML, WAV mp3, mp4, jpeg, gif, etc.)?                                                                     | UNE-71362<br>Kurilovas, 2011<br>eQNet<br>EDIA-CEDEC, 2020 |
| <b>OER Development</b> | Ensure that no system functions (e.g., zoom, text-to-speech) are disabled.                                                                                                                                                                                                                    | Are system functions (e.g., zoom, text-to-speech) disabled?                                                                                                                                                                                                                 | Affordable Learning Georgia                               |
| <b>OER Development</b> | Ensure that the virtual learning environment runs on a suitable server, which guarantees its stability.                                                                                                                                                                                       | Does the virtual learning environment run on a suitable server, which guarantees its stability?                                                                                                                                                                             | Ehlers, 2012                                              |



| ASPECT                             | RECOMMENDATION                                                                                                                                                                                                                                          | QUESTION                                                                                                                                                                                                                                     | SOURCE                                        |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <b>OER Development</b>             | Ensure that the virtual learning environment includes learning platforms and management systems, provision of e-learning material, reporting requirements, monitoring and updating of the e-learning system, online assessment and alternative formats. | Does the virtual learning environment include learning platforms and management systems, e-learning material provision, reporting requirements, monitoring and updating of the e-learning system, online assessment and alternative formats? | Williams et al., 2012                         |
| <b>OER Development</b>             | Ensure that the virtual learning environment is accessible through different browsers and operating systems.                                                                                                                                            | Is the virtual learning environment accessible through different browsers and operating systems?                                                                                                                                             | Ehlers, 2012                                  |
| <b>OER Development</b>             | Ensures that students can easily access the technologies required in the course.                                                                                                                                                                        | Can students easily access the technologies required in the course?                                                                                                                                                                          | Quality Matters Program, 2011<br>Ehlers, 2012 |
| <b>OER Development</b>             | Provide learners and teachers with the necessary technology for OER.                                                                                                                                                                                    | Do students and teachers have the necessary technology for OER?                                                                                                                                                                              | Ehlers, 2012<br>Latchem, 2012<br>QLT, 2022    |
| <b>Institution-OER Development</b> | Ensure the institution provides an adequate and reliable media/technology infrastructure to develop, deliver and manage OER.                                                                                                                            | Does the institution provide an adequate and reliable technological and media infrastructure to develop, deliver and manage OER?                                                                                                             | Jung & Lee, 2014                              |
| <b>Institution-OER Development</b> | Ensure the institution regularly evaluates the quality and use of media/technology and infrastructure.                                                                                                                                                  | Does the institution regularly evaluate the quality and uses of media/technology and infrastructure?                                                                                                                                         | Jung & Lee, 2014                              |
| <b>Institution-OER Development</b> | Ensure the institution takes into account that OER are developed in a way that is appropriate to students' computer systems, network                                                                                                                    | Does the institution ensure that OER are developed in a way that is appropriate to students' computer systems, network speeds, etc.?                                                                                                         | Jung & Lee, 2014                              |



| ASPECT                             | RECOMMENDATION                                                                                                                                                                                            | QUESTION                                                                                                                                                                                                       | SOURCE                                               |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
|                                    | speeds, etc.                                                                                                                                                                                              |                                                                                                                                                                                                                |                                                      |
| <b>Institution-OER Development</b> | Ensure the institution develops forms of OER (e.g., modules, learning objects, videos, audios, tests, software, full courses, learning materials, etc.) appropriate to learners' needs and circumstances. | Does the institution develop forms of OER (e.g., modules, learning objects, videos, audios, quizzes, software, full courses, learning materials, etc.) appropriate to the needs and circumstances of learners? | Jung & Lee, 2014                                     |
| <b>Institution-OER Development</b> | Ensure the institution achieves the best possible use of available courses and learning materials through the design, adoption or adaptation of OER.                                                      | Does the institution make the best possible use of available courses and learning materials by designing, adopting or adapting OER?                                                                            | Jung & Lee, 2014                                     |
| <b>Institution-OER Development</b> | Ensure the institution ensures that all online education students, regardless of where they are located, have access to appropriate library/learning resources to support the courses they are taking.    | Does the institution ensure that all online education students, have access to adequate library/learning resources to support the courses they are taking regardless of their location?                        | Sloan, 2013                                          |
| <b>Institution-OER Development</b> | Ensure the institution provides technical support and course management assistance to students, teachers and coordinators.                                                                                | Does the institution offer technical support and course management assistance to students, faculty and coordinators?                                                                                           | Bakken & Bridges, 2011                               |
| <b>Technological Re-usability</b>  | Ensure that an OER can be modified, e.g., from a configuration file, from a plain text file or because it is provided together with its source code or an authoring tool.                                 | Can an OER be modified, e.g., from a configuration file, from a plain text file or because it is provided together with its source code or an authoring tool?                                                  | Kurilovas, 2011<br>EDIA-CEDEC, 2020<br>Latchem, 2012 |



| ASPECT                            | RECOMMENDATION                                                                                                                                                                   | QUESTION                                                                                                                                                                | SOURCE                                                |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>Technological Re-usability</b> | Structure the OER by separating data, application logic and presentation.                                                                                                        | Is the OER structured in order to separate data, presentation and application logic?                                                                                    | Kurilovas, 2011                                       |
| <b>Technological Re-usability</b> | Flexibility and scalability of the formative module and/or learning materials from a technical point of view.                                                                    | Is there flexibility and scalability in the training module and/or learning materials from a technical point of view?                                                   | Camilleri & Tannhäuser, 2012                          |
| <b>Technological Re-usability</b> | Ensure that the technical aspect of the learning resources can be adapted and customised to take into account the user's background, language, country of origin, special needs, | Can the technical aspects of learning resources be adapted and customised to take into account the user's background, language, country of origin, special needs, etc.? | Camilleri & Tannhäuser, 2012                          |
| <b>Technological Re-usability</b> | Ensure the availability of content in different languages, helpdesk, technical and didactic support.                                                                             | Are the contents available in different languages, helpdesk, technical and didactic support?                                                                            | Camilleri & Tannhäuser, 2012                          |
| <b>Technological Re-usability</b> | Share best practices to prevent reinventing the wheel.                                                                                                                           | Can best practices be shared to prevent reinventing the wheel?                                                                                                          | Jung et al., 2016                                     |
| <b>Technological Re-usability</b> | Ensure that the course architecture allows the online instructor to add content, activities and assessments to extend learning opportunities.                                    | Does the course architecture allow the online instructor to add content, activities and assessments to expand learning opportunities?                                   | Bakken & Bridges, 2011                                |
| <b>Formation</b>                  | Ensure that faculty, staff and students are supported in the development and use of new technologies and skills.                                                                 | Are faculty, staff and students supported in the development and use of new technologies and skills?                                                                    | Sloan, 2013<br>Mhlanga, 2010<br>Williams et al., 2012 |
| <b>Formation</b>                  | Ensure that links to or explanations of technical assistance are available in the                                                                                                | Are technical support links or explanations available in the                                                                                                            | Sloan, 2013                                           |



| ASPECT                  | RECOMMENDATION                                                                                                                                           | QUESTION                                                                                                                                                           | SOURCE                                         |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
|                         | course (i.e., each course provides suggested solutions to possible technical problems and/or links to technical assistance).                             | course (i.e., each course provides suggested solutions to possible technical problems and/or links to technical assistance)?                                       |                                                |
| <b>Formation</b>        | Train teachers and learners in the use of equipment, facilities and communication and information systems.                                               | Are teachers and students trained in the use of communication and information equipment, facilities and systems?                                                   | Latchem, 2012                                  |
| <b>Learning support</b> | Ensure that course tools and media support learner engagement and guide the learner to become active.                                                    | Do the course tools and media encourage student participation and help them become active learners?                                                                | Quality Matters Program, 2011<br>Haughey, 2005 |
| <b>Learning support</b> | Ensure that the tools and media facilitate the achievement of the learning objectives/outcomes of the course.                                            | Do the tools and means facilitate the achievement of the learning objectives/outcomes of the course?                                                               | QLT, 2022<br>Sloan, 2013                       |
| <b>Learning support</b> | Use available technology to improve the quality of learning and that learners are given sufficient support to make the best use of available technology. | Is the available technology used to improve the quality of learning and that learners are sufficiently supported to make the best use of the available technology? | Mhlanga, 2010                                  |
| <b>Learning support</b> | Ensure that the technology tools and resources used in the course enable learner participation and active learning.                                      | Do the technological tools and resources used in the course allow for student participation and active learning?                                                   | QLT, 2022<br>Quality Matters Program, 2011     |





### 3.2 Usability

**Usability** measures the effectiveness, efficiency, and satisfaction with which a specific user in a specific context can use a product/design to achieve a goal.

| ASPECT                   | RECOMMENDATION                                                                                                              | QUESTION                                                                        | SOURCE                                            |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------|
| <b>Personalisation</b>   | Give the user the option to customise font type.                                                                            | Is there an option to customise the font type used in the OER?                  | UNE-71362                                         |
| <b>Personalisation</b>   | Give the user the option to customise font size.                                                                            | Is there an option to customise the font size used in the OER?                  | UNE-71362                                         |
| <b>Personalisation</b>   | Allow customisation of font colour.                                                                                         | Is there an option to customise the font colour used in the OER?                | UNE-71362                                         |
| <b>Personalisation</b>   | Allow customisation of background colour.                                                                                   | Is there an option to customise the background colour used in the OER?          | UNE-71362                                         |
| <b>Personalisation</b>   | Allow customisation of additional interface aspects.                                                                        | Is there an option to customise other aspects of the OER?<br><br>Which aspects? | UNE-71362                                         |
| <b>Format and design</b> | Ensure an intuitive interface (e.g., clear instructions are easy to locate and allow understanding of learning objectives). | Is the interface intuitive for users to understand learning objectives?         | UNE-71362                                         |
| <b>Format and design</b> | Provide clear instructions for using the learning object.                                                                   | Are clear instructions for using the learning object provided?                  | UNE-71362<br>EDIA-CEDEC, 2020,<br>Kurilovas, 2011 |
| <b>Format and design</b> | Use a language familiar to the users.                                                                                       | Is the language used familiar to users?                                         | Haughey, 2005<br>MERLOT                           |
| <b>Format and design</b> | Ensure that the design of the                                                                                               | Is the user interface design                                                    | UNE-71362<br>EDIA-CEDEC, 2020                     |



| ASPECT                   | RECOMMENDATION                                                                            | QUESTION                                                                                | SOURCE                   |
|--------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------------------------|
|                          | user interface reflects the hierarchy of information.                                     | clearly defined and reflective of information hierarchy?                                | Kurilovas, 2011          |
| <b>Format and design</b> | Maintain a clear and easy to use interface.                                               | Is the user interface clear and easy to use?                                            | JISC Tiger Project, 2011 |
| <b>Format and design</b> | Ensure high-quality OER design.                                                           | Are high-quality OER design in place?                                                   | MERLOT                   |
| <b>Format and design</b> | Include help functions for common user issues.                                            | Are there help functions identifying common user problems and solutions?                | Kurilovas, 2011          |
| <b>Format and design</b> | Avoid, as far as possible, an abusive use of scrolling on each screen.                    | Is there an abusive use of scrolling on each screen?                                    | Rubio Pulido, 2018       |
| <b>Format and design</b> | Provide quick, visible and audible responses to user actions.                             | Are responses to user actions quick, visible and audible?                               | Kurilovas, 2011          |
| <b>Format and design</b> | Enable quick recovery from errors, informing users of errors and solutions if applicable. | Can users quickly recover from errors, and are they informed of the error and solution? | Kurilovas, 2011          |
| <b>Format and design</b> | Ensure that OER are interactive.                                                          | Is the OER interactive?                                                                 | MERLOT                   |
| <b>Format and design</b> | Ensure that the interactive element is relevant to learning.                              | Is the interactive element purposeful and directly related to learning?                 | Achieve Inc., 2011       |
| <b>Format and design</b> | Allow the user to exit at any point.                                                      | Is the user allowed to leave at any time?                                               | Kurilovas, 2011          |
| <b>Visual design</b>     | Ensure that the design of the user interface is clear and coherent.                       | Is the user interface design clear and coherent?                                        | Jung et al., 2016        |



| ASPECT        | RECOMMENDATION                                                                                              | QUESTION                                                                                           | SOURCE                          |
|---------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|---------------------------------|
| Visual design | Ensure that the design of the user interface is visually appealing and intuitive                            | Is the user interface design visually appealing and intuitive?                                     | Sreedher, 2009                  |
| Visual design | Ensure that layouts are clearly defined and reflect information hierarchy.                                  | Is the screen layout clearly defined and does it reflect the hierarchy of information?             | Sreedher, 2009                  |
| Visual design | Ensure that the screen layout of OER is suited to the learners' experience, knowledge and abilities.        | Is the screen layout of OER suited to the learners' experience, knowledge and abilities?           | Sreedher, 2009                  |
| Visual design | Ensure that the screen layout of OER helps learners comprehend the content and avoids distracting features. | Does the screen layout of OER help learners understand the content and avoid distracting elements? | Jung & Lee, 2014                |
| Visual design | Match UI aesthetics to the resource's theme.                                                                | Are the UI aesthetics compatible and appropriate to the resource's theme?                          | Jung & Lee, 2014                |
| Visual design | Minimise visual noise and unnecessary information.                                                          | Are UI aesthetics free of visual noise and unnecessary information?                                | UNE-71362, Rubio Pulido, 2018   |
| Visual design | Arrange UI components (e.g., buttons, menus) coherently.                                                    | Are UI components arranged coherently to aid navigation?                                           | UNE-71362<br>Rubio Pulido, 2018 |
| Visual design | Ensure that fonts are legible and visually appealing.                                                       | Are the fonts legible and visually appealing?                                                      | Jung & Lee, 2014                |
| Visual design | Ensure that OER is visually appealing.                                                                      | Is the OER visually appealing?                                                                     | MERLOT                          |
| Visual design | Match presentation and methods to learners' knowledge and skills.                                           | Are the presentation and methods used aligned with learners' knowledge and                         | Jung et al., 2016               |



| ASPECT               | RECOMMENDATION                                                                                                                                                                           | QUESTION                                                                                                                                                                       | SOURCE                                                                               |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|                      |                                                                                                                                                                                          | skills?                                                                                                                                                                        |                                                                                      |
| <b>Visual design</b> | Use visual and auditory design to enhance learning: Aesthetics; Navigation; User-friendly interface; Structured information; Personalisation.                                            | Does visual and auditory information design enhance learning and efficient mental processing?                                                                                  | Kurilovas 2011<br>LORI                                                               |
| <b>Navigation</b>    | Ensure that navigation is predictable and efficient.                                                                                                                                     | Is navigation predictable and efficient?                                                                                                                                       | Bakken & Bridges, 2011<br>Jung et al., 2016<br>QLT, 2024<br>Alberta University, 2014 |
| <b>Navigation</b>    | Ensure that navigation is logical.                                                                                                                                                       | Is navigation logical?                                                                                                                                                         | Alberta University, 2014<br>QLT, 2022                                                |
| <b>Navigation</b>    | Ensure that navigation guidance systems (e.g., breadcrumb trail and site map) are integrated into the OER site so that learners know where they are in relation to the rest of the site. | Are navigational guidance systems (e.g., breadcrumb trail and site map) integrated into the OER site so that learners know where they are in relation to the rest of the site? | Bakken & Bridges, 2011<br>Jung et al., 2016<br>QLT, 2022<br>Alberta University, 2014 |
| <b>Navigation</b>    | Ensure that navigation is clear and consistent throughout the course.                                                                                                                    | Is the navigation clear and consistent throughout the course?                                                                                                                  | Jung & Lee, 2014                                                                     |
| <b>Navigation</b>    | Ensure that undone navigation actions are available.                                                                                                                                     | Can navigation actions be undone?                                                                                                                                              | Kurilovas, 2011                                                                      |



### 3.3 Platforms

**Platforms:** Repositories are storage systems or databases where educational resources are stored and accessed.

| RECOMMENDATION                                                                                                                                                                                                                                 | QUESTION                                                                                                                                                     | SOURCE                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Ensure that a wide audience can find OER that are accessible to all learners. It is important to help teachers locate OER that are accessible to all learners, including those with disabilities and those with diverse needs and preferences. | Does it ensure that a wide audience can find OER accessible to all learners, including those with disabilities and those with diverse needs and preferences? | Hasley et al., 2021, UNESCO, 2021                      |
| Ensure that learners are helped to find their way through the repository and to find other websites where they can find appropriate OER.                                                                                                       | Are learners helped to find their way around the repository and to find other appropriate OER for the learner on other websites?                             | Jung & Lee, 2014                                       |
| Ensure tagging for ease of searching.                                                                                                                                                                                                          | Is there tagging to facilitate the search?                                                                                                                   | Leicester University, 2010<br>JISC Tiger Project, 2011 |
| Ensure OER are available in different formats for 'as is' use.                                                                                                                                                                                 | Are OER available in different formats for use "as is"?                                                                                                      | Leicester University, 2010                             |
| Ensure that OER are easy to download.                                                                                                                                                                                                          | Are OER easy to download?                                                                                                                                    | JISC Tiger Project, 2011<br>Leicester University, 2010 |
| Ensure that there are clear mechanisms for the end-user to submit revised or adapted versions of downloaded OER.                                                                                                                               | Is there a clear mechanism for the end-user to submit revised or reused versions of downloaded OER?                                                          | JISC Tiger Project, 2011                               |
| Ensure that internal testing can be performed on the repository.                                                                                                                                                                               | Is it possible to test the OER repository internally?                                                                                                        | JISC Tiger Project, 2011<br>Leicester University, 2010 |
| Ensure regular system backups for data availability.                                                                                                                                                                                           | Is the repository regularly backed up to ensure data availability?                                                                                           | Sloan, 2013                                            |



| RECOMMENDATION                                                                                                                                                                                                                                                                                                                                                                           | QUESTION                                                                                                                                      | SOURCE              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Include a recommender system (software which assists users to select resources based on specific user needs and characteristics) to ensure that content is appropriate for the users. Including a recommender system would also prompt developers towards greater awareness of accessibility features. The recommender system may not, however, be appropriate for all OER repositories. | Does the repository have a recommender system (software that helps users select resources based on their specific needs and characteristics)? | Brahim et al., 2020 |

### 3.4 Metadata

**Metadata:** A set of data that describes and gives information about the OER purpose, contents, etc.

| RECOMMENDATION                                                                       | QUESTION                                                                                  | SOURCE                                                                                     |
|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Ensure that OERs have appropriate and relevant metadata associated with them.        | Does the OER have adequate and relevant metadata associated with it?                      | UNE-71362<br>REA-Andalucía<br>Leicester University,<br>2010<br>JISC Tiger Project,<br>2011 |
| Use international standards (e.g., LOM, LOM-ES, Dublin Core) to create OER metadata. | Are international standards (e.g., LOM, LOM-ES, Dublin Core) used to create OER metadata? | UNE-71362                                                                                  |
| Ensure that there are metadata tags to indicate accessibility features.              | Does the OER have metadata tags indicating accessibility features?                        |                                                                                            |
| Ensure that there are metadata tags to indicate inclusion features.                  | Does the OER have metadata tags to indicate inclusion features?                           |                                                                                            |

## LOM (Learning Object Metadata)

**Category** Metadata elements associated to the OER



Co-funded by  
the European Union

Funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or EACEA. Neither the European Union nor the granting authority can be held responsible for them.

## LOM (Learning Object Metadata)

|                      |                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General</b>       | <ul style="list-style-type: none"> <li>1.1. Identifier</li> <li>1.2. Title</li> <li>1.3 Language</li> <li>1.4. Description</li> <li>1.5. Keyword</li> <li>1.6. Coverage</li> <li>1.7. Structure</li> <li>1.8. Aggregation level</li> </ul>                                                                                                                                               |
| <b>Life Cycle</b>    | <ul style="list-style-type: none"> <li>2.1. Version</li> <li>2.2. Status</li> <li>2.3. Contribute</li> </ul>                                                                                                                                                                                                                                                                             |
| <b>Meta-Metadata</b> | <ul style="list-style-type: none"> <li>3.1. Identifier</li> <li>3.2. Contribute</li> <li>3.3. Metadata Schema</li> <li>3.4 Language</li> </ul>                                                                                                                                                                                                                                           |
| <b>Technical</b>     | <ul style="list-style-type: none"> <li>4.1. Format</li> <li>4.2. Size</li> <li>4.3. Location</li> <li>4.4. Requirement</li> <li>4.5 Installation Remarks</li> <li>4.6. Other Platform Requirements</li> <li>4.7. Duration</li> </ul>                                                                                                                                                     |
| <b>Educational</b>   | <ul style="list-style-type: none"> <li>5.1. Interactivity type</li> <li>5.2. Learning resource type</li> <li>5.3. Interactivity level</li> <li>5.4. Semantic Density</li> <li>5.5. Intended End User Role</li> <li>5.6. Context</li> <li>5.7. Typical age range</li> <li>5.8. Difficult</li> <li>5.9 Typical learning time</li> <li>5.10. Description</li> <li>5.11. Language</li> </ul> |
| <b>Rights</b>        | <ul style="list-style-type: none"> <li>6.1. Cost</li> <li>6.2. Copyright and other restrictions</li> <li>6.3. Description</li> </ul>                                                                                                                                                                                                                                                     |
| <b>Relation</b>      | <ul style="list-style-type: none"> <li>7.1. Kind</li> <li>7.2. Resource</li> </ul>                                                                                                                                                                                                                                                                                                       |



## LOM (Learning Object Metadata)

**Annotation**

- 8.1. Entity
- 8.2 Date
- 8.3 Description

**Classification**

- 9.1. Purpose
- 9.2. Taxon path
- 9.3. Description
- 9.4 Keyword

## 393339999LOM -ES V1.0 (Norma UNE-71361:2010)

**Category** Metadata elements associated to the OER

**General**

- 1.1. Identifier
- 1.2. Title
- 1.3 Language
- 1.4. Description
- 1.5. Keyword
- 1.6. Coverage
- 1.7. Structure
- 1.8. Aggregation level

**Life Cycle**

- 2.1. Version
- 2.2. Status
- 2.3. Contribute

**Meta-Metadata**

- 3.1. Identifier
- 3.2. Contribute
- 3.3. Metadata Schema
- 3.4 Language

**Technical**

- 4.1. Format
- 4.2. Size
- 4.3. Location
- 4.4. Requirement
- 4.5 Installation Remarks
- 4.6. Other Platform Requirements
- 4.7. Duration

**Educational**

- 5.1. Interactivity type
- 5.2. Learning resource type
- 5.3. Interactivity level



Co-funded by  
the European Union

Funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or EACEA. Neither the European Union nor the granting authority can be held responsible for them.



## 393339999LOM -ES V1.0 (Norma UNE-71361:2010)

|                       |                                                                                                                                                                                                                 |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | 5.4. Semantic Density<br>5.5. Intended End User Role<br>5.6. Context<br>5.7. Typical age range<br>5.8. Difficult<br>5.9 Typical learning time<br>5.10. Description<br>5.11. Language<br>5.12. Cognitive process |
| <b>Rights</b>         | 6.1. Cost<br>6.2. Copyright and other restrictions<br>6.3. Description<br>6.4 Access                                                                                                                            |
| <b>Relation</b>       | 7.1. Kind<br>7.2. Resource                                                                                                                                                                                      |
| <b>Annotation</b>     | 8.1. Entity<br>8.2 Date<br>8.3 Description                                                                                                                                                                      |
| <b>Classification</b> | 9.1. Purpose<br>9.2. Taxon path<br>9.3. Description<br>9.4 Keyword                                                                                                                                              |

## Dublin Core

### METADATA ELEMENTS ASSOCIATED TO THE OER

|                    |                                                                                     |
|--------------------|-------------------------------------------------------------------------------------|
| <b>identifier</b>  | An unambiguous reference to the resource in a given context.                        |
| <b>title</b>       | Resource name.                                                                      |
| <b>creator</b>     | Person or organisation responsible for the creation of the content of the resource. |
| <b>contributor</b> | Person or organisation responsible for contributing to the content of the resource. |
| <b>publisher</b>   | Entity responsible for the publication of the resource.                             |



| Dublin Core        |                                                                                              |
|--------------------|----------------------------------------------------------------------------------------------|
| <b>subject</b>     | Subject of the content of the resource.                                                      |
| <b>description</b> | Information on the content of the resource.                                                  |
| <b>coverage</b>    | Scope or domain of the content of the resource (geographic, temporal or jurisdictional).     |
| <b>format</b>      | Physical or digital manifestation of the resource.                                           |
| <b>type</b>        | Nature or type of resource.                                                                  |
| <b>date</b>        | The date or time period associated with any type of event in the life cycle of the resource. |
| <b>relation</b>    | Reference to a resource related to the described resource.                                   |
| <b>source</b>      | Reference to a resource from which the current resource is derived.                          |
| <b>rights</b>      | Information on the rights associated with the resource.                                      |
| <b>language</b>    | Language of the content of the resource.                                                     |

### 3.5 Licensing

**Licensing:** this subdimension deals with permissions and restrictions governing the use of IA-OERs. It specifies how resources can be shared, reused, or modified.

| RECOMMENDATION                                                                                                                                                   | QUESTION                                                                                                                                        | SOURCE                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Ensure that the OER is provided under an open license, free to use or open code, and that this is specified clearly in the OER to declare its conditions of use. | Is the OER provided under an open license, free to use or open code, and is this clearly specified in the OER to declare its conditions of use? | Kurilovas, 2011<br>Halsey et al., 2021                             |
| If the license of the OER is Creative Commons, it must be at least CC BY or CC BY-SA to enable reuse or share.                                                   | If your OER license is Creative Commons, is it at least CC BY or CC BY-SA to enable reuse or share?                                             | Pereira et al., 2017<br>Bakken & Bridges, 2011<br>OER Africa, 2015 |
| The creation and inclusion of different elements (texts, videos, audio files,                                                                                    | Do you create or include different elements in the OER ensuring that                                                                            | Pereira et al., 2017<br>UNE 71362                                  |



| RECOMMENDATION                                                                                                                                                                                                                                                                                            | QUESTION                                                                                                                                                                                                      | SOURCE                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| and other materials) must abide by the OER license or other similar/compatible licenses such as CC0 or public domain                                                                                                                                                                                      | their licenses are similar or compatible with your OER?                                                                                                                                                       |                                                                    |
| All embedded elements in the OER must be published by the OER's authors (original work) or by those who hold the rights for their publication and dissemination, with their corresponding author citations.                                                                                               | Have all embedded elements in the OER been published by the OER's authors (original work) or by those who hold the rights for their publication and dissemination, with their corresponding author citations? | Pereira et al., 2017<br>Bakken & Bridges, 2011<br>OER Africa, 2015 |
| Links to external content (web pages, blogs, articles, articles/documents, presentations, videos, audio files, etc.) should direct users to the original resource or to links where the author and/or authors have authorised publication and dissemination, likewise authorising the use of the content. | Do you link external contents (web pages, blogs, articles, articles/documents, presentations, videos, audio files, etc.) to redirect users to the original resource?                                          | Pereira et al., 2017<br>Rubio Pulido, 2018                         |
| At least one of the stakeholders who created or used the OER must be in charge of managing the license of use and permissions.                                                                                                                                                                            | Is one of the stakeholders who created or uses an OER in charge of managing the license of use and permissions?                                                                                               | UNESCO, 2019                                                       |
| If the students are asked to create materials in a task, they should receive information about author's rights and licenses of use.                                                                                                                                                                       | If the students are asked to create materials in a task, do they receive information about author's rights and licenses of use?                                                                               | EDIA-CEDEC, 2020<br>Wakin, 2020                                    |
| The creator of an OER should investigate how other users are using the OER, and if they are respecting the licenses.                                                                                                                                                                                      | If you have created an OER, do you investigate how other users are using the OER, and if they are respecting the licenses?                                                                                    | JISC Tiger Project, 2011                                           |



## 4 Dimension: Diversity

This dimension refers to the personal and sociocultural circumstances of potential users of the IA-OER. It aims to detect possible education and adjust the IA-OER accordingly IA-OERs.

### 4.1 General

**General:** this refers to how the OER considers individual learning needs, avoiding bias, and the need to include specific content about inclusion, thus preventing violence or promoting a healthy lifestyle.

| ASPECT                           | RECOMMENDATION                                                                                                                                                                                              | QUESTION                                                                                                                                                                                                 | SOURCE                                                                                                                                        |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Individual Learning needs</b> | Ensure that the target group's learning needs and interest are taken into account to develop or adapt/reuse the contents of the OER, considering diverse fields of study, perspectives and formats.         | Are the target group's learning needs and interest taken into account to develop or adapt/reuse the contents of the OER, considering diverse fields of study, perspectives and formats?                  | Ehlers, 2012<br>Latchem, 2012<br>QLT, 2022,<br>Leicester University, 2010<br>Jung et al., 2016<br>Baya'a et al., 2009<br>Politis et al., 2014 |
| <b>Individual Learning needs</b> | Ensure that the target group's learning needs and interest are taken into account to develop or adapt/reuse the methodology, strategies, and pedagogical principles of the OER.                             | Are the target group's learning needs and interest taken into account to develop or adapt/reuse the methodology, institutional strategies, or pedagogical principles of the OER?                         | Ehlers, 2012<br>Alberta University, 2014                                                                                                      |
| <b>Individual Learning needs</b> | Ensure that the target group's learning needs are taken into account to adapt the evaluation methods.                                                                                                       | Are the target group's learning needs taken into account to adapt the evaluation methods?                                                                                                                | UNESCO, 2016                                                                                                                                  |
| <b>Individual Learning needs</b> | Ensure that the students' profiles include their strengths, weaknesses and preferences to identify possible barriers and opportunities, and to prepare the materials and methods to be included in the OER. | Do students' profiles include their strengths, weaknesses and preferences, in order to identify possible barriers and opportunities, and to prepare the materials and methods to be included in the OER? | Ávila et al., 2016                                                                                                                            |



| ASPECT                           | RECOMMENDATION                                                                                                                                                                                                                                                       | QUESTION                                                                                                                                                                                                                                      | SOURCE                                 |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>Individual Learning needs</b> | Ensure that learner profiles include at least this minimum information: demographic, technology profile, special educational needs, records of prior learning experiences and achievements, and language profile. Ensure that this information is regularly updated. | Do learner profiles include at least some information about demographics, technology profile, special educational needs, records of prior learning experiences and achievements, and language profile? Is this information regularly updated? | Mhlanga, 2010                          |
| <b>Individual Learning needs</b> | Ensure that learners are provided with clear information on how to use the OER and create 'personal learning environments' by remixing, manipulating, adding and sharing content according to their particular needs and interests.                                  | Are learners provided with clear information on how to use the OER and create 'personal learning environments' by remixing, manipulating, adding and sharing content according to their particular needs and interests?                       | Jung & Lee, 2014                       |
| <b>Individual Learning needs</b> | Ensure that the OER program is flexible, accessible and responsive to changing needs and circumstances of the learners.                                                                                                                                              | Is the OER program flexible, accessible and responsive to changing needs and circumstances of the learners?                                                                                                                                   | Latchem, 2012<br>Kurilovas, 2011       |
| <b>Individual Learning needs</b> | Ensure that someone in the faculty is in charge of analysing the learning support needs for different user groups or individuals.                                                                                                                                    | Is there anybody in the faculty in charge of analysing the learning support needs for different user groups or individuals?                                                                                                                   | Williams et al., 2012<br>Mhlanga, 2010 |
| <b>No bias</b>                   | Ensure that the OER contents are open and accessible for a wider community of teachers and learners.                                                                                                                                                                 | Are the OER contents open and accessible for a wider community of teachers and learners?                                                                                                                                                      | Jung et al., 2016                      |
| <b>No bias</b>                   | Ensure that planners and providers work with target                                                                                                                                                                                                                  | Do planners and providers work with target groups to assess their                                                                                                                                                                             | Latchem, 2012                          |



| ASPECT                          | RECOMMENDATION                                                                                                                                                                                                                                                     | QUESTION                                                                                                                                                                                                                                               | SOURCE                           |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
|                                 | groups to assess their knowledge needs, abilities, culture and circumstances.                                                                                                                                                                                      | knowledge needs, abilities, culture and circumstances?                                                                                                                                                                                                 |                                  |
| <b>No bias</b>                  | Ensure that materials reflect equity in illustrations, characters, especially heroes, lifestyle and language.                                                                                                                                                      | Do materials reflect equity in their illustrations, characters, especially heroes, lifestyle and language?                                                                                                                                             | UNESCO, 2015                     |
| <b>No bias</b>                  | Ensure that all students in a class are always present and integrated in all activities and learning situations.                                                                                                                                                   | Are all students in the class always present and integrated in all activities and learning situations?                                                                                                                                                 | Mhlanga, 2010<br>Murillo, 2020   |
| <b>Contents about inclusion</b> | Ensure that contents are objective and respectful, without ideological bias.                                                                                                                                                                                       | Is the content objective and respectful, without ideological bias?                                                                                                                                                                                     | UNE-71362<br>Hasley et al., 2021 |
| <b>Contents about inclusion</b> | Ensure that the OER syllabus is aligned with policies of diversity, equity, and inclusion, and the institution offering the OER provides means by which students can address violations of these policies and ideals (microaggressions or disrespectful comments). | Is the OER syllabus aligned with policies of diversity, equity, and inclusion? Does the institution offering the OER provide means by which students can address violations of these policies and ideals (microaggressions or disrespectful comments)? | QLT, 2022                        |
| <b>Contents about inclusion</b> | Ensure that the OER objectives achieve long-term impacts in the community, such as improving employment, social inclusion, security and health care, and lower corruption, pollution and conflicts,                                                                | Do the OER objectives achieve long-term impacts in the community, such as improving employment, social inclusion, security and health care, and lower corruption, pollution and conflicts?                                                             | Latchem, 2012                    |
| <b>Contents about inclusion</b> | Create OER that include life skills to learn, such as self-esteem, cooperation, communication, decision making,                                                                                                                                                    | Does the OER include contents about life skills to learn, such as self-esteem, cooperation, communication, decision making, problem solving,                                                                                                           | UNESCO, 2015                     |



| ASPECT                          | RECOMMENDATION                                                                                                                                                                                                                        | QUESTION                                                                                                                                                                                                                                          | SOURCE       |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|                                 | problem solving, critical thinking, respect and concern for others, managing feelings and stress, etc.                                                                                                                                | critical thinking, respect and concern for others, managing feelings and stress, etc?                                                                                                                                                             |              |
| <b>Contents about inclusion</b> | Create OER that include healthy lifestyles to learn, such as improving hygiene and healthy nutrition, promotion of mental health, and prevention of violence, substance abuse, early pregnancy, and illnesses such as STIs, HIV/AIDS. | Does the OER include contents about healthy lifestyles to learn, such as improving hygiene and healthy nutrition, promotion of mental health, and prevention of violence, substance abuse, early pregnancy, and illnesses such as STIs, HIV/AIDS? | UNESCO, 2015 |
| <b>Contents about inclusion</b> | Create OER that include contents about how protect children from harm, violence, and abuse.                                                                                                                                           | Does the OER include contents about how protect children from harm, violence, and abuse?                                                                                                                                                          | UNESCO, 2015 |

## 4.2 Gender, Language, Cultura, Race/Ethnicity, Age, Socio-economic

Socio-cultural and Socio-economic subdimensions:

- **Gender:** refers to how the OER has to consider the sexual identity declared by learners in accessing IA-OERs
- **Language:** refers to how the OER has to consider different languages used by learners in accessing IA-OERs
- **Culture:** refers to how the OER has to consider learners with different cultural identity, including religion.
- **Race/Ethnicity:** refers to how the OER has to consider learners with a different race or ethnicity
- **Age:** refers to how the OER has to take into account the varying age groups of learners.
- **Socio-economic:** refers to how the OER has to consider the academic level, profession, social class, family conditions, financial resources and place of residence of the learners in accessing IA-OERs

| ASPECT        | RECOMMENDATION                                                                       | QUESTION                                                                     | SOURCE               |
|---------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------|
| <b>Gender</b> | Ensure that content, visuals, and the instructional strategies are gender sensitive. | Are the content, visuals, and the instructional strategies gender sensitive? | Pereira et al., 2017 |



| ASPECT        | RECOMMENDATION                                                                                                                                                   | QUESTION                                                                                                                                                                                  | SOURCE                               |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| <b>Gender</b> | Ensure equality in pictures: include each sex in equal proportions.                                                                                              | Is there equality in pictures, including each sex in equal proportions?                                                                                                                   | Pereira et al., 2017                 |
| <b>Gender</b> | Ensure that male supremacy is not conveyed in group images.                                                                                                      | Is male supremacy not conveyed in group images?                                                                                                                                           | Pereira et al., 2017                 |
| <b>Gender</b> | Eliminate gender stereotypes in written language or images when depicting professional activities : male doctor / female nurse.                                  | Have stereotypes been eliminated in written language or images when depicting professional activities such as male doctor / female nurse?                                                 | Pereira et al., 2017                 |
| <b>Gender</b> | Take into account the recommendations from guides on non-sexist language and images in terms of clothing depicted, accessories, attitudes, use of space, etc.    | Have you taken into account the recommendations from guides on non-sexist language and images in terms of clothing depicted, accessories, attitudes, use of space, etc when creating OER? | Pereira et al., 2017                 |
| <b>Gender</b> | Ensure equality in written language that avoids the use of masculine generalisations and singular masculine terms to refer to collectives made up of both sexes. | Is there equality in written language that avoids the use of masculine generalisations and singular masculine terms to refer to collectives made up of both sexes?                        | Pereira et al., 2017                 |
| <b>Gender</b> | Use nouns that are not gender specific –such as “officer”– instead of gender-specific nouns like “policeman” or “police-woman”.                                  | Do you use nouns that are not gender specific –such as “officer”– instead of gender-specific nouns like “policeman” or “policewoman”?                                                     | Pereira et al., 2017                 |
| <b>Gender</b> | Try to ensure the visibility of the contributions made by women to the different fields of knowledge, adding positive role models.                               | Do you ensure the visibility of the contributions made by women to the different fields of knowledge, adding positive role models?                                                        | Pereira et al., 2017                 |
| <b>Gender</b> | Offer activities that promote                                                                                                                                    | Do you offer activities that pro-                                                                                                                                                         | Pereira et al., 2017<br>UNESCO, 2015 |





| ASPECT          | RECOMMENDATION                                                                                                                                                                                    | QUESTION                                                                                                                                                                                  | SOURCE                                                                                                     |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
|                 | the critical analysis of inequalities throughout the entire project.                                                                                                                              | mote the critical analysis of inequalities throughout the entire project?                                                                                                                 |                                                                                                            |
| <b>Gender</b>   | Introduce a more flexible curriculum and self-directed learning materials, since some girls may have many demands on their time, such as domestic work and care for siblings.                     | Have you introduced a more flexible curriculum and self-directed learning materials, since some girls may have many demands on their time, such as domestic work and care for siblings?   | UNESCO, 2015                                                                                               |
| <b>Gender</b>   | Make Mathematics, Science, and Language meaningful for All, including girls, linking their learning with daily life activities.                                                                   | Do you make Mathematics, Science, and Language meaningful for All, including girls, linking their learning with daily life activities?                                                    | UNESCO, 2015, pp. 55-56                                                                                    |
| <b>Language</b> | Ensure that written and iconic language is inclusive, without bias of age, culture, ethnicity, religion, sexual orientation, gender, disability, ability levels or socio-economic considerations. | Is the written and iconic language inclusive, without bias of age, culture, ethnicity, religion, sexual orientation, gender, disability, ability levels or socio-economic considerations? | CEDEC, 2020<br>Alberta University, 2014<br>CEDEC<br>Sreedher, 2009<br>UNESCO, 2019<br>Pereira et al., 2017 |
| <b>Language</b> | Ensure that the language used in the materials is accessible to all students: try to take into consideration language independence or low language dependence or multilingualism.                 | Is the language used in the materials accessible to all students, taking into consideration the use of language independence or low language dependence or multilingualism?               | Kurilovas, 2011<br>UNESCO, 2016<br>Elias et al., 2020                                                      |
| <b>Culture</b>  | Ensure that the teaching and learning strategies used are culturally appropriate and inclusive.                                                                                                   | Are the teaching and learning strategies used culturally appropriate and inclusive?                                                                                                       | Latchem, 2012<br>UNESCO, 2019                                                                              |
| <b>Culture</b>  | Ensure that the contents, tasks and instructional strategies reflect and respect the different cultural backgrounds of the students.                                                              | Do the contents, tasks and instructional strategies reflect and respect the different cultural backgrounds of the students?                                                               | Bakken & Bridges, 2011<br>Ehlers, 2012<br>Jung & Lee, 2014<br>Mhlanga, 2010<br>QLT, 2022<br>CEDEC, 2020    |



| ASPECT                     | RECOMMENDATION                                                                                                                                                                                                                                                           | QUESTION                                                                                                                                                                                                            | SOURCE                          |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
|                            |                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                     | Open Textbook rubric            |
| <b>Race/<br/>Ethnicity</b> | Ensure that there is visual representation of people with different physical features and/or that resemble the reality of the students.                                                                                                                                  | Is there a visual representation of people with different physical features and/or that resemble the reality of the students?                                                                                       | CEDEC, 2020<br>Jung & Lee, 2014 |
| <b>Race/<br/>Ethnicity</b> | Ensure that the text uses examples that are inclusive of a variety of races, ethnicities and backgrounds.                                                                                                                                                                | Does the text use examples that are inclusive of a variety of races, ethnicities and backgrounds?                                                                                                                   | Open Textbook rubric            |
| <b>Race/<br/>Ethnicity</b> | Ensure that content, visuals, and instructional strategies are racially sensitive.                                                                                                                                                                                       | Are the content, visuals, and instructional strategies racially sensitive?                                                                                                                                          | Sreedher, 2009<br>UNESCO, 2019  |
| <b>Age</b>                 | Ensure that the objectives and contents of an OER are based on the age and the entry level competence of the learners.                                                                                                                                                   | Are the objectives and contents of an OER based on the age and the entry level competence of the learners?                                                                                                          | Mhlanga, 2010<br>Daniela, 2022  |
| <b>Age</b>                 | Ensure that the learning environment helps learners develop the necessary discipline for increasingly independent learning and good work habits, especially for younger learners. Ensure that for older learners, peer support and collaborative learning are available. | Does the learning environment help learners develop the necessary discipline for increasingly independent learning and good work habits? For older learners, are peer support and collaborative learning available? | Mhlanga, 2010                   |
| <b>Age</b>                 | Ensure that the level of support is based on the age and entry level competence of the learners, and it varies for different types of learners in the programme and/or at different stages of the programme considered in the OER.                                       | Is the level of support based on the age and the entry level competence of the learners? Does the level of support change depending on the type of learners and the stages of the programme considered in the OER?  | Mhlanga, 2010<br>Daniela, 2022  |



| ASPECT                           | RECOMMENDATION                                                                                                                                                                                                                                                                          | QUESTION                                                                                                                                                                                                                                                                               | SOURCE                            |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>Socio-economic background</b> | Ensure that OER takes into account students who have a disadvantaged socioeconomic background and whose family members cannot provide them with access quality education, the opportunity to attend educational classes of interest, or the materials needed to learn specific content. | Does the OER take into account students who have a disadvantaged socioeconomic background and whose family members cannot provide them with access to quality education, the opportunity to attend educational classes of interest, or the materials needed to learn specific content? | Daniela, 2022                     |
| <b>Socio-economic background</b> | Provide support for geographically, socially or economically excluded students, non-traditional students, work-based learners, etc.                                                                                                                                                     | Does your institution provide support for geographically, socially or economically excluded students, non-traditional students, work-based learners, etc.?                                                                                                                             | Jung et al., 2016<br>UNESCO, 2019 |
| <b>Socio-economic background</b> | Ensure that the costs of study are affordable for the community and all the learners.                                                                                                                                                                                                   | Are the costs of study affordable for the community and all learners?                                                                                                                                                                                                                  | Latchem, 2012                     |

### 4.3 Disability and specific educational needs

**Disability and specific educational needs:** this refers to how the OER has to consider the physical, functional and psychological circumstances of the learner in accessing IA-OERs and their specific educational needs in the case of students presenting difficulties in learning or that require reasonable adjustments.

| RECOMMENDATION                                                                                                                                                                                              | QUESTION                                                                                                                                                                                        | SOURCE                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Ensure that you take into account learners with disabilities and specific educational needs, including those with learning disabilities from the beginning, i.e., the designing and methodology of the OER. | Do you take into account learners with disabilities and specific educational needs, including those with learning disabilities from the beginning, i.e., the design and methodology of the OER? | Mhlanga, 2010<br>Project Floe<br>Ehlers, 2012<br>UNESCO, 2019<br>Daniela, 2022<br>Haughey, 2005 |



| RECOMMENDATION                                                                                                                                                                                                                                                                                                                                                   | QUESTION                                                                                                                                                                                                                                                                                                               | SOURCE                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| Ensure that people with disabilities or other learning barriers participate in the co-creation and auditing of OER to assure their accessibility.                                                                                                                                                                                                                | Do people with disabilities or other learning barriers participate in the co-creation and auditing of OER, assuring their accessibility?                                                                                                                                                                               | UNESCO, 2019                               |
| Design course materials and activities to provide appropriate access to all students. Develop the course with universal design principles in mind.                                                                                                                                                                                                               | Have you designed the course materials and activities to provide appropriate access to all students? Has the OER been developed with universal design principles in mind?                                                                                                                                              | Bakken & Bridges, 2011                     |
| Ensure that the learning methodologies offer opportunities for learners to determine their learning pace, and time to do tasks and answer their questions.                                                                                                                                                                                                       | Do learning methodologies offer opportunities for learners to determine their learning pace, and time to do tasks and answer questions?                                                                                                                                                                                | Ehlers, 2012<br>UNESCO, 2015               |
| Ensure that the material respects one or several different learning styles (visual-verbal, active-reflexive, procedural-intuitive, sequential-global).                                                                                                                                                                                                           | Does the material respect one or several different learning styles (visual-verbal, active-reflexive, procedural-intuitive, sequential-global)?                                                                                                                                                                         | UNE-71362                                  |
| Provide different kinds of tutorials for different students, for both remedial and enrichment purposes, i.e., for fast learners who need enrichment in various areas of learning as well as learners who need their learning reinforced.                                                                                                                         | Does the OER provide different kinds of tutorials for different students, for both remedial and enrichment purposes?                                                                                                                                                                                                   | Mhlanga, 2010                              |
| Make reasonable adjustments to facilitate specific students their access to education, such as location in the classroom, presence of a sign language interpreter, personal assistant or guide dog, anticipation of the material, extra time and alternative formats in tests, pen grips, adapted PE equipment, assistive technologies, and accessible software. | Have you made reasonable adjustments to facilitate specific students their access to the education, such as location in the classroom, presence of a sign language interpreter, a personal assistant or guide dog? Do you anticipate the material? Do you give extra time? Do you provide alternative formats, adapted | Equality and Human Rights Commission, 2019 |

| RECOMMENDATION | QUESTION | SOURCE |
|----------------|----------|--------|
|----------------|----------|--------|

PE equipment, assistive technologies, and accessible software?

|                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                         |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Ensure that the syllabus includes links to the disability support services resources and policy related to the accessibility of courses and/or instructional materials, so that students can clearly ascertain the role of the instructor in providing support for those officially registered. | Does the syllabus include links to the campus' disability support services resources and policy related to the accessibility of courses and/or instructional materials? Can students clearly ascertain the role of the instructor in providing support for those officially registered? | QLT, 2022 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|

#### 4.4 Guidelines based on UDL

**Guidelines based on UDL:** it details the guidelines of the Universal Design for Learning.

| ASPECT                           | SUB-ASPECT | RECOMMENDATION                                                                                                                                                 | QUESTION                                                                                                                                                                             | SOURCE                                                                                                                        |
|----------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Multiple means of representation | PERCEPTION | Incorporate multiple means of perceptive representation: audio, video, sign language, graphics, text descriptions, Easy language, Braille, auditory keys, etc. | Does the OER incorporate multiple means of perceptive representation, such as audio, video, sign language, graphics, text descriptions, Easy language, Braille, auditory keys, etc.? | QLT, 2022<br>Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |



Co-funded by  
the European Union

Funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or EACEA. Neither the European Union nor the granting authority can be held responsible for them.

| ASPECT                                  | SUB-ASPECT        | RECOMMENDATION                                                                                                                                                                                                                                                                                                                                                                                                                                                        | QUESTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | SOURCE                                                                          |
|-----------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>Multiple means of representation</b> | <b>PERCEPTION</b> | Display information in a flexible format so that the following perceptual features can be varied: font, colour, contrast, visual layout, sound volume, video and sound speed, and size of text, images, graphs, tables, or other visual content.                                                                                                                                                                                                                      | Does the OER display information in a flexible format so that the following perceptual features can be varied: font, colour, contrast, visual layout, sound volume, video and sound speed, and size of text, images, graphs, tables, or other visual content?                                                                                                                                                                                                                      | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| <b>Multiple means of representation</b> | <b>PERCEPTION</b> | Provide alternatives for auditory information: captions, automated speech-to-text (voice recognition) or written transcriptions for spoken language; visual diagrams, charts, notations of music or sound; sign Language; visual analogues to represent emphasis and prosody (e.g., emoticons, symbols, or images); visual or tactile (e.g., vibrations) equivalents for sound effects or alerts; and Visual and/or emotional description for musical interpretation. | Does the OER provide alternatives for auditory information: captions, automated speech-to-text (voice recognition) or written transcriptions for spoken language; visual diagrams, charts, notations of music or sound; sign Language; visual analogues to represent emphasis and prosody (e.g., emoticons, symbols, or images); visual or tactile (e.g., vibrations) equivalents for sound effects or alerts; and Visual and/or emotional description for musical interpretation? | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |



| ASPECT                           | SUB-ASPECT           | RECOMMENDATION                                                                                                                                                                                                                                                                                                                                                                               | QUESTION                                                                                                                                                                                                                                                                                                                                                                                                                    | SOURCE                                                                                                                        |
|----------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Multiple means of representation | PERCEPTION           | Provide alternatives for visual information: Descriptions (text or spoken) for all images, graphics, videos, or animations;<br><br>touch equivalents (tactile graphics or objects of reference) for key visuals that represent concepts; Physical objects and spatial models to convey perspective or interaction; and Auditory cues for key concepts and transitions in visual information. | Does the OER provide alternatives for visual information:<br>(a) Descriptions (text or spoken) for all images, graphics, videos, or animations;<br>(b) Touch equivalents (tactile graphics or objects of reference) for key visuals that represent concepts;<br>(c) Physical objects and spatial models to convey perspective or interaction; and (d) Auditory cues for key concepts and transitions in visual information? | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a>                                               |
| Multiple means of representation | PERCEPTION           | Ensure that audiovisual contents are not decorative and do not hinder or slow down learning.                                                                                                                                                                                                                                                                                                 | Does the OER avoid using audiovisual contents that are decorative and hinder or slow down learning?                                                                                                                                                                                                                                                                                                                         | UNE-71362                                                                                                                     |
| Multiple means of representation | LANGUAGE AND SYMBOLS | Incorporate multiple options for language, mathematical expressions, and symbols: vocabulary, icons, labels, conceptual maps, pictograms, symbols, charts, diagrams, graphical representation of formulas and problems.                                                                                                                                                                      | Does the OER incorporate multiple options for language, mathematical expressions, and symbols: vocabulary, icons, labels, conceptual maps, pictograms, symbols, charts, diagrams, graphical representation of formulas and problems?                                                                                                                                                                                        | QLT, 2022<br>Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of representation | LANGUAGE AND SYMBOLS | Promote understanding across languages, e.g., use of different languages, electronic trans-                                                                                                                                                                                                                                                                                                  | Does the OER promote understanding across languages, e.g., use of different languages, electronic                                                                                                                                                                                                                                                                                                                           | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a>                                               |



| ASPECT                           | SUB-ASPECT           | RECOMMENDATION                                                                                                                                                             | QUESTION                                                                                                                                                                                | SOURCE                                                                          |
|----------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
|                                  |                      | lation tools or multilingual glossaries.                                                                                                                                   | translation tools or multilingual glossaries?                                                                                                                                           |                                                                                 |
| Multiple means of representation | LANGUAGE AND SYMBOLS | Include embed visual non-linguistic supports for vocabulary clarification (pictures, videos, etc).                                                                         | Does the OER use embed visual non-linguistic support for vocabulary clarification (pictures, videos, etc)?                                                                              | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of representation | LANGUAGE AND SYMBOLS | Clarify vocabulary and symbols, e.g., pre-teach them to promote connection to the learners' experience and prior knowledge.                                                | Does the OER clarify vocabulary and symbols, e.g., pre-teaching them to promote connection to the learners' experience and prior knowledge?                                             | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of representation | LANGUAGE AND SYMBOLS | Highlight how complex terms, expressions, or equations are composed of simpler words or symbols.                                                                           | Does the OER highlight how complex terms, expressions, or equations are composed of simpler words or symbols?                                                                           | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of representation | LANGUAGE AND SYMBOLS | Use embed support for vocabulary and symbols within the text (e.g., hyperlinks or footnotes to definitions, explanations, illustrations, previous coverage, translations). | Does the OER use embed support for vocabulary and symbols within the text (e.g., hyperlinks or footnotes to definitions, explanations, illustrations, previous coverage, translations)? | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |



| ASPECT                           | SUB-ASPECT           | RECOMMENDATION                                                                                                                                                                                                                                                                                                                                    | QUESTION                                                                                                                                                                                                                                                                                                                                                        | SOURCE                                                                          |
|----------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Multiple means of representation | LANGUAGE AND SYMBOLS | Use embed support for unfamiliar references within the text (e.g., domain specific notation, lesser-known properties and theorems, idioms, academic language, figurative language, mathematical language, jargon, archaic language, colloquialisms, and dialects).                                                                                | Does the OER use embed support for unfamiliar references within the text (e.g., domain specific notation, lesser-known properties and theorems, idioms, academic language, figurative language, mathematical language, jargon, archaic language, colloquialisms, and dialects)?                                                                                 | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of representation | LANGUAGE AND SYMBOLS | Clarify unfamiliar syntax (in language or in math formulas) or underlying structure (in diagrams, graphs, illustrations, extended expositions or narratives) through alternatives that highlight structural relations or make them more explicit, e.g., highlighting the transition words in an essay, links between ideas in a concept map, etc. | Does the OER clarify unfamiliar syntax (in language or in math formulas) or underlying structure (in diagrams, graphs, illustrations, extended expositions or narratives) through alternatives that highlight structural relations or make them more explicit, e.g., highlighting the transition words in an essay, links between ideas in a concept map, etc.? | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of representation | LANGUAGE AND SYMBOLS | Incorporate multiple options for comprehension, e.g., help options, no distracting elements, etc..                                                                                                                                                                                                                                                | Does the OER incorporate multiple options for comprehension, e.g., help options, no distracting elements, etc.?                                                                                                                                                                                                                                                 | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of representation | LANGUAGE AND SYMBOLS | Use templates, schematics, graphic organisers and routine/time organisers, or exploration                                                                                                                                                                                                                                                         | Does the OER use templates, schematics, graphic organisers and rou-                                                                                                                                                                                                                                                                                             | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |



| ASPECT                           | SUB-ASPECT           | RECOMMENDATION                                                                                                                         | QUESTION                                                                                                                                                        | SOURCE                                                                                                           |
|----------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
|                                  |                      | guidelines.                                                                                                                            | time/time organisers, or exploration guidelines?                                                                                                                |                                                                                                                  |
| Multiple means of representation | LANGUAGE AND SYMBOLS | Provide checklists, organisers, sticky notes, or electronic reminders.                                                                 | Does the OER provide checklists, organisers, or sticky notes, electronic reminders?                                                                             | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a>                                  |
| Multiple means of representation | LANGUAGE AND SYMBOLS | Highlight or emphasise key elements in text, graphics, diagrams, or formulas.                                                          | Does the OER highlight or emphasise key elements in text, graphics, diagrams, or formulas?                                                                      | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a>                                  |
| Multiple means of representation | LANGUAGE AND SYMBOLS | Use multiple cues, prompts, examples, analogies, metaphors and theoretical explanations to present the information in real contexts.   | Does the OER use multiple cues, prompts, examples, analogies, metaphors and theoretical explanations to present the information in real contexts?               | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a>                                  |
| Multiple means of representation | LANGUAGE AND SYMBOLS | Allow the review and repetition of each of the tasks, and the generalisation of learning. Include links for consultation or extension. | Does the OER allow the review and repetition of each of the tasks and the generalisation of learning? Does the OER include links for consultation or extension? | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a>                                  |
| Multiple means of representation | LANGUAGE AND SYMBOLS | Offer opportunities over time to revisit key ideas and linkages between ideas.                                                         | Does the OER offer opportunities over time to revisit key ideas and linkages between ideas?                                                                     | Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of ac-            | PHYSICAL ACTION      | Provide alternatives in the requirements for rate, timing, speed, and                                                                  | Does the OER provide alternatives in the requirements for rate, timing,                                                                                         | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a>                                  |



| ASPECT                                  | SUB-ASPECT                 | RECOMMENDATION                                                                                                                                                             | QUESTION                                                                                                                                                                      | SOURCE                                                                          |
|-----------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| tion and expression                     |                            | range of motor action required to interact with instructional materials, physical manipulatives, and technologies.                                                         | speed, and range of motor action required to interact with instructional materials, physical manipulatives, and technologies?                                                 |                                                                                 |
| Multiple means of action and expression | PHYSICAL ACTION            | Provide alternatives for physically responding or indicating selections (e.g., alternatives to marking with pen and pencil, alternatives to mouse control).                | Does the OER provide alternatives for physically responding or indicating selections (e.g., alternatives to marking with pen and pencil, alternatives to mouse control)?      | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of action and expression | PHYSICAL ACTION            | Provide alternatives for physically interacting with materials by hand, voice, single switch, joystick, keyboard, or adapted keyboard.                                     | Does the OER provide alternatives for physically interacting with materials by hand, voice, single switch, joystick, keyboard, or adapted keyboard?                           | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of action and expression | PHYSICAL ACTION            | Customise overlays for touch screens and keyboards.                                                                                                                        | Does the OER customise overlays for touch screens and keyboards?                                                                                                              | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of action and expression | EXPRESSION & COMMUNICATION | Compose your OER in multiple media such as text, speech, drawing, illustration, comics, storyboards, design, film, music, dance/movement, visual art, sculpture, or video. | Is the OER composed by multiple media such as text, speech, drawing, illustration, comics, storyboards, design, film, music, dance/movement, visual art, sculpture, or video? | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of action and expression | EXPRESSION & COMMUNICATION | Use physical manipulatives (e.g., blocks, 3D models, base-ten blocks).                                                                                                     | Are there physical manipulatives for users (e.g., blocks, 3D models, base-ten blocks)?                                                                                        | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |



| ASPECT                                  | SUB-ASPECT                 | RECOMMENDATION                                                                                                                                                       | QUESTION                                                                                                                                                                          | SOURCE                                                                          |
|-----------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Multiple means of action and expression | EXPRESSION & COMMUNICATION | Use social media and interactive web tools (e.g., discussion forums, chats, web design, annotation tools, storyboards, comic strips, animation presentations).       | Does the OER use social media and interactive web tools (e.g., discussion forums, chats, web design, annotation tools, storyboards, comic strips, animation presentations)?       | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of action and expression | EXPRESSION & COMMUNICATION | Include problem solving using a variety of strategies.                                                                                                               | Does the OER include problem solving using a variety of strategies?                                                                                                               | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of action and expression | EXPRESSION & COMMUNICATION | Provide spell checkers, grammar checkers, word prediction software, text-to-speech software (voice recognition), human dictation, recording.                         | Does the OER provide spell checkers, grammar checkers, word prediction software, text-to-speech software (voice recognition), human dictation, or recordings?                     | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of action and expression | EXPRESSION & COMMUNICATION | Provide Computer-Aided-Design (CAD), music notation (writing) software, or mathematical notation software.                                                           | Does the OER provide Computer-Aided-Design (CAD), music notation (writing) software, or mathematical notation software?                                                           | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of action and expression | EXPRESSION & COMMUNICATION | Provide calculators, graphing calculators, geometric sketchpads, or pre-formatted graph paper, and mathematics manipulatives (e.g., base-10 blocks, algebra blocks). | Does the OER provide calculators, graphing calculators, geometric sketchpads, or pre-formatted graph paper, and mathematics manipulatives (e.g., base-10 blocks, algebra blocks)? | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of action and expression | EXPRESSION & COMMUNICATION | Provide sentence starters or sentence strips.                                                                                                                        | Does the OER provide sentence starters or sentence strips?                                                                                                                        | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |



| ASPECT                                  | SUB-ASPECT                 | RECOMMENDATION                                                                                                                                     | QUESTION                                                                                                                                                        | SOURCE                                                                          |
|-----------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Multiple means of action and expression | EXPRESSION & COMMUNICATION | Provide differentiated models to emulate (i.e., models that demonstrate the same outcomes but use differing approaches, strategies, skills, etc.). | Does the OER provide differentiated models to emulate (i.e., models that demonstrate the same outcomes but use differing approaches, strategies, skills, etc.)? | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of action and expression | EXPRESSION & COMMUNICATION | Provide differentiated mentors (i.e., teachers/tutors who use different approaches to motivate, guide, feedback or inform) when using the OER.     | Are differentiated mentors provided (i.e., teachers/tutors who use different approaches to motivate, guide, feedback or inform) when using the OER?             | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of action and expression | EXPRESSION & COMMUNICATION | Supplement the teacher's job with activities and homework, involving parents, classmates or others to assist in instruction.                       | Is the teacher's job supplemented with activities and homework, involving parents, classmates or others to assist in instruction?                               | UNESCO, 2015                                                                    |
| Multiple means of action and expression | EXPRESSION & COMMUNICATION | Provide differentiated feedback (i.e., feedback that is accessible because it can be customised to individual learners).                           | Does the OER provide differentiated feedback (i.e., feedback that is accessible because it can be customised to individual learners)?                           | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of action and expression | EXECUTIVE FUNCTIONS        | Provide prompts and scaffolds to estimate effort, resources, and difficulty.                                                                       | Does the OER provide prompts and scaffolds to estimate effort, resources, and difficulty?                                                                       | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of action and expression | EXECUTIVE FUNCTIONS        | Provide models or examples of the process or product for goal setting.                                                                             | Does the OER provide models or examples of the process or product for goal setting?                                                                             | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |



| ASPECT                                  | SUB-ASPECT          | RECOMMENDATION                                                                                                                                 | QUESTION                                                                                                                                                    | SOURCE                                                                          |
|-----------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Multiple means of action and expression | EXECUTIVE FUNCTIONS | Provide guides and checklists for scaffolding goal setting.                                                                                    | Does the OER provide guides and checklists for scaffolding goal setting?                                                                                    | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of action and expression | EXECUTIVE FUNCTIONS | Post goals, objectives, and schedules in an obvious place.                                                                                     | Does the OER post goals, objectives, and schedules in an obvious place?                                                                                     | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of action and expression | EXECUTIVE FUNCTIONS | Provide checklists and project planning templates for understanding the problem, setting up prioritisation, sequences, and schedules of steps. | Does the OER provide checklists and project planning templates for understanding the problem, setting up prioritisation, sequences, and schedules of steps? | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of action and expression | EXECUTIVE FUNCTIONS | Provide guides for breaking long-term goals into reachable short-term objectives.                                                              | Does the OER provide guides for breaking long-term goals into reachable short-term objectives?                                                              | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of action and expression | EXECUTIVE FUNCTIONS | Break down the activities, sequencing them from easy to difficult.                                                                             | Does the OER break down the activities, sequencing them from easy to difficult?                                                                             | UNESCO, 2015                                                                    |
| Multiple means of action and expression | EXECUTIVE FUNCTIONS | Provide graphic organisers and templates for data collection and for organising information.                                                   | Does the OER provide graphic organisers and templates for data collection and for organising information?                                                   | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |



| ASPECT                                  | SUB-ASPECT          | RECOMMENDATION                                                                                                                                                                                                                                                                                                                                                                                 | QUESTION                                                                                                                                                                                                                                                                                                                                                                                                     | SOURCE                                                                                                           |
|-----------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Multiple means of action and expression | EXECUTIVE FUNCTIONS | Provide formative feedback that allows learners to monitor their own progress effectively and to use that information to guide their own effort and practice: e.g., show representations of progress (e.g., before and after photos, graphs and charts showing progress over time, process portfolios), or use templates, rubrics or checklist that guide self-reflection and self-monitoring. | Does the OER provide formative feedback that allows learners to monitor their own progress effectively and to use that information to guide their own effort and practice (e.g., show representations of progress (e.g., before and after photos, graphs and charts showing progress over time, process portfolios), or use templates, rubrics or checklist that guide self-reflection and self-monitoring)? | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a>                                  |
| Multiple means of engagement            | RECRUITING INTEREST | Offer students options for recruiting their interest: intrinsic/extrinsic motivation, allowing students to choose the learning contents, the objectives to reach, the level of difficulty, the time spent completing the task and the colour and design of their texts.                                                                                                                        | Does the OER offer students options for recruiting their interest: intrinsic/extrinsic motivation, allowing students to choose the learning contents, the objectives to reach, the level of difficulty, the time spent completing the task and the colour and design of their texts?                                                                                                                         | Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of engagement            | RECRUITING INTEREST | Divide objectives in sub-objectives and ensure that there are different means to reach them.                                                                                                                                                                                                                                                                                                   | Does the OER divide objectives in sub-objectives and ensure that there are different means to reach them?                                                                                                                                                                                                                                                                                                    | Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |



| ASPECT                       | SUB-ASPECT          | RECOMMENDATION                                                                                                                                                                                                                                   | QUESTION                                                                                                                                                                                                                                                       | SOURCE                                                                                                           |
|------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Multiple means of engagement | RECRUITING INTEREST | Give users enough time to read and use the content. Activities should be contextualised in real life and appropriate for the age.                                                                                                                | Does the OER give users enough time to read and use the content? Are activities contextualised in real life and appropriate for the age?                                                                                                                       | Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of engagement | RECRUITING INTEREST | Provide options that optimise what is relevant, valuable, important and motivating for each student.                                                                                                                                             | Does the OER provide options that optimise what is relevant, valuable, important and motivating for each student?                                                                                                                                              | Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of engagement | RECRUITING INTEREST | Allow learners to participate in the design of classroom activities and academic tasks.                                                                                                                                                          | Does the OER allow learners to participate in the design of classroom activities and academic tasks?                                                                                                                                                           | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a>                                  |
| Multiple means of engagement | RECRUITING INTEREST | Vary activities and sources of information so that they can be personalised and contextualised to learners' lives, socially relevant and responsive, and appropriate for age, ability and different racial, cultural, ethnic, and gender groups. | Do activities and sources of information vary in the OER so that they can be personalised and contextualised to learners' lives, socially relevant and responsive, and appropriate for age, ability and different racial, cultural, ethnic, and gender groups? | <a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a>                                  |
| Multiple means of engagement | RECRUITING INTEREST | Create a climate of support and acceptance in the classroom, offering options that reduce levels of uncertainty, feelings of insecurity, perception of threats and distractions, and offering different levels of sensory stimulation.           | Does the OER create a climate of support and acceptance in the classroom, offering options that reduce levels of uncertainty, feelings of insecurity, perception of threats and distractions, and offering different levels of sensory stimulation?            | Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |





| ASPECT                       | SUB-ASPECT                      | RECOMMENDATION                                                                                                                                                                                                                                                                    | QUESTION                                                                                                                                                                                                                                                                                                                 | SOURCE                                                                                                           |
|------------------------------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Multiple means of engagement | RECRUITING INTEREST             | Vary the level of novelty or risk. Increase the predictability of daily activities with charts, calendars, schedules, visible timers, cues, alerts, etc, and creating class routines. In contrast, maximise the unexpected, surprising, or novel in highly routinised activities. | Does the OER vary the level of novelty or risk? Does the OER increase the predictability of daily activities with charts, calendars, schedules, visible timers, cues, alerts, etc, and creating class routines? In contrast, does the OER maximise the unexpected, surprising, or novel in highly routinised activities? | Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of engagement | SUSTAINING EFFORT & PERSISTENCE | Indicate and variate the degree of difficulty or complexity in core activities to be completed.                                                                                                                                                                                   | Does the OER indicate and variate the degree of difficulty or complexity in core activities to be completed?                                                                                                                                                                                                             | Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of engagement | SUSTAINING EFFORT & PERSISTENCE | Vary the degrees of acceptable performance.                                                                                                                                                                                                                                       | Does the OER vary the degrees for acceptable performance?                                                                                                                                                                                                                                                                | Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of engagement | SUSTAINING EFFORT & PERSISTENCE | Provide reminders to perform tasks.                                                                                                                                                                                                                                               | Does the OER provide reminders to perform tasks?                                                                                                                                                                                                                                                                         | Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of engagement | SUSTAINING EFFORT & PERSISTENCE | Offer a variety of resources to perform activities.                                                                                                                                                                                                                               | Does the OER offer a variety of resources to perform activities?                                                                                                                                                                                                                                                         | Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of engagement | SUSTAINING EFFORT & PERSISTENCE | Offer group activities, indicating rules, goals, roles, and responsibilities for working as a                                                                                                                                                                                     | Does the OER offer group activities, indicating rules, goals, roles, and responsibilities for working as a                                                                                                                                                                                                               | Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |



| ASPECT                              | SUB-ASPECT                                 | RECOMMENDATION                                                                                                                                                       | QUESTION                                                                                                                                                                          | SOURCE                                                                                                           |
|-------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
|                                     |                                            | group.                                                                                                                                                               | group?                                                                                                                                                                            | <a href="https://udl-guide-lines.cast.org">guide-lines.cast.org</a>                                              |
| <b>Multiple means of engagement</b> | <b>SUSTAINING EFFORT &amp; PERSISTENCE</b> | Provide instruction or guidance for individuals or small groups of students, giving step-by-step prompts or instructions.                                            | Does the OER provide instruction or guidance for individuals or small groups of students, giving step-by-step prompts or instructions?                                            | UNESCO, 2015                                                                                                     |
| <b>Multiple means of engagement</b> | <b>SUSTAINING EFFORT &amp; PERSISTENCE</b> | Ensure that students know when to ask for help.                                                                                                                      | Does students know when to ask for help?                                                                                                                                          | Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| <b>Multiple means of engagement</b> | <b>SUSTAINING EFFORT &amp; PERSISTENCE</b> | Provide automatic and corrective feedback in some activities oriented towards improvement, identifying patterns of errors and wrong answers, and emphasising effort. | Does the OER provide automatic and corrective feedback in some activities oriented towards improvement, identifying patterns of errors and wrong answers, and emphasising effort? | Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| <b>Multiple means of engagement</b> | <b>SUSTAINING EFFORT &amp; PERSISTENCE</b> | Provide feedback that is frequent, timely, specific, and informative, rather than comparative or competitive.                                                        | Does the OER provide feedback that is frequent, timely, specific, and informative, rather than comparative or competitive?                                                        | Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| <b>Multiple means of engagement</b> | <b>SUSTAINING EFFORT &amp; PERSISTENCE</b> | Includes rewards and incentive elements, such as badges, prizes, points, exchange coins, mentions, recognition diplomas, etc.                                        | Does the OER include rewards and incentive elements, such as badges, prizes, points, exchange coins, mentions, recognition diplomas, etc?                                         | Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |



| ASPECT                       | SUB-ASPECT      | RECOMMENDATION                                                                                                                                                                        | QUESTION                                                                                                                                                                                            | SOURCE                                                                                                           |
|------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Multiple means of engagement | SELF-REGULATION | Offer activities that encourage self-reflection and the identification of personal goals.                                                                                             | Does the OER offer activities that encourage self-reflection and the identification of personal goals?                                                                                              | Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of engagement | SELF-REGULATION | Provide support and teach adaptive strategies to control emotional responses and managing frustration.                                                                                | Does the OER provide support and teach adaptive strategies to control emotional responses and managing frustration?                                                                                 | Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of engagement | SELF-REGULATION | Use real situations or simulations to demonstrate skills to cope with everyday life problems.                                                                                         | Does the OER use real situations or simulations to demonstrate skills to cope with everyday life problems?                                                                                          | Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of engagement | SELF-REGULATION | Facilitate the recognition of their own progress in the right moment, allowing self-evaluation and co-evaluation, both of the learning processes and of the resource itself.          | Does the OER facilitate the recognition of their own progress in the right moment, allowing self-evaluation and co-evaluation, both of the learning processes and of the resource itself?           | Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| Multiple means of engagement | SELF-REGULATION | Offer different evaluation instruments, such as checklist, learning target, questionnaires or tests, estimation scales, evaluation rubrics with different levels of achievement, etc. | Does the OER offer different evaluation instruments, such as checklist, learning target, questionnaires or tests, estimation scales, evaluation rubrics with different levels of achievement, etc.? | Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |



| ASPECT                              | SUB-ASPECT             | RECOMMENDATION                                                                                                                                                                          | QUESTION                                                                                                                                                                                             | SOURCE                                                                                                           |
|-------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Multiple means of engagement</b> | <b>SELF-REGULATION</b> | Offer devices, aids, or charts to assist individuals in learning to collect, chart and display data from their own behaviour for the purpose of monitoring changes in those behaviours. | Does the OER offer devices, aids, or charts to assist individuals in learning to collect, chart and display data from their own behaviour for the purpose of monitoring changes in those behaviours? | Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |
| <b>Multiple means of engagement</b> | Self-Regulation        | Include actions to enhance reflection on learning (learning journal or portfolio, thinking skills and routines, or other metacognitive strategies).                                     | Does the OER include actions to enhance reflection on learning (learning journal or portfolio, thinking skills and routines, or other metacognitive strategies)?                                     | Ayuso-del Puerto et al., 2022<br><a href="https://udl-guide-lines.cast.org">https://udl-guide-lines.cast.org</a> |



## 5 Dimension: Accessibility

**Accessibility**, which involves designing educational resources that people with disabilities can use, is a fundamental aspect of inclusion.

### 5.1 Accessibility for blind and partially sighted learners. Audio description

**Audio description (AD)** is a form of accessibility, which provides access to people with visual impairments. Audio description can make videos and static images accessible. It includes two sub dimensions: Audio Description of videos and Audio Description of images

#### Audio Description of videos

| ASPECT              | RECOMMENDATION                                                                                                                                                                                                                                                             | QUESTION                                                                                                                                                                                           | SOURCE                 |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| General guide-lines | Ensure that the description is elaborated mainly on the needs of those who most require this service, i.e., totally blind people, which in turn would also help those with low vision.                                                                                     | Has the audio description been elaborated taking totally blind people as the target audience?                                                                                                      | Norma 153020 AENOR UNE |
| General guide-lines | Ensure that the script takes into account the plot of the dramatic action and the environments and plastic data of the image.                                                                                                                                              | Does the script take into account the plot of the dramatic action and the environments and plastic data of the image?                                                                              | Norma 153020 AENOR UNE |
| General guide-lines | Apply the spatiotemporal rule of clarifying 'when', 'where', 'who', 'what' and 'how' of the situations to be audio described.                                                                                                                                              | Is the spatiotemporal rule of clarifying the 'when', 'where', 'who', 'what' and 'how' of the situations being audio described applied?                                                             | Norma 153020 AENOR UNE |
| General guide-lines | Ensure that the audio describer never offers his or her opinion or interpretation of the images described so that listeners have the opportunity to create their own opinions about them. He/she should only describe them without showing emotions or feelings about what | Is the audio description professional's opinion or interpretation of the images described offered?<br><br>Does the audio describer show his or her emotions or feelings about what he or she sees? | Chica, 2019            |

| ASPECT                     | RECOMMENDATION                                                                                                                                                                                                                                                                                                                                              | QUESTION                                                                                                                                                                                                                                             | SOURCE                                |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
|                            | he/she sees. The strategy for describing elements that are unpleasant, for example, should be to use adjectives that indicate their actual condition. You can say, for example, that they look 'unkempt', 'neglected', 'un-gainly', etc.                                                                                                                    | Are terms and adjectives used that indicate the actual condition of the images?                                                                                                                                                                      |                                       |
| <b>General guide-lines</b> | Ensure that the style of writing in the script is fluent with simple phrases.                                                                                                                                                                                                                                                                               | Is the style of writing in the script fluent and with simple phrases? Do the syntactic constructions include subordinate clauses?                                                                                                                    | Norma AENOR UNE 153020<br>Chica, 2019 |
| <b>General guide-lines</b> | Use specific terminology, appropriate for each work to be described.                                                                                                                                                                                                                                                                                        | Does the terminology used in the script adapt to the film genre?                                                                                                                                                                                     | Chica, 2019                           |
| <b>General guide-lines</b> | Ensure that the type of vocabulary used is rich enough so as not to create a text with few idiomatic and stylistic resources. In this sense, it is recommended that different synonyms be used to refer to the same element (thief, robber, crook, thug, delinquent, etc.), as long as this does not generate confusion about the referent being described. | Does the audio describer use a rich enough vocabulary to create a text with great idiomatic and stylistic resources?<br><br>Do the audio description use different synonyms to refer to the same element, as long as it does not generate confusion? | Chica, 2019                           |
| <b>General guide-lines</b> | Ensure that the descriptor tries to be rigorous in the way it refers to objects or processes that have a specific name, i.e., it will use precise terms and avoid more generic ones. For example, it is recommended to use the term 'foremast' instead of 'fore stick' or 'fore pole' when                                                                  | Does the audio describer use rigorous terms to refer to objects or processes that have a specific name?<br><br>Does the audio describer use rigorous terms to refer to objects or processes that have a specific designation?                        | Chica, 2019                           |



| ASPECT                     | RECOMMENDATION                                                                                                                                                                                           | QUESTION                                                                                                                                                                                            | SOURCE                             |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
|                            | referring to a ship.                                                                                                                                                                                     |                                                                                                                                                                                                     |                                    |
| <b>General guide-lines</b> | Ensure that the information included in the description is appropriate for the type of target audience for a specific show/film. The language/ choice of vocabulary used must suit the tone of the film. | Is the information included in the description appropriate for the type of target audience for a specific show/film?<br><br>Does the language/ choice of vocabulary used suit the tone of the film? | Norma 153020 AENOR UNE             |
| <b>General guide-lines</b> | Ensure that the describer takes into consideration the intonation of the dialogue and the mood of the film, in order to find the right language for the descriptions as well.                            | Does the describer take into consideration the intonation of the dialogue and the mood of the film?                                                                                                 | German AD standards                |
| <b>General guide-lines</b> | Ensure that the use of technical terms (i.e., camera angles) is not used and only used very sparingly.                                                                                                   | Are film of photography technical terms used in the description?                                                                                                                                    | ITC Ofcom guidelines               |
| <b>General guide-lines</b> | Avoid causing the visually impaired listener to become tired due to saturation of information or anxiety due to a lack of information.                                                                   | Could the visually impaired listener become tired or anxious due to saturation or lack of information?                                                                                              | Norma 153020 AENOR UNE             |
| <b>General guide-lines</b> | Ensure that plot events are not revealed, nor situations of tension, suspense or mystery are broken.                                                                                                     | Are plot events uncovered or are situations of tension, suspense or mystery broken?                                                                                                                 | Norma 153020 AENOR UNE             |
| <b>General guide-lines</b> | Ensure that you only describe something before it appears on the screen if there is no pause when they actually make an appearance or just immediately after.                                            | Are some image elements or events described before they appear on the screen?<br><br>Is it done because there is no pause when they actually                                                        | ADI guidelines German AD standards |



| ASPECT                     | RECOMMENDATION                                                                                                                                                                                                                                                                                                                                                                                             | QUESTION                                                                                                                                                                                                                                                                             | SOURCE                                         |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
|                            |                                                                                                                                                                                                                                                                                                                                                                                                            | make an appearance or just immediately after?                                                                                                                                                                                                                                        |                                                |
| <b>General guide-lines</b> | Omit a less significant description of what is onscreen in order to highlight a critical piece of description.                                                                                                                                                                                                                                                                                             | Is a less significant description omitted in order to highlight a critical piece of description?                                                                                                                                                                                     | ADI guidelines<br>German AD standards          |
| <b>General guide-lines</b> | Ensure that the describer do not censor the description and describes what is on screen. The information provided by the image must be respected, neither censoring nor cutting out any supposed excesses or complementing any supposed deficiencies.                                                                                                                                                      | Has the describer censored any type of information on screen?<br><br>Has the describer complemented any supposed deficiencies?                                                                                                                                                       | ITC Ofcom guidelines<br>Norma AENOR UNE 153020 |
| <b>General guide-lines</b> | Description cannot and need not convey every visual image on display. Listeners should be allowed to hear actors' voices, sound effects, music, and ambience—or experience silence throughout the description.                                                                                                                                                                                             | To what degree the description conveys every visual image on display? (1 = Not at all, 5 = Very much)<br>Are listeners allowed to hear actors' voices, sound effects, music, and ambience—or experience silence throughout the description?                                          | ADI guidelines                                 |
| <b>General guide-lines</b> | A review and correction of the audio description script is mandatory. This process should be carried out by someone other than the person who has been in charge of writing the script. This person should have sufficient experience as a professional audio describer. The changes made during this process will be incorporated into the final script for the recording of the audio description voice. | Has the script been reviewed and proofread?<br>Was it done by a person other than the person who wrote the script? Does he/she have sufficient experience as a professional audio describer?<br>Have the changes been incorporated into the final script before the voice recording? | Norma AENOR UNE 153020<br>Chica, 2019          |





| ASPECT                     | RECOMMENDATION                                                                                                                                                                                                                                                                                                                                                                                                                                                            | QUESTION                                                                                                                                                                                                                                                                                                                                                                                                             | SOURCE                             |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| <b>General guide-lines</b> | After the audio description script has been corrected, it will be recorded and voiced in a professional recording studio. Professionals will be commissioned with voices that possess certain qualities: that they are pleasant to listen to, that the diction is correct in terms of intonation, rhythm and vocalisation, and that the prosody is relatively neutral but not monotonous.                                                                                 | Has the audio description been voiced in a professional recording studio? Do the voices of the speakers have the appropriate qualities?                                                                                                                                                                                                                                                                              | Norma AENOR UNE 153020 Chica, 2019 |
| <b>General guide-lines</b> | The type of voices (male, female, adult or young) shall be selected according to the characteristics of the work and in order to be clear to the listener. In particular, for works intended for a children's audience, it is advisable for the speaker to use an intonation that is somewhat more expressive and suitable for children. On the other hand, when female performers predominate in the work, the voice of the main speaker should be male, and vice versa. | <p>Has the type of voices (male, female, adult or young) been selected according to the characteristics of the work and in order to be clear to the listener?</p> <p>In works intended for a children's audience, does the speaker use a slightly more expressive intonation suitable for children?</p> <p>If the work is dominated by female performers, is the main speaker's voice masculine, and vice versa?</p> | Norma AENOR UNE 153020 Chica, 2019 |
| <b>General guide-lines</b> | The last step in the process of creating an audio description of an audiovisual product is to edit it on the medium chosen for distribution. During the audio mixing of the original product and the AD, the volumes, equalisations and ambience effects of                                                                                                                                                                                                               | Have the volumes, equalisations and ambient effects of the audio description track been matched to the original soundtrack, adapting the soundtrack to the acoustic conditions of the audiovisual product described at each                                                                                                                                                                                          | Norma AENOR UNE 153020             |

| ASPECT            | RECOMMENDATION                                                                                                                                                                     | QUESTION                                                                                                                                                             | SOURCE      |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                   | the audio description soundtrack must be matched to the original soundtrack, adapting the soundtrack at all times to the acoustic conditions of the audiovisual product described. | moment?                                                                                                                                                              |             |
| <b>Characters</b> | In general, the physical appearance of the main characters should be described.                                                                                                    | Is the physical appearance of the main characters described?                                                                                                         | Chica, 2019 |
| <b>Characters</b> | Normally, the main characters are described in more detail than the other characters.                                                                                              | Are the main characters described in more detail than the other characters?                                                                                          | Chica, 2019 |
| <b>Characters</b> | The height (tall, short, medium height, etc.) of the characters should be described.                                                                                               | Is the height of the characters described?                                                                                                                           | Chica, 2019 |
| <b>Characters</b> | The physical appearance (small, big, chubby, thin, etc.) of the characters must be described.                                                                                      | Is the physical appearance of the characters described?                                                                                                              | Chica, 2019 |
| <b>Characters</b> | Describe how the characters are dressed.                                                                                                                                           | Are the main characters' dresses described in more detail than those of the rest of the characters?                                                                  | Chica, 2019 |
| <b>Characters</b> | Describe the colour and type of hairstyle of the characters (for women, the type of haircut and hairstyle; for men, the type of beard or moustache, etc.).                         | Is the characters' colour and type of hairstyle described (for women, the type of haircut and hairstyle; for men, the type of beard or moustache, etc.)?             | Chica, 2019 |
| <b>Characters</b> | The facial expression (eyes, mouth, facial expressions etc.) adopted by the characters should be described in detail so that it informs about their attitudes and emotions.        | Is the facial expression (eyes, mouth, facial expressions etc.) adopted by the characters described in detail so that it informs about their attitudes and emotions? | Chica, 2019 |

| ASPECT            | RECOMMENDATION                                                                                                                                                                                                                                                   | QUESTION                                                                                                                                                                                                                                  | SOURCE               |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>Characters</b> | Describe the age of the character if known or provide an estimate age.                                                                                                                                                                                           | Is the age of the character described?                                                                                                                                                                                                    | Chica, 2019          |
| <b>Characters</b> | Mention the race or ethnicity of the character only if necessary.                                                                                                                                                                                                | Is the race or ethnicity of the character described if that were necessary?                                                                                                                                                               | ITC Ofcom guidelines |
| <b>Characters</b> | Identifying and describing characters is vital to effective AD. Identify the characters by their name in the narration.                                                                                                                                          | Are characters identified by their name?                                                                                                                                                                                                  | ITC Ofcom guidelines |
| <b>Characters</b> | Give supplementary information about the characters who will be important as the film progresses. If possible, they should be described in more detail than others.                                                                                              | Is supplementary information given about characters who will be important as the film progresses?<br><br>Are they described in more detail than others?                                                                                   | German AD standards  |
| <b>Characters</b> | The describer should uncover details about the character that can be considered relevant to the plot, such as attitudes, thoughts or emotions displayed or experienced by the characters that are not readily apparent from the dialogue.                        | Does the describer uncover details about the character that can be considered relevant to the plot, such as attitudes, thoughts or emotions displayed or experienced by the characters that are not readily apparent from their dialogue? | Chica, 2019          |
| <b>Characters</b> | It is not advisable to explicitly reveal the type of emotion or thought of the character, e.g., 'she/he is sad', 'she/he is happy', 'she/he is in love', etc. in order not to give information or subjective interpretations away to blind and low vision users. | Does the describer explicitly reveal the type of emotion or thought of the character, e.g., 'she/he is sad', 'she/he is happy', 'she/he is in love', etc.?                                                                                | Chica, 2019          |

| ASPECT                         | RECOMMENDATION                                                                                                                                                                                                                                                                                                                                        | QUESTION                                                                                                                                                                                                                                                                                  | SOURCE      |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Characters</b>              | Perceptual verbs (visual or auditory) such as 'to look' allow describing mental and emotional states if they are accompanied by specific adjectives, adverbs or adverbial locutions that specify the way in which the character performs certain actions (look + attentive / scared / moved / smiling / excited / serious / teary-eyed / grim-faced). | Does the describer use perceptual verbs accompanied by specific adjective, adverb or adverbial locutions to describe characters 'mental and emotional states when performing certain actions (look + attentive / scared / moved / smiling / excited / serious / teary-eyed / grim-faced)? | Chica, 2019 |
| <b>Space and time settings</b> | The description of the spatial-temporal settings should be provided especially when there are scene changes in which location and time in the narration are also changed. It is very important to keep the user informed of these issues so she/he does not lose the thread of the action.                                                            | Are space and time settings described when there are scene changes in which location and time in the narration also change?                                                                                                                                                               | Chica, 2019 |
| <b>Space and time settings</b> | Whenever possible, the describer should describe where the actions take place to capture the most salient features of the setting ('in a Berlin square decorated with flags', 'next to a piano', 'on the floor of a living room', etc.).                                                                                                              | Does the describer mention where the action takes place to capture the most salient features of the setting ('in a Berlin square decorated with flags', 'next to a piano', 'on the floor of a living room', etc.)?                                                                        | Chica, 2019 |
| <b>Space and time settings</b> | If the physical location of the place is not revealed, describe the objects, lighting, surfaces or colours that provide contextual information ('in a room lit in red', 'in front of a wardrobe', etc.).                                                                                                                                              | If the physical location of the place is not revealed, are objects, lighting, surfaces or colours that can provide contextual information described?                                                                                                                                      | Chica, 2019 |

| ASPECT                         | RECOMMENDATION                                                                                                                                                                                                                          | QUESTION                                                                                                                                                                                                                          | SOURCE      |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Space and time settings</b> | It is also useful to describe changes in weather conditions when these cannot be inferred from the ambient sound of the soundtrack ('hail on the roofs').                                                                               | Are weather conditions described when these cannot be inferred from the ambient sound of the soundtrack ('hail on the roofs')?                                                                                                    | Chica, 2019 |
| <b>Space and time settings</b> | Spatial information is usually placed at different points in the description and is often found at the beginning and at the end of a sentence as a circumstantial complement of place.                                                  | Is the spatial information placed at the beginning or at the end of the sentence as a circumstantial complement of place?                                                                                                         | Chica, 2019 |
| <b>Space and time settings</b> | As a general rule, time setting information is placed at the beginning of the audio description excerpt, unless the focus of the shot forces to describe another visual element before it.                                              | Is the time setting information placed at the beginning of the audio description excerpt?                                                                                                                                         | Chica, 2019 |
| <b>Space and time settings</b> | The description of the time setting should inform about the time of day when the scene takes place ('in the morning', 'at night', 'at dawn', etc.).                                                                                     | Does the description of the time setting inform about the time of day when the scene takes place ('in the morning', 'at night', 'at dawn', etc.)?                                                                                 | Chica, 2019 |
| <b>Space and time settings</b> | In sequences showing a temporal progression over several moments of a given period of time (hours, days, weeks...), the passage of time can be described with adverbs or adverbial temporal locutions such as 'then', 'later', 'later'. | Is the passage of time described with adverbs or adverbial temporal locutions such as 'then', 'later', 'later in sequences showing a temporal progression over several moments of a given period of time (hours, days, weeks...)? | Chica, 2019 |
| <b>Space and time settings</b> | If there is no space in the message after a change of shot to                                                                                                                                                                           | Is the end of the previous shot used to introduce the temporal description if there                                                                                                                                               | Chica, 2019 |



| ASPECT         | RECOMMENDATION                                                                                                                                                                                                                                                                                                        | QUESTION                                                                                                                                                            | SOURCE                               |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
|                | introduce the temporal description, the end of the previous shot can be used to introduce this information.                                                                                                                                                                                                           | is no space in the message after a change of shot?                                                                                                                  |                                      |
| <b>Actions</b> | It is essential to convey the most relevant actions of the story in a meticulous way, making use of a varied lexicon without using a register that is too complex or difficult to follow for the users.                                                                                                               | Are the most relevant actions of the story conveyed in a meticulous way, making use of a varied lexicon without using a register that is too complex or the users?  | Chica, 2019                          |
| <b>Actions</b> | In general, verbal actions are expressed in the script in the present tense and in the third person, as the descriptions translate images that are taking place at the moment of listening to the audiovisual product.                                                                                                | Are the actions that take place at the moment of listening to the audiovisual product described with verbs in the present tense and in the third person?            | Chica, 2019                          |
| <b>Actions</b> | It is also common to use descriptions with verbs in the past tense to refer to actions that have taken place in the audiovisual work at an earlier time.                                                                                                                                                              | Are actions that have taken place in the audiovisual work at an earlier time described with verbs in the past tense?                                                | Chica, 2019                          |
| <b>Actions</b> | The analysis of a corpus of audio description scripts concluded that the semantic fields of actions (verb forms) that are most frequently activated in audio description are mainly those of movement, perception, feeling, state, possession and change. A balanced description must take these fields into account. | Does the description of the audiovisual product include verbs forms belonging to the semantic fields of movement, perception, feeling, state, possession or change? | Chica, 2019<br>Jiménez Hurtado, 2010 |
| <b>Actions</b> | Another relevant point about                                                                                                                                                                                                                                                                                          | Are verbs with specific mean-                                                                                                                                       | Chica, 2019                          |



| ASPECT                         | RECOMMENDATION                                                                                                                                                                                                                                                                                                                      | QUESTION                                                                                                                                                                                                            | SOURCE      |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                                | the use of verbs in audio description is that variety is important to enrich the text. Whenever possible, avoid verbs with general or imprecise meaning and use those with specific meaning, appropriate for the action the character is performing (e.g., 'carves a block of marble' instead of 'hits a block of stone').          | ings appropriate to the action the character is performing privileged over those with a more general or imprecise meaning (e.g., 'carves a block of marble' instead of 'hits a block of stone')?                    |             |
| <b>Sounds, music and songs</b> | Describe only those diegetic sounds (generated from the action of the film itself) whose origin is not clear or can be easily deduced from the context or dialogue of the audiovisual product. In other words, it is not necessary to describe sounds that are obvious.                                                             | Are diegetic sounds (generated from the action of the film itself) whose origin is not clear or can be easily deduced from the context or dialogue of the audiovisual product described?                            | Chica, 2019 |
| <b>Sounds, music and songs</b> | When an audiovisual work uses extradiegetic sounds (those that are included in the soundtrack, i.e., outside the audiovisual development), they usually carry a clear intention (tension, fear or suspense). Avoid introducing audio descriptions over them as much as possible. If necessary, describe before or after the sounds. | <p>The use of audio descriptions over extradiegetic sounds used to mark a clear intention in the audiovisual product is avoided?</p> <p>Is the description introduced before or after the extradiegetic sounds?</p> | Chica, 2019 |
| <b>Sounds, music and songs</b> | In cases where the music or sound are the prelude to a relevant event (horror or suspense scenes), describe after them so as not to advance information and thus not to ruin the surprise factor.                                                                                                                                   | Is the description postponed in cases where the music or sound are the prelude to a relevant event (horror or suspense scenes)?                                                                                     | Chica, 2019 |



| ASPECT                         | RECOMMENDATION                                                                                                                                                                                                                                                                                                 | QUESTION                                                                                                                                                                          | SOURCE                 |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>Sounds, music and songs</b> | As a general rule, when you come across a song, try to let the listeners enjoy the music. However, if some important action occurs that needs to be described, the description will be brief and will not step on the music too much. Take advantage of the downbeats or the parts without lyrics to describe. | Does the description allow listeners to enjoy songs and music?<br><br>If important actions occur while the music sounds, is the description brief? Does it not step on the music? | Chica, 2019            |
| <b>Sounds, music and songs</b> | When it is essential to describe over the music, it is technically possible to lower the soundtrack's audio level and introduce the description so it is clearly understood. This audio description procedure is particularly relevant in musical films.                                                       | When it is essential to describe over the music, is the soundtrack's audio level lowered a little to introduce the description (particularly in musical films)?                   | Chica, 2019            |
| <b>On-screen information</b>   | Occasional subtitle information, signs, notices and/or credits should be included in summary form where the gap is short to allow for verbatim audio description.                                                                                                                                              | Are subtitle information, signs, notices and/or credits included in summary form where the gap is short to allow for verbatim audio description?                                  | Norma 153020 AENOR UNE |
| <b>On-screen information</b>   | Credits usually appear at the beginning or end of the film. Those referring to the studio, production company, lead actors, director, screenwriter, cinematographer and producer(s) should be included.                                                                                                        | Are credits referring to the studio, production company, lead actors, director, screenwriter, cinematographer and producer(s) included in the script?                             | Chica, 2019            |



| ASPECT                       | RECOMMENDATION                                                                                                                                                                                                                                                                                                             | QUESTION                                                                                                                                                                                                                                                      | SOURCE      |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>On-screen information</b> | When describing the texts in signs or inscriptions, the voice tone should be different from the rest of the AD. Credits and other texts should be voiced by a different (secondary) voice.                                                                                                                                 | Is the voice tone used when describing texts in signs or inscriptions different from the rest of the AD?<br>Are the credits and texts voiced by a different (secondary) voice?                                                                                | Chica, 2019 |
| <b>On-screen information</b> | Texts appearing on labels, inscriptions or inserts should also be included in the audio description script. The voice recording procedure is the same as for the credits.                                                                                                                                                  | Are texts appearing on labels, inscriptions or inserts included in the audio description script? Is the voice recording procedure for labels the same as that of credits?                                                                                     | Chica, 2019 |
| <b>On-screen information</b> | The subtitles appearing in the audiovisual work as a result of the translation of a dialogue in a language other than the main language must be transferred to the audio description script as texts, respecting their literal meaning if possible or in a summarised form. They will also be read by the secondary voice. | Are the subtitles that appear in the audiovisual work as a result of the translation of a dialogue in a language other than the main language transferred to the audio description script as texts, respecting their literal meaning or in a summarised form? | Chica, 2019 |
| <b>On-screen information</b> | Sometimes there is a more or less detailed description of the logos of the studio and the production company or companies of the work. This happens when there is a contractual obligation to do so. In most cases it is sufficient to simply read the names of these companies and leave out their description.           | Are the logos of the studio and production companies described in a more or less detailed way?<br><br>Are their names simply read, leaving out the description of the logos?                                                                                  | Chica, 2019 |



## Audio Description of images

| ASPECT             | RECOMMENDATION                                                                                                            | QUESTION                                                                   | SOURCE                                                                              |
|--------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Users</b>       | Adjust the language of the audio description for a young audience.                                                        | Is the language appropriate for a young audience?                          | AENOR, 2005<br>Remael et al., 2015<br>Descrivedendo, 2021<br>RNIB & VocalEyes, 2003 |
| <b>Users</b>       | Adapt the length of the audio description for a young audience.                                                           | Is the duration of the audio description appropriate for a young audience? | AENOR, 2005<br>Remael et al., 2015<br>Descrivedendo, 2021<br>RNIB & VocalEyes, 2003 |
| <b>Description</b> | Mention what type of resource is being described. For example: photos, drawings, illustrations, diagrams, slides or maps. | Is the type of resource being described indicated?                         | AENOR, 2005<br>Remael et al., 2015<br>Descrivedendo, 2021<br>RNIB & VocalEyes, 2003 |
| <b>Description</b> | Describe the most representative visual information so the listener can understand the described image.                   | Is the most representative visual information of the image described?      | AENOR, 2005<br>Remael et al., 2015<br>Descrivedendo, 2021<br>RNIB & VocalEyes, 2003 |
| <b>Description</b> | Describe the visual elements in a specific order: from top to bottom and from left to right, if the image allows it.      | Does the verbal description follow the recommended order?                  | AENOR, 2005<br>Remael et al., 2015<br>Descrivedendo, 2021<br>RNIB & VocalEyes, 2003 |
| <b>Description</b> | Provide details of the image, clarifying the “when” “where”, “who”, “what” and “how” of each image.                       | Are the most important details of the image explained?                     | AENOR, 2005<br>Remael et al., 2015<br>Descrivedendo, 2021<br>RNIB & VocalEyes, 2003 |
| <b>Description</b> | Do not add personal opinions or interpretations.                                                                          | Is the audio description neutral?                                          | AENOR, 2005<br>Remael et al., 2015<br>Descrivedendo, 2021<br>RNIB & VocalEyes, 2003 |
| <b>Description</b> | Describe additional information with vivid details. Provide additional information about the image using more details.    | Are vivid details provided?                                                | AENOR, 2005<br>Remael et al., 2015<br>Descrivedendo, 2021<br>RNIB & VocalEyes, 2003 |



| ASPECT             | RECOMMENDATION                                                                                                                                                                                                        | QUESTION                                  | SOURCE                                                                              |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Description</b> | Describe the colour. Provide a specific audio description of the colours and tones in the image.                                                                                                                      | Is the colour described specifically?     | AENOR, 2005<br>Remael et al., 2015<br>Descrivedendo, 2021<br>RNIB & VocalEyes, 2003 |
| <b>Description</b> | Include personality and behaviour. If describing people, include evident attitudes, emotions, or thoughts to help students understand, as long as the image requires it. For example, "The child on the left is sad". | Are personality and behaviour described?  | AENOR, 2005<br>Remael et al., 2015<br>Descrivedendo, 2021<br>RNIB & VocalEyes, 2003 |
| <b>Description</b> | Mention other senses like touch and smell if the image allows it.                                                                                                                                                     | Are other senses referred to?             | AENOR, 2005<br>Remael et al., 2015<br>Descrivedendo, 2021<br>RNIB & VocalEyes, 2003 |
| <b>Description</b> | Use sounds. Add music or sound effects to create a context or help identify the image before or during the AD.                                                                                                        | Are sounds used?                          | AENOR, 2005<br>Remael et al., 2015<br>Descrivedendo, 2021<br>RNIB & VocalEyes, 2003 |
| <b>Description</b> | Review the quality. Ask someone with experience to review and correct the audio description script to ensure quality. This reviewer could be a person with visual impairment.                                         | Has a reviewer been involved?             | AENOR, 2005<br>Remael et al., 2015<br>Descrivedendo, 2021<br>RNIB & VocalEyes, 2003 |
| <b>Language</b>    | Use simple language. Write clear and simple sentences. Each sentence conveys one idea. Avoid overly complex sentences and subordination.                                                                              | Are clear and simple sentences used?      | AENOR, 2005<br>Remael et al., 2015<br>Descrivedendo, 2021<br>RNIB & VocalEyes, 2003 |
| <b>Language</b>    | Use the present tense to make the description easier to understand.                                                                                                                                                   | Is the present tense used in the AD?      | AENOR, 2005<br>Remael et al., 2015<br>Descrivedendo, 2021<br>RNIB & VocalEyes, 2003 |
| <b>Language</b>    | Transcribe the text. If an image includes text, transcribe it and                                                                                                                                                     | Is the text within the image transcribed? | AENOR, 2005<br>Remael et al., 2015                                                  |



| ASPECT          | RECOMMENDATION                                                                                                                                                                                               | QUESTION                                               | SOURCE                                                                              |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------|
|                 | mention it in the AD.                                                                                                                                                                                        |                                                        | Descrivedendo, 2021<br>RNIB & VocalEyes, 2003                                       |
| <b>Language</b> | Avoid automatically generated descriptions. Do not rely on descriptions generated by artificial intelligence or unreliable resources. They may not be suitable and could compromise students' understanding. | Are automatically generated descriptions avoided?      | AENOR, 2005<br>Remael et al., 2015<br>Descrivedendo, 2021<br>RNIB & VocalEyes, 2003 |
| <b>Language</b> | Vary the vocabulary. Use a rich vocabulary with synonyms to avoid repetition, as long as it doesn't cause confusion. The vocabulary should be suitable for the type of image and the young audience.         | Is varied vocabulary used?                             | AENOR, 2005<br>Remael et al., 2015<br>Descrivedendo, 2021<br>RNIB & VocalEyes, 2003 |
| <b>Language</b> | Use specific vocabulary. Use specific words for objects and actions, avoiding vague descriptions or figurative language (e.g., "to fly into a range", "to take on the world").                               | Is specific vocabulary used?                           | AENOR, 2005<br>Remael et al., 2015<br>Descrivedendo, 2021<br>RNIB & VocalEyes, 2003 |
| <b>Language</b> | Avoid complicated terms. Use terminology only if the students know it. If it is an unfamiliar concept, explain it.                                                                                           | Are complicated terms avoided or explained?            | AENOR, 2005<br>Remael et al., 2015<br>Descrivedendo, 2021<br>RNIB & VocalEyes, 2003 |
| <b>Language</b> | Use analogies and metaphors. Use analogies and metaphors with elements from the students' reality, such as: "The building's shape looks like a shoebox" or "The planet is round like a football ball".       | Are analogies used?                                    | AENOR, 2005<br>Remael et al., 2015<br>Descrivedendo, 2021<br>RNIB & VocalEyes, 2003 |
| <b>Voice</b>    | Choose the appropriate voice. Select a voice (male or female,                                                                                                                                                | Does the voice help improve understanding of the audio | AENOR, 2005<br>Remael et al., 2015<br>Descrivedendo, 2021                           |



| ASPECT | RECOMMENDATION                                                                                                                                                                                               | QUESTION                                                                             | SOURCE                 |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------------------|
|        | adult or child) that suits the image. Use a human voice with an expressive tone to keep the students' attention. Modulate the rhythm to enhance understanding. If possible, contact a professional narrator. | description and maintain attention?<br>Likert Scale: 1 = not at all, 5 = completely. | RNIB & VocalEyes, 2003 |

## 5.2 Accessibility for deaf and hard of hearing learners

In includes two sub subdimensions: Subtitles for the deaf and hard of hearing and Sign Language Interpreting.

### Subtitles for the deaf and hard of hearing

| ASPECT                  | SUB-ASPECT | RECOMMENDATION                                                                                                                                                                                                                                               | QUESTION                                                         | SOURCE                                      |
|-------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------|
| General recommendations |            | Multimedia/Video/Audio<br><br>Use closed captions (CC) and/or transcripts to relay content.                                                                                                                                                                  | Have you completed the Closed Captions (CC) of the entire video? | UNESCO, 2021<br>Affordable Learning Georgia |
| General recommendations |            | Avoid automatically generated subtitles.                                                                                                                                                                                                                     | Have you avoided automatically generated subtitles?              |                                             |
| Technical parameters    | POSITION   | Subtitles should appear at the bottom of the screen.<br><br>The subtitles can be moved from the bottom if they cover relevant information.<br><br>The use of the subtitle box is obsolete. Nevertheless, many free subtitling programs include it by default | Do the CC appear at the bottom of the screen?                    | UNE 153010, 2012                            |



| ASPECT               | SUB-ASPECT                      | RECOMMENDATION                                                                                                                                                                                                                                                                                                                                                                    | QUESTION                                                                            | SOURCE                            |
|----------------------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------|
| Technical parameters | CONTRAST AND/OR COLOUR          | <p>Character and background colours should be assigned that are most legible, producing greater contrast and less visual fatigue (yellow font on a black background).</p> <p>Almost all programs allow the user to choose the wallpaper, the font type, etc.</p>                                                                                                                  | Is the CC easily readable against the background of the audiovisual product?        | UNE 153010, 2012                  |
| Technical parameters | CONTRAST AND/OR COLOUR          | If the colour of the character matches the background, the subtitle can be framed in a coloured box to obtain sufficient contrast.                                                                                                                                                                                                                                                | Is the CC easily readable against the background of the audiovisual product?        | UNE 153010, 2012                  |
| Technical parameters | CONTRAST AND/OR COLOUR          | <p>If the colour of the character matches the background, the subtitle can be framed in a coloured box to obtain sufficient contrast.</p> <p>We would remove it because it is too complex to accomplish for amateur subtitlers.</p>                                                                                                                                               | If the CC is too complex to accomplish for an amateur subtitler, can you remove it? |                                   |
| Technical parameters | FONT AND TYPOGRAPHY INFORMATION | <p>The choice of font must respond to criteria of maximum legibility.</p> <p>For example, in Netflix (2024):</p> <ul style="list-style-type: none"> <li>- Font style: Arial as a generic placeholder for proportional SansSerif.</li> <li>- Font size: relative to video resolution and ability to fit 42 characters across the screen.</li> <li>- Font colour: White.</li> </ul> | Is the font type and size clearly visible?                                          | UNE 153010, 2012<br>Netflix, 2024 |



| ASPECT               | SUB-ASPECT           | RECOMMENDATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | QUESTION                                                                                  | SOURCE                            |
|----------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------|
| Technical parameters | SYNCHRONISATION      | <p>The inputs and outputs of the subtitles must coincide with the lip movement, the speech and/or the sound information.</p> <p>Some subtitling programs also use words per minute (wpm). In some countries, it is done this way. Typically, it ranges between 12 and 16 characters per second (cps) or between 145 and 220 words per minute (wpm) (Szarkowska, 2016). The subtitle speed gradually increases, and platforms like Netflix allow speeds of up to 13 cps for children (Netflix, 2024).</p> | Are the subtitles synchronised with the audiovisual product?                              | UNE 153010, 2012<br>Netflix, 2024 |
| Technical parameters | READING SPEED LIMITS | The exposition speed of the subtitle must allow it to be read without difficulty. The maximum recommended speed is 12 characters per second.                                                                                                                                                                                                                                                                                                                                                             | Is the time of exposition speed enough for the CC to be read without difficulty?          | UNE 153010, 2012                  |
| Technical parameters | CHARACTER LIMITATION | The maximum number of characters per line is 37 although this number has been increased, e.g., Netflix (2024) includes 42 characters per line.                                                                                                                                                                                                                                                                                                                                                           | Does the CC exceed 42 characters per line?                                                | UNE 153010, 2012                  |
| Technical parameters | LINES                | Subtitles must occupy two lines and, exceptionally, three for educational material.                                                                                                                                                                                                                                                                                                                                                                                                                      | Have you used more than two lines in the CC when it is intended for educational material? | UNE 153010, 2012                  |





| ASPECT              | SUB-ASPECT                     | RECOMMENDATION                                                                                                                                                               | QUESTION                                                                                                      | SOURCE           |
|---------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|------------------|
| Language parameters | <b>ORTHOGRAPHY AND GRAMMAR</b> | Subtitles must be spelling and grammatically correct.                                                                                                                        | Do the CC follow the spelling and grammar rules of the language of your country?                              | UNE 153010, 2012 |
| Language parameters | <b>ORTHOGRAPHY AND GRAMMAR</b> | Inaccuracies in speech should be corrected except when they are relevant to the argument. In that case they must be enclosed in quotes.                                      | Have you found any language errors?<br><br>If you have, have they been enclosed in quotes?                    | UNE 153010, 2012 |
| Language parameters | <b>ABBREVIATIONS AND UNITS</b> | Natural numbers from zero to ten, inclusive, must be written in letters, except when accompanied by abbreviations, signs, or symbols.                                        | Have you used the rules of the language of your country to subtitle abbreviations and numbers?                | UNE 153010, 2012 |
| Language parameters | <b>ABBREVIATIONS AND UNITS</b> | The use of abbreviations should be avoided unless there are space limitation or they are very common for personal titles, such as: Mr, Ms, Mrs, Dr. preceding a proper noun. | Do you avoid using abbreviations? If you use abbreviations are they the most commonly used (Mr, Ms, Mrs, Dr)? | Netflix, 2024    |



| ASPECT              | SUB-ASPECT | RECOMMENDATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | QUESTION                                                                            | SOURCE        |
|---------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------|
| Language parameters | ACRONYMS   | <p>Acronyms are written without periods or spaces: ONU, FBI</p> <p>Do not use accents if they are written in all caps: CIA, NATO.</p> <p>Acronyms that are pronounced the way they are written (i.e., not spelled) should be capitalized (e.g., UNICEF, UNESCO) if they are proper names and have more than four letters. If they are common nouns and have become part of the daily lexicon (e.g., UFO, AIDS) they are written in lowercase; if this is the case, they need accents following the Spanish accent rules (e.g., laser).</p>                                                                                                                                                      | Have you used the rules of the language of your country to subtitle acronyms?       | Netflix, 2024 |
| Language parameters | ITALICS    | <p>Italicise the following (Netflix, 2024):</p> <ul style="list-style-type: none"> <li>- Titles of works of art (e.g., sculptures, paintings), albums, books, periodicals, films, TV/radio shows, video games, etc. (use quotation marks for song titles and episode titles).</li> <li>- Foreign words, unless they are proper names (e.g., a company name or a vessel name) or part of regular usage (e.g., sushi, pizza). This is not always in line with official regulatory recommendations. Thus, linguists should follow their best judgment, considering that italics should not interfere with legibility.</li> <li>- Song lyrics when sung, not quoted (if rights have been</li> </ul> | Have you used the rules of the language of your country to italicise words or text? | Netflix, 2024 |



| ASPECT              | SUB-ASPECT  | RECOMMENDATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | QUESTION                                                                            | SOURCE        |
|---------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------|
|                     |             | granted).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                     |               |
|                     |             | <ul style="list-style-type: none"> <li>- Recited poetry.</li> <li>- Dialogue that is heard through electronic media, such as a phone, television, computer, loudspeaker, non-sentient robots, robotic voices or AI, etc. This use of italics applies only when the speaker is not in the scene(s), not merely off screen or off camera.</li> </ul>                                                                                                                                                                                                                                                    |                                                                                     |               |
| Language parameters | ITALICS     | <ul style="list-style-type: none"> <li>- Voice-over, narration, the voice of a visible character expressing unspoken thought or inner monologue. This use of italics applies only when the speaker is not in the scene(s), not merely off screen or off camera. Additionally, italics are not needed for subtitle events that belong to a single continuous sentence, divided by a shot change, in which the speaker is not in the scene before the shot change, but is visible after the shot change or vice versa.</li> <li>- Do not use italics to indicate emphasis on specific words.</li> </ul> |                                                                                     |               |
| Language parameters | PUNCTUATION | <p>Do not use semicolons (;).</p> <p>Do not use exclamation and question marks together (!?), please pick the one that suits the intonation or the meaning best.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                  | Have you used the rules of punctuation of the language of your country to subtitle? | Netflix, 2024 |



| ASPECT              | SUB-ASPECT     | RECOMMENDATION                                                                                                                                                                                                                                                                                                                      | QUESTION                                                                                                                                       | SOURCE                            |
|---------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| Language parameters | SEGMENTATION   | <p>When dividing a subtitle into lines, you should: take advantage of speech pauses and silences; take advantage of grammatical pauses or punctuation marks; Write the conjunctions and connections on the bottom line.</p> <p>The most important thing is that the subtitle segmentation is done in complete units of meaning.</p> | Have you segmented the subtitles, respecting the units of meaning as much as possible? Is the subtitle segmented in complete units of meaning? | UNE 153010, 2012                  |
| Language parameters | SEGMENTATION   | Syllables of the same word; nominal, verbal and prepositional phrases should not be separated into two lines.                                                                                                                                                                                                                       | Have syllables of the same word, nominal verbs and prepositional phrases been separated into two lines?                                        | UNE 153010, 2012                  |
| Language parameters | SEGMENTATION   | Ellipses should only be used in accordance with grammatical rules and not to divide sentences into multiple subheadings.                                                                                                                                                                                                            | Have you avoided the use of ellipsis according to the grammatical rules of the language of your country?                                       | UNE 153010, 2012<br>Netflix, 2024 |
| Language parameters | ON-SCREEN TEXT | <p>- Forced narrative titles for on-screen text should only be included if plot pertinent.</p> <p>- When on-screen text and dialogue overlap, precedence should be given to the most plot pertinent message. Avoid over truncating or severely reducing reading speed in order to include both dialogue and on-screen text.</p>     | Have you translated on-screen text?                                                                                                            | Netflix, 2024                     |

| ASPECT                 | SUB-ASPECT               | RECOMMENDATION                                                                                                                                                                                                                                                                                                                                                                                                                  | QUESTION                                                                                                                                               | SOURCE                                       |
|------------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
|                        |                          | - The duration of the FN subtitle should as much as possible mimic the duration of the on-screen text, except for cases where reading speed and/or surrounding dialogue takes precedence.                                                                                                                                                                                                                                       |                                                                                                                                                        |                                              |
| Translation parameters | SOUND EFFECTS AND MUSIC  | <p>The sound effects (music, background noises, sounds of animals or things) that are relevant for the understanding of the video must be described, having them differentiated from the rest of the dialogue.</p> <p>The information should be provided in parentheses, with the first letter in capital letters and the rest in lower case, for example: (Doorbell).</p> <p>The trend nowadays is to use square brackets.</p> | <p>Have you described the most relevant sound effects and music in the audiovisual product?</p> <p>Have they been described using square brackets?</p> | <p>UNE 153010, 2012</p>                      |
| Translation parameters | CHARACTER IDENTIFICATION | <p>The different characters involved in the audiovisual work must be identified, using colours or with their name in parentheses and in capital letters to avoid confusion.</p> <p>Colours are only used if the characters are recurring. It's better to use characters in white and distinguish them with a hyphen.</p> <p>- Use a hyphen followed by a</p>                                                                    | <p>Is it clear when one character speaks from another?</p>                                                                                             | <p>UNE 153010, 2012</p> <p>Netflix, 2024</p> |

| ASPECT                 | SUB-ASPECT                 | RECOMMENDATION                                                                                                                                                                                                                                                                                                                             | QUESTION                                                                           | SOURCE           |
|------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------|
|                        |                            | <p>space to indicate two speakers in one subtitle, with a maximum of one speaker per line.</p> <p>- Text in each line in a dual speaker subtitle should ideally be a contained sentence and should not carry into the preceding or subsequent subtitle. Creating shorter sentences and timing appropriately helps to accommodate this.</p> |                                                                                    |                  |
| Translation parameters | CHARACTER IDENTIFICATION   | In dialogues, scripts should be used only when there is a risk of confusion between characters, and they cannot be differentiated by colours or labels.                                                                                                                                                                                    | Are there characters who speak simultaneously and are difficult to distinguish?    | UNE 153010, 2012 |
| Translation parameters | CHARACTER IDENTIFICATION   | In dialogues, a new line must be started for each participant.                                                                                                                                                                                                                                                                             | Have you used different lines for each character?                                  | UNE 153010, 2012 |
| Translation parameters | VOICE-OVER                 | Voice-overs must be subtitled, that is, those emitted by characters who do not appear on screen.                                                                                                                                                                                                                                           | Have you used a label or italics to indicate the voice-overs?                      | UNE 153010, 2012 |
| Translation parameters | VOICE-OVER                 | Voice-over subtitles must be identified with the same style as that of the characters.                                                                                                                                                                                                                                                     | Are voice-over subtitles identified with the same style as that of the characters? | UNE 153010, 2012 |
| Translation parameters | PARALINGUISTIC INFORMATION | Contextual information must be included in parentheses and in capital letters before the text to which it applies and on the same line. Examples of contextual information                                                                                                                                                                 | Is the contextual information included appropriately?                              | UNE 153010, 2012 |



| ASPECT                 | SUB-ASPECT                             | RECOMMENDATION                                                                                                                                                                                            | QUESTION                                                                                                                | SOURCE                            |
|------------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
|                        |                                        | are: (SHOUTS) (SARCASTIC)<br>(FRENCH ACCENT) (BOTH)<br>(ALL).                                                                                                                                             |                                                                                                                         |                                   |
| Translation parameters | LITERAL TRANSLATION                    | Subtitles must be literal.<br><br>Include as much of the original content as possible.<br><br>Do not simplify, neutralise or water down the original dialogue.                                            | Have you subtitled the content literally? Have you translated the information that appears in another foreign language? | UNE 153010, 2012<br>Netflix, 2024 |
| Translation parameters | OTHER LINGUISTIC ADAPTATION TECHNIQUES | When the dialogue is longer, and due to space-time limitations it cannot be fully reflected, techniques of simplification (lexical or syntactic), condensation, reduction, or elimination should be used. | Have you had to use any type of linguistic adaptation of the dialogue in the audiovisual product?                       | Martínez-Martínez, 2015           |

### Sign Language Interpreting

| ASPECT   | RECOMMENDATION                                                                                         | QUESTION                                                                                                                               | SOURCE    |
|----------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Language | Provide a sign language version of the text to understand key information, ideas and processes.        | Have any key texts related to key information, ideas, and processes been translated into sign language?                                | W3C. G160 |
| Language | Inform in which sign language the content is offered.                                                  | Have you informed which sign language the content is provided in?                                                                      | W3C. G54  |
| Language | Choose the sign language of the main audience and, if there are several audiences, consider using more | Has the sign language of the majority audience or international signs been used if there are people speaking different sign languages? | W3C. G54  |



| ASPECT          | RECOMMENDATION                                                                                                                                      | QUESTION                                                                                                                            | SOURCE                       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
|                 | than one or include international signs as well.                                                                                                    |                                                                                                                                     |                              |
| <b>Language</b> | If the original text is not adapted for the target audience, provide an adapted signed version.                                                     | Is an adapted signed version offered for the target audience?                                                                       | W3C. G54                     |
| <b>Visual</b>   | When using the finger-spelling, synchronise the subtitle of the spelt word with the sign language video on the screen.                              | Has the subtitle been synchronised when there is a finger-spelling to display them at the same time on the screen?                  | Vinayagamoorthy et al., 2019 |
| <b>Visual</b>   | Provide a synchronised video of the sign language interpreter that can be displayed in a different viewport or overlaid on the image by the player. | Is the sign language video displayed on different devices, or is it overlaid on the image by the video player?                      | W3C. G81                     |
| <b>Visual</b>   | The sign language interpreter should be clearly displayed and available at full screen format.                                                      | Can the sign language interpreter be seen clearly and in full screen format?                                                        | W3C. G54                     |
| <b>Visual</b>   | Synchronise the sign language translation to the audio.                                                                                             | Is the sign language translation synchronised with the audio or is there a lag between the audio and the sign language translation? | CNLSE, 2017                  |
| <b>Visual</b>   | If visual elements such as titles, subtitles, etc., are included, consider the placement of the interpreter on the screen.                          | Has the design of the format considered the place of the interpreter and other visual elements such as subtitles, titles, etc.?     | CNLSE, 2017                  |
| <b>Visual</b>   | If a chroma with a transparent background is used,                                                                                                  | Is there enough contrast between the interpreter and the chosen background?                                                         | Arrufat et al., 2021         |



| ASPECT              | RECOMMENDATION                                                                                                                                        | QUESTION                                                                         | SOURCE                         |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------|
|                     | consider background elements for an adequate contrast.                                                                                                |                                                                                  |                                |
| <b>Visual</b>       | The clothing of the signing person should have a high contrast to their skin colour, uniform colour and texture, without any elements that stand out. | Do the interpreter's clothes contrast appropriately with his or her skin colour? | UNE 139804, 2007               |
| <b>Visual</b>       | Provide medium or medium-long framing                                                                                                                 | Is the framing medium or medium long?                                            | UNE 139804, 2007               |
| <b>Visual</b>       | Good illumination of the face and body of the person signing should be ensured to avoid excessive light and shadows on the person and the background. | Is the lighting adequate?                                                        | UNE 139804, 2007               |
| <b>Stakeholders</b> | Consider hiring a deaf translator                                                                                                                     | Have you considered hiring a deaf translator?                                    | W3C. G54                       |
| <b>Stakeholders</b> | Collaboration with the deaf community in the co-creation of content.                                                                                  | Have you collaborated with the deaf community for the co-creation of content?    | Portugal et al., 2023          |
| <b>Stakeholders</b> | Conduct user testing with sign language students to evaluate the material.                                                                            | Have tests been conducted with students to evaluate assess to the content?       | Rodríguez-Fuentes et al., 2022 |
| <b>Stakeholders</b> | If guides for parents and educators are produced, consider an accessible version for deaf parents.                                                    | Have deaf parents been considered in guides for parents and educators?           | Arrufat et al., 2021           |



### 5.3 Cognitive accessibility

**Cognitive accessibility:** Cognitive accessibility focuses on making content understandable and navigable for learners with cognitive disabilities. It includes two sub-dimensions: Simplified Language and Easy Language and Visual Aids and Pictograms

#### Simplified Language

| ASPECT             | SUB-ASPECT     | RECOMMENDATION                                                                                                          | QUESTION                                                                                                                | SOURCE                                                       |
|--------------------|----------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Macrotextual level | END USERS      | Adapt or create the text to meet the needs of the document and its users. Incorporate multimodal elements as necessary. | Does the text meet the needs of the document and its users?<br><br>Does it incorporate multimodal elements?             | ISO/IEC 23859                                                |
| Macrotextual level | END USERS      | Make sure the language level is appropriate for a wide variety of ability levels and cognitive needs.                   | Is the language level appropriate for a wide variety of ability levels and cognitive needs?                             | UNESCO, 2021<br>Affordable Learning Georgia<br>ISO/IEC 23859 |
| Macrotextual level | END USERS      | Make sure the content can be translated easily (or it is already available in multiple languages).                      | Is the content already available in multiple languages?<br>Can the content be easily translated?                        | UNESCO, 2021<br>Affordable Learning Georgia                  |
| Macrotextual level | END USERS      | Facilitators should check document comprehension with end users.                                                        | Has document comprehension been checked with end users?                                                                 | UNE<br>153101:2018 EX                                        |
| Macrotextual level | END USERS      | Avoid using childish language if the text is not intended for children.                                                 | Does the text use childish language to address adult users?                                                             | ISO/IEC 23859                                                |
| Macrotextual level | TEXT STRUCTURE | Arrange information in the easiest-to-understand order.                                                                 | How well does the document arrange information in a logical and easy-to-understand order? (1= Not at all, 5= Very well) | Politis et al., 2014<br>CEDEC, 2020                          |



| ASPECT             | SUB-ASPECT     | RECOMMENDATION                                                                                                                                                                                                               | QUESTION                                                                                                                                                                                              | SOURCE                           |
|--------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| Macrotextual level | TEXT STRUCTURE | Information on the same topic should be grouped into text blocks. Line spacing or design elements may be used to separate blocks from one another.                                                                           | Are related pieces of information grouped together into distinct text blocks?<br><br>Does the document use line spacing or other design elements to clearly separate different blocks of information? | UNE 153101:2018 EX ISO/IEC 23859 |
| Macrotextual level | TEXT STRUCTURE | A timeline explaining the order of events should be maintained. Adverbs or expressions of time such as “before,” “during,” “after,” “next morning,” “during the night,” “in the beginning,” “first,” or “later” can be used. | How well does the document maintain a clear and understandable timeline of events? (1= Not at all, 5= Very well)<br><br>Are temporal adverbs or expressions used to indicate the sequence of events?  | UNE 153101:2018 EX               |
| Macrotextual level | TEXT STRUCTURE | Spatial information should be conveyed to locate the action.                                                                                                                                                                 | How well does the document convey spatial information to clarify the location of actions or events? (1= Not at all, 5= Very well)                                                                     | UNE 153101:2018 EX               |
| Macrotextual level | SUBDIVISIONS   | Divisions should be established in the text using titles, subtitles, clauses, subclauses, etc., to strengthen the logical structure.                                                                                         | Does the document use titles, subtitles, clauses, and subclauses to create a clear logical structure?                                                                                                 | UNE 153101:2018 EX               |
| Macrotextual level | SUBDIVISIONS   | Titles should reference or anticipate the content. They should be clearly differentiated                                                                                                                                     | Do the titles accurately reference or anticipate the content that follows?                                                                                                                            | UNE 153101:2018 EX ISO/IEC 23859 |



| ASPECT                    | SUB-ASPECT                     | RECOMMENDATION                                                                      | QUESTION                                                                                                                  | SOURCE                                               |
|---------------------------|--------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
|                           |                                | from the main text.                                                                 | Are the titles clearly differentiated from the main text?                                                                 |                                                      |
| <b>Macrotextual level</b> | <b>SUBDIVISIONS</b>            | Complex structures with over three division levels should be avoided.               | Does the document avoid using complex structures with more than three division levels?                                    | UNE 153101:2018 EX                                   |
| <b>Macrotextual level</b> | <b>INFERENCES AND COHESION</b> | Inferences should be avoided.                                                       | Does the document avoid making inferences that are not explicitly supported by the text?                                  | UNE 153101:2018 EX<br>ISO/IEC 23859                  |
| <b>Macrotextual level</b> | <b>INFERENCES AND COHESION</b> | Textual cohesion and coherence should be maintained throughout the entire document. | How well does the document maintain textual cohesion and coherence from beginning to end? (1 = Not at all, 5 = Very well) | UNE 153101:2018 EX                                   |
| <b>Macrotextual level</b> | <b>STYLE</b>                   | Avoid unnecessarily long texts.                                                     | Is the text concise and free from unnecessary length?                                                                     | ISO/IEC 23859                                        |
| <b>Macrotextual level</b> | <b>STYLE</b>                   | Texts with dialogues should reflect a theatrical style.                             | Does the document use a theatrical style for texts with dialogues?                                                        | UNE 153101:2018 EX                                   |
| <b>Vocabulary</b>         | <b>GENERAL RECOMMENDATIONS</b> | Simple, frequently used language should be used.                                    | How effectively does the document use simple and frequently used language? (1 = Not at all effective, 5 = Very effective) | Politis et al., 2014<br>CEDEC, 2020<br>ISO/IEC 23859 |
| <b>Vocabulary</b>         | <b>GENERAL RECOMMENDATIONS</b> | A vocabulary suited to the end user and text type of the document should be used.   | Is the vocabulary in the document appropriate for the intended audience and the type of text?                             | UNE 153101:2018 EX<br>ISO/IEC 23859                  |
| <b>Vocabulary</b>         | <b>GENERAL REC-</b>            | Use the same word to name the same object                                           | Does the document consistently use the same                                                                               | UNE 153101:2018 EX<br>ISO/IEC 23859                  |



| ASPECT            | SUB-ASPECT                         | RECOMMENDATION                                                                                                                                                                                                                | QUESTION                                                                                                               | SOURCE                              |
|-------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
|                   | <b>RECOMMENDATIONS</b>             | or reference throughout a text.                                                                                                                                                                                               | word to refer to the same object or reference?                                                                         |                                     |
| <b>Vocabulary</b> | <b>GENERAL RECOMMENDATIONS</b>     | Abstract, technical or complex terms should be limited. Use them only to convey essential information.                                                                                                                        | Does the document limit the use of abstract, technical, or complex terms to essential information only?                | UNE 153101:2018 EX ISO/IEC 23859    |
| <b>Vocabulary</b> | <b>GENERAL RECOMMENDATIONS</b>     | Vague and ambiguous words should be avoided.                                                                                                                                                                                  | Does the document avoid using vague and ambiguous words?                                                               | ISO/IEC 23859                       |
| <b>Vocabulary</b> | <b>GENERAL RECOMMENDATIONS</b>     | Words that do not add information and lengthen the text should be avoided.                                                                                                                                                    | Are words avoided in the text that do not add information and unnecessarily lengthen the text avoided in the document? | UNE 153101:2018 EX ISO/IEC 23859    |
| <b>Vocabulary</b> | <b>GENERAL RECOMMENDATIONS</b>     | When dealing with compound words, use a simpler synonym or when possible, break the word up.                                                                                                                                  | Are simpler synonyms used for compound words, or are compound words broken up when possible?                           | ISO/IEC 23859                       |
| <b>Vocabulary</b> | <b>EXPLANATIONS AND GLOSSARIES</b> | If you need to use difficult words, explain them clearly. Use examples when needed.                                                                                                                                           | Are difficult words clearly explained with examples provided if necessary?                                             | Politis et al., 2014<br>CEDEC, 2020 |
| <b>Vocabulary</b> | <b>EXPLANATIONS AND GLOSSARIES</b> | If there are several terms that are hard to understand, a glossary should be prepared or glosses (i.e., brief explanatory notes positioned close to the word or expression being defined) should be included in the document. | Is a glossary or glosses included in the document to explain hard-to-understand terms?                                 | UNE 153101:2018 EX                  |



| ASPECT     | SUB-ASPECT                                    | RECOMMENDATION                                                                                                                                                                           | QUESTION                                                                                                       | SOURCE                                                                  |
|------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Vocabulary | <b>HOMOPHONES, ACRONYMS AND FOREIGN WORDS</b> | If words with the same sound or the same spelling (homophones or homographs) are used with several meanings, the text should be written to unequivocally facilitate their comprehension. | Does the document clearly differentiate the meanings of homophones or homographs to avoid confusion?           | UNE 153101:2018 EX                                                      |
| Vocabulary | <b>HOMOPHONES, ACRONYMS AND FOREIGN WORDS</b> | The use of words from other languages should be avoided if these are not common.                                                                                                         | Does the document avoid using uncommon words from other languages?                                             | UNE 153101:2018 EX ISO/IEC 23859                                        |
| Vocabulary | <b>HOMOPHONES, ACRONYMS AND FOREIGN WORDS</b> | Acronyms, initials and abbreviations should be avoided. If they are used, they should be spelled-out or explained or both.                                                               | Are acronyms, initials, and abbreviations either avoided or clearly spelled out and explained in the document? | Politis et al., 2014<br>CEDEC, 2020<br>UNE 153101:2018 EX ISO/IEC 23859 |
| Vocabulary | <b>ABSTRACTION AND FIGURATIVE LANGUAGE</b>    | Words of undetermined content should be avoided, such as “thing,” something,” or “matter.”                                                                                               | Does the document avoid using words with undetermined content like “thing” or “something”?                     | UNE 153101:2018 EX                                                      |
| Vocabulary | <b>ABSTRACTION AND FIGURATIVE LANGUAGE</b>    | Figurative expressions should be limited. Use them considering text type and intention and explain their meaning when necessary.                                                         | Are figurative expressions used sparingly and explained for clarity when necessary?                            | Politis et al., 2014<br>CEDEC, 2020<br>UNE 153101:2018 EX ISO/IEC 23859 |
| Syntax     | <b>GENERAL RECOMMENDATIONS</b>                | Use short and simple sentences.                                                                                                                                                          | Are sentences short and simple?                                                                                | Politis et al., 2014<br>CEDEC, 2020<br>UNE 153101:2018 EX               |



| ASPECT | SUB-ASPECT              | RECOMMENDATION                                                                                                                  | QUESTION                                                                                                         | SOURCE                                                                     |
|--------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Syntax | GENERAL RECOMMENDATIONS | Address the reader directly when possible.                                                                                      | Does the document address the reader directly?                                                                   | Politis et al., 2014<br>CEDEC, 2020<br>UNE 153101:2018 EX                  |
| Syntax | GENERAL RECOMMENDATIONS | Write no more than two ideas per sentence. If possible, write only one idea per sentence and line.                              | Does the document limit sentences to no more than two ideas, preferably one?                                     | Politis et al., 2014<br>CEDEC, 2020<br>UNE 153101:2018 EX<br>ISO/IEC 23859 |
| Syntax | GENERAL RECOMMENDATIONS | If a sentence is longer than one line, cut it where a natural pause would be appropriate.                                       | Are longer sentences broken up at natural pauses?                                                                | Politis et al., 2014<br>CEDEC 2.020                                        |
| Syntax | GRAMMAR                 | Affirmative sentences are preferable, except for cases where negations are clearer and more direct (e.g., simple prohibitions). | Does the document favour affirmative sentences, using negations only when they are clearer and more direct?      | UNE 153101:2018 EX                                                         |
| Syntax | GRAMMAR                 | Avoid double negations.                                                                                                         | Does the document avoid the use of double negations?                                                             | UNE 153101:2018 EX                                                         |
| Syntax | GRAMMAR                 | Do not replace names or subjects with pronouns unless the reference can be easily identified.                                   | Does the document avoid replacing names or subjects with pronouns unless the reference can be easily identified? | UNE 153101:2018 EX                                                         |
| Syntax | GRAMMAR                 | The use of appositions and explanations between commas that pose a digression should be avoided.                                | Does the document avoid appositions and explanations between commas that cause digressions?                      | UNE 153101:2018 EX                                                         |



| ASPECT | SUB-ASPECT | RECOMMENDATION                                                                                                                   | QUESTION                                                           | SOURCE                                                    |
|--------|------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------|
| Syntax | GRAMMAR    | Complex connectors between phrases, such as “therefore,” “notwithstanding,” “consequently,” or “nevertheless” should be avoided. | Does the document avoid using complex connectors between phrases?  | UNE 153101:2018 EX                                        |
| Syntax | GRAMMAR    | The use of nominal sentences should be avoided.                                                                                  | Does the document avoid nominal sentences?                         | UNE 153101:2018 EX                                        |
| Syntax | VERBS      | Use the active voice.                                                                                                            | Does the document primarily use the active voice?                  | Politis et al., 2014<br>CEDEC, 2020<br>UNE 153101:2018 EX |
| Syntax | VERBS      | Use simple forms of verbs (preferably in present tense) when possible.                                                           | Is the present tense used throughout the document?                 | UNE 153101:2018 EX<br>ISO/IEC 23859                       |
| Syntax | VERBS      | Compound or seldom-used verbal tenses (e.g., conditional, subjunctive) should be avoided.                                        | Are compound or seldom-used verbal tenses avoided in the document? | UNE 153101:2018 EX                                        |
| Syntax | VERBS      | Impersonal sentences should be avoided.                                                                                          | Does the document avoid the use of impersonal sentences?           | UNE 153101:2018 EX                                        |
| Syntax | VERBS      | Sentences using gerunds should be avoided.                                                                                       | Are sentences using gerunds avoided in the document?               | UNE 153101:2018 EX                                        |
| Syntax | VERBS      | Subordinate clauses should be avoided.                                                                                           | Does the document avoid the use of subordinate clauses?            | ISO/IEC 23859                                             |
| Syntax | VERBS      | The consecutive use of two or more verbs                                                                                         | Is the consecutive use of two or more verbs                        | UNE 153101:2018 EX                                        |





| ASPECT             | SUB-ASPECT | RECOMMENDATION                                                                                                                                            | QUESTION                                                                                                   | SOURCE                                                    |
|--------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
|                    |            | should be avoided, except for modal verbs.                                                                                                                | avoided in the document except for modal verbs?                                                            |                                                           |
| <b>Punctuation</b> |            | Punctuation should follow the grammar rules of each language.                                                                                             | Does the document follow the grammar rules of the language used?                                           | ISO/IEC 23859                                             |
| <b>Punctuation</b> |            | Use full stops to divide phrases and sentences containing different ideas.                                                                                | Are full stops used to divide phrases and sentences that contain different ideas?                          | UNE 153101:2018 EX                                        |
| <b>Punctuation</b> |            | Use paragraphs to join related ideas.                                                                                                                     | Are paragraphs used to join related ideas?                                                                 | UNE 153101:2018 EX                                        |
| <b>Punctuation</b> |            | Use commas to divide related elements in a single sentence.                                                                                               | Are commas used correctly to divide related elements within a single sentence?                             | UNE 153101:2018 EX                                        |
| <b>Punctuation</b> |            | The colon (:) should precede lists with more than three elements.                                                                                         | Are colons used to precede lists with more than three elements?                                            | UNE 153101:2018 EX                                        |
| <b>Punctuation</b> |            | The use of semicolons (;) should be avoided.                                                                                                              | Does the document avoid the use of semicolons?                                                             | UNE 153101:2018 EX                                        |
| <b>Punctuation</b> |            | The use of parentheses, brackets and infrequent punctuation signs and special characters (% , & / , #) should be avoided.                                 | Are parentheses, brackets, and infrequent punctuation signs or special characters avoided in the document? | Politis et al., 2014<br>CEDEC, 2020<br>UNE 153101:2018 EX |
| <b>Punctuation</b> |            | Unclear expressions including “etcetera”, and ellipses should not be used. These may be replaced by expressions such as “among other things” or “and many | Does the document avoid using “etcetera” or ellipses, replacing them with clear expressions?               | UNE 153101:2018 EX                                        |

| ASPECT             | SUB-ASPECT | RECOMMENDATION                                                                                                                                                                                        | QUESTION                                                                                                           | SOURCE                                               |
|--------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
|                    |            | more”.                                                                                                                                                                                                |                                                                                                                    |                                                      |
| <b>Punctuation</b> |            | Quotation marks should be avoided. When used, (e.g., in verbatim quotations), they should be accompanied by an explanation.                                                                           | Are quotation marks avoided unless necessary, or accompanied by an explanation when used?                          | UNE 153101:2018 EX                                   |
| <b>Numbers</b>     |            | Numbers should be written in digits.                                                                                                                                                                  | Are numbers written in digits?                                                                                     | Politis et al., 2014<br>CEDEC, 2020                  |
| <b>Numbers</b>     |            | Use non-breaking spaces when chunking large numbers (e.g., 1234 5678).                                                                                                                                | Are non-breaking spaces used when chunking large numbers?                                                          | ISO/IEC 23859                                        |
| <b>Numbers</b>     |            | If numbers with many digits are used, write them out, use qualitative comparisons or substitute them for terms such as “several,” “thousands,” or “millions” when possible, to enhance comprehension. | Are large numbers written out, compared qualitatively, or substituted with simpler terms to enhance comprehension? | Politis et al., 2014<br>CEDEC, 2020<br>ISO/IEC 23859 |
| <b>Numbers</b>     |            | Fractions and percentages should be avoided. If they are necessary, their equivalence should be described, or an explanation should be given.                                                         | Are fractions and percentages avoided, or clearly explained in the document if necessary?                          | Politis et al., 2014<br>CEDEC, 2020                  |
| <b>Numbers</b>     |            | Avoid using Roman numerals in general. However, for centuries, monarchs, and popes, it                                                                                                                | Are Roman numerals avoided unless referencing centuries, monarchs, or popes?                                       | Politis et al., 2014<br>CEDEC, 2020                  |



| ASPECT         | SUB-ASPECT                     | RECOMMENDATION                                                                                                                                                 | QUESTION                                                                                                        | SOURCE                                               |
|----------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
|                |                                | is recommended to use the Roman numeral along with an indication of its pronunciation.                                                                         | Are they accompanied by pronunciation when used?                                                                |                                                      |
| <b>Numbers</b> |                                | Dates written with hyphens or slashes should be avoided. Provide the full date only when it is essential for understanding the text.                           | Does the document avoid using dates with hyphens or slashes, providing the full date only when necessary?       | Politis et al., 2014<br>CEDEC, 2020<br>ISO/IEC 23859 |
| <b>Numbers</b> |                                | Telephone numbers should be separated into blocks.                                                                                                             | Are telephone numbers separated into blocks?                                                                    | UNE 153101:2018 EX                                   |
| <b>Numbers</b> |                                | Ordinal numbers (1st) should be avoided and substituted by cardinal numbers (1).                                                                               | Are ordinal numbers avoided, with cardinal numbers used instead?                                                | UNE 153101:2018 EX                                   |
| <b>Numbers</b> |                                | Writing the time in 24-hour format should be avoided.                                                                                                          | Is the 24-hour time format avoided in favour of a more readable format?                                         | UNE 153101:2018 EX                                   |
| <b>Numbers</b> |                                | When using graphs or tables, keep numbers simple and explain them well.                                                                                        | Are numbers in graphs or tables kept simple and explained well?                                                 | Politis et al., 2014<br>CEDEC, 2020                  |
| <b>Design</b>  | <b>GENERAL RECOMMENDATIONS</b> | A universal design approach shall be taken into account when designing content in all types of formats, so that it is accessible to the widest range of users. | Is a universal design approach taken into account to ensure content is accessible to the widest range of users? | ISO/IEC 23859                                        |
| <b>Design</b>  | <b>GENERAL RECOMMENDATIONS</b> | Alternative formats to written text, such as auditory, haptic, visual and                                                                                      | Are alternative formats to written text, such as auditory, haptic, and visual                                   | ISO/IEC 23859                                        |

| ASPECT | SUB-ASPECT              | RECOMMENDATION                                                                                                                        | QUESTION                                                                                        | SOURCE                                                                     |
|--------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
|        |                         | other suitable means shall be provided where possible.                                                                                | means, provided where possible?                                                                 |                                                                            |
| Design | GENERAL RECOMMENDATIONS | Individualisation shall be offered where possible (e.g., removable subtitles and audio description)                                   | Is individualisation offered where possible (e.g., removable subtitles and audio description)?  | ISO/IEC 23859                                                              |
| Design | PAGE DESIGN             | Margins should be wide enough to allow for comfortable reading and possible addition of glosses.                                      | Are the margins wide enough to allow for comfortable reading and the addition of glosses?       | ISO/IEC 23859                                                              |
| Design | PAGE DESIGN             | Do not use a background that makes the text difficult to read. Instead, use a flat colour that has good contrast with the foreground. | Does the document use a flat background colour with good contrast to the text?                  | Politis et al., 2014<br>CEDEC, 2020<br>UNE 153101:2018 EX<br>ISO/IEC 23859 |
| Design | PAGE DESIGN             | Watermarks shall be avoided.                                                                                                          | Are watermarks avoided in the document?                                                         | UNE 153101:2018 EX                                                         |
| Design | PAGE DESIGN             | All the pages of the document should have the same orientation, whether horizontal or vertical.                                       | Do all pages of the document have the same orientation (either all horizontal or all vertical)? | UNE 153101:2018 EX                                                         |
| Design | PAGE DESIGN             | Vertical titles or text layouts should be avoided.                                                                                    | Are vertical titles and text layouts avoided in the document?                                   | UNE 153101:2018 EX                                                         |
| Design | PAGE DESIGN             | Number the pages of the text and use a larger font than for the rest of the words of the text.                                        | Are the pages numbered with a font larger than the main text?                                   | Politis et al., 2014<br>CEDEC, 2020<br>ISO/IEC 23859                       |



| ASPECT | SUB-ASPECT          | RECOMMENDATION                                                                                                                                             | QUESTION                                                                                                                                 | SOURCE                                                                     |
|--------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Design | PAGE DESIGN         | Descriptions such as “continues on the next page” should be added where relevant.                                                                          | Are descriptions like “continues on the next page” included where relevant?                                                              | ISO/IEC 23859                                                              |
| Design | PAGE DESIGN         | Never use footnotes.                                                                                                                                       | Are footnotes avoided in the document?                                                                                                   | Politis et al., 2014<br>CEDEC, 2020                                        |
| Design | PAGE DESIGN         | The division of a text block into several pages shall be avoided and lines should not be isolated from their context.                                      | Is the division of text blocks across pages avoided to ensure that lines are not isolated from their context?                            | UNE 153101:2018 EX                                                         |
| Design | ALIGNMENT AND LISTS | Avoid writing in columns.                                                                                                                                  | Are columns avoided in the document?                                                                                                     | Politis et al., 2014<br>CEDEC, 2020<br>ISO/IEC 23859                       |
| Design | ALIGNMENT AND LISTS | Justify the text so that it is easy to navigate. Use left justification for Latin alphabet languages and full justification for character-based languages. | Is the text justified appropriately (left justification for Latin alphabet languages, full justification for character-based languages)? | Politis et al., 2014<br>CEDEC, 2020<br>UNE 153101:2018 EX<br>ISO/IEC 23859 |
| Design | ALIGNMENT AND LISTS | A homogeneous line length should be maintained whenever possible.                                                                                          | Is a homogeneous line length maintained throughout the document?                                                                         | UNE 153101:2018 EX                                                         |
| Design | ALIGNMENT AND LISTS | A minimum line spacing of 1.5, adapted to the font size and the size of the support used to display the text, should be used.                              | Is a minimum line spacing of 1.5 used, adapted to the font size and display medium?                                                      | UNE 153101:2018 EX                                                         |

| ASPECT | SUB-ASPECT          | RECOMMENDATION                                                                                                                                                                               | QUESTION                                                                                                         | SOURCE                                               |
|--------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| Design | ALIGNMENT AND LISTS | Use bullet points for lists.                                                                                                                                                                 | Are bullet points used for lists in the document?                                                                | Politis et al., 2014<br>CEDEC, 2020                  |
| Design | FONT                | Use a clear, easy-to-read font (e.g., Arial, Tahoma, Ubuntu).                                                                                                                                | Does the document have a clear, easy-to-read font such as Arial, Tahoma, or Ubuntu?                              | Politis et al., 2014<br>CEDEC, 2020                  |
| Design | FONT                | Avoid italics, underlining, shadows, contour and relief.                                                                                                                                     | Are italics, underlining, shadows, contour, and relief avoided in the document?                                  | Politis et al., 2014<br>CEDEC, 2020<br>ISO/IEC 23859 |
| Design | FONT                | Use bold text only to highlight important words or words which are explained in glosses or in the glossary.                                                                                  | Is bold text limited to highlighting important words or those explained in glosses or in the glossary?           | UNE<br>153101:2018 EX<br>ISO/IEC 23859               |
| Design | FONT                | Use a font size between 12 and 16 points, with 14 points being the standard. However, the optimal size may vary depending on the font type.                                                  | Is the font size between 12 and 16 points (ideally 14), or adjusted to font type?                                | Politis et al., 2014<br>CEDEC, 2020                  |
| Design | FONT                | In languages which use uppercase and lowercase letters, do not write everything in capital letters. Limit them to the beginning of a paragraph or title, after a period, or in proper nouns. | Is the use of capital letters limited to the beginning of paragraphs, titles, after periods, or in proper nouns? | Politis et al., 2014<br>CEDEC, 2020<br>ISO/IEC 23859 |
| Design | FONT                | Use underlining carefully.                                                                                                                                                                   | Is underlining used carefully in the document?                                                                   | Politis et al., 2014<br>CEDEC, 2020                  |



| ASPECT               | SUB-ASPECT                  | RECOMMENDATION                                                                                                                   | QUESTION                                                                                                                        | SOURCE                              |
|----------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| Design               | COLOURS                     | Avoid writing in colour.                                                                                                         | Is writing in colour avoided in the document?                                                                                   | Politis et al., 2014<br>CEDEC, 2020 |
| Design               | COLOURS                     | Cultural differences in the use of colours should be taken into account.                                                         | Are cultural differences in the use of colours considered in the document?                                                      | ISO/IEC 23859                       |
| Design               | COLOURS                     | The meaning of colours should also be conveyed by alternative means to be accessible to all users.                               | Is the meaning of colours conveyed by alternative means to ensure accessibility for all users?                                  | ISO/IEC 23859                       |
| Design               | COLOURS                     | Avoid using many different colours.                                                                                              | Does the document avoid using many different colours?                                                                           | ISO/IEC 23859                       |
| Design               | COLOURS                     | High contrast ratios should be used to enhance readability in the document.                                                      | Are high contrast ratios used to enhance readability in the document?                                                           | ISO/IEC 23859                       |
| Paratextual elements | VERBAL PARATEXTUAL ELEMENTS | Tables of contents, glosses, glossaries, summaries, and pre- and post-reading activities can enhance comprehension.              | Are tables of contents, glosses, glossaries, summaries, and pre- and post-reading activities provided to enhance comprehension? | ISO/IEC 23859                       |
| Paratextual elements | VERBAL PARATEXTUAL ELEMENTS | Use a table of contents at the beginning of long texts. The page number should be easy to associate with the respective section. | Is there a table of contents at the beginning of long texts?<br><br>Does it have easily associated page numbers?                | ISO/IEC 23859                       |



| ASPECT               | SUB-ASPECT                  | RECOMMENDATION                                                                                                                                                        | QUESTION                                                                                                        | SOURCE                              |
|----------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------|
| Paratextual elements | VERBAL PARATEXTUAL ELEMENTS | A gloss may be used when complex words or expressions are used and cannot be understood in the text. It shall include an explanation and, where relevant, an example. | Are glosses provided for complex words or expressions, including explanations and examples where relevant?      | ISO/IEC 23859                       |
| Paratextual elements | VERBAL PARATEXTUAL ELEMENTS | Glosses should be positioned close to the words being defined, generally on the margin and always on the same page.                                                   | Are glosses positioned close to the words being defined (e.g., on the margin and on the same page)?             | ISO/IEC 23859                       |
| Paratextual elements | VERBAL PARATEXTUAL ELEMENTS | If a glossary is provided, it should be easy to find and have a heading.                                                                                              | Is the glossary easy to find?<br><br>Does it have a clear heading?                                              | UNE 153101:2018 EX<br>ISO/IEC 23859 |
| Paratextual elements | VERBAL PARATEXTUAL ELEMENTS | The words and expressions in the glossary should be presented in alphabetical order.                                                                                  | Are the words and expressions in the glossary presented in alphabetical order?                                  | UNE 153101:2018 EX                  |
| Paratextual elements | VERBAL PARATEXTUAL ELEMENTS | When organising the glossary, highlight the word or expressions to be defined by using bold or a larger font.                                                         | Are the words or expressions to be defined in the glossary highlighted using bold or a larger font?             | UNE 153101:2018 EX                  |
| Paratextual elements | VERBAL PARATEXTUAL ELEMENTS | If summaries are used, locate them at the end of the text to which they refer. Make sure they can be distinguished from the main text.                                | Are summaries located at the end of the text they refer to?<br><br>Are they distinguishable from the main text? | UNE 153101:2018 EX                  |





| ASPECT               | SUB-ASPECT | RECOMMENDATION                                                                                                                             | QUESTION                                                                                                                  | SOURCE                           |
|----------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| Paratextual elements | IMAGES     | The use of images (photos, drawings, illustrations) should help or complement the comprehension of the text.                               | Do images used in the document help or complement the comprehension of the text?                                          | UNE 153101:2018 EX ISO/IEC 23859 |
| Paratextual elements | IMAGES     | Images should be placed close to the text without affecting readability. Locate them on the margin or in the upper part of the text block. | Are images placed close to the text, on the margin or in the upper part of the text block, without affecting readability? | UNE 153101:2018 EX ISO/IEC 23859 |
| Paratextual elements | IMAGES     | A concept should be represented by the same image throughout the document.                                                                 | Is each concept consistently represented by the same image throughout the document?                                       | UNE 153101:2018 EX ISO/IEC 23859 |
| Paratextual elements | IMAGES     | High-resolution colour images should be used.                                                                                              | Are high-resolution colour images used in the document?                                                                   | UNE 153101:2018 EX ISO/IEC 23859 |
| Paratextual elements | IMAGES     | Images should be clear, without complex frames or abstract details.                                                                        | Are images clear and free of complex frames or abstract details?                                                          | UNE 153101:2018 EX ISO/IEC 23859 |
| Paratextual elements | IMAGES     | The background of the images should be flat, without elements that distract attention.                                                     | Do images have flat backgrounds that avoid distractions?                                                                  | UNE 153101:2018 EX               |
| Paratextual elements | IMAGES     | When an image description is provided, it should be written in an easy manner so that it does not interfere with reading.                  | Are image descriptions written clearly and in a way that does not interfere with reading?                                 | ISO/IEC 23859                    |

| ASPECT                               | SUB-ASPECT               | RECOMMENDATION                                                                                            | QUESTION                                                                                                               | SOURCE                              |
|--------------------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| Paratextual elements                 | IMAGES                   | Alternative text shall be used for images in digital content.                                             | Is alternative text provided for images in digital content?                                                            | ISO/IEC 23859                       |
| Paratextual elements                 | OTHER GRAPHICAL ELEMENTS | Other graphical elements should contain a heading, a description, and, when applicable, alternative text. | Do other graphical elements contain a heading and a description?<br><br>Do they have alternative text (if applicable)? | ISO/IEC 23859                       |
| Paratextual elements                 | OTHER GRAPHICAL ELEMENTS | If graphs, maps or plans are used, explain or adapt them and caption them.                                | Are graphs, maps, or plans explained, adapted and captioned?                                                           | UNE 153101:2018 EX                  |
| Paratextual elements                 | OTHER GRAPHICAL ELEMENTS | Limit the number of variables in graphs to two.                                                           | Are the number of variables in graphs limited to two?                                                                  | UNE 153101:2018 EX                  |
| Paratextual elements                 | OTHER GRAPHICAL ELEMENTS | Avoid tables when possible.                                                                               | Are tables avoided whenever possible?                                                                                  | ISO/IEC 23859                       |
| Audiovisual texts (where applicable) | VIDEO                    | Informative videos should be no longer than 20-30 minutes.                                                | Are informative videos no longer than 20-30 minutes?                                                                   | Politis et al., 2014<br>CEDEC, 2020 |
| Audiovisual texts (where applicable) | VIDEO                    | The video and sound should be clear and of very high quality.                                             | Are the video and sound clear and of very high quality?                                                                | Politis et al., 2014<br>CEDEC, 2020 |
| Audiovisual texts (where applicable) | VIDEO                    | The screen should not be too dark or too light.                                                           | Is the screen neither too dark nor too light?                                                                          | Politis et al., 2014<br>CEDEC, 2020 |
| Audiovisual texts (where applicable) | VIDEO                    | Sound and full-screen icons should be easy to find.                                                       | Are sound and full-screen icons easy to find?                                                                          | Politis et al., 2014<br>CEDEC, 2020 |
| Audiovisual                          | VIDEO                    | Avoid slow and fast motion effects.                                                                       | Are slow and fast motion effects avoided in the                                                                        | Politis et al., 2014                |



| ASPECT                               | SUB-ASPECT | RECOMMENDATION                                                                                             | QUESTION                                                                               | SOURCE                              |
|--------------------------------------|------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------|
| texts (where applicable)             |            |                                                                                                            | video?                                                                                 | CEDEC, 2020                         |
| Audiovisual texts (where applicable) | VIDEO      | If you are using voiceover, only talk about the things that people see on screen.                          | Does the voiceover only describe what it is visible on the screen?                     | Politis et al., 2014<br>CEDEC, 2020 |
| Audiovisual texts (where applicable) | VIDEO      | If you use subtitles, make them removable and easy to read.                                                | Are the subtitles removable?<br><br>Are they easy to read?                             | Politis et al., 2014<br>CEDEC, 2020 |
| Audiovisual texts (where applicable) | VIDEO      | If you use audio description, make it removable and use easy-to-understand words.                          | Is the audio description removable?<br><br>Does it use easy-to-understand words?       | Politis et al., 2014<br>CEDEC, 2020 |
| Audiovisual texts (where applicable) | VIDEO      | Textual description and transcription of the video should be provided.                                     | Is a textual description and transcription of the video provided?                      | Politis et al., 2014<br>CEDEC, 2020 |
| Audiovisual texts (where applicable) | VIDEO      | When using audio and textual description, make sure to explain the recording location whenever it changes. | Are changes in the recording location explained in the audio and textual descriptions? | Politis et al., 2014<br>CEDEC, 2020 |
| Audiovisual texts (where applicable) | AUDIO      | An audio alternative of a written text should be provided.                                                 | Is an audio alternative to the written text provided?                                  | ISO/IEC 23859                       |
| Audiovisual texts (where applicable) | AUDIO      | Make sure the quality of the sound is high and at the right volume.                                        | Is the sound quality high and at the correct volume?                                   | Politis et al., 2014<br>CEDEC, 2020 |
| Audiovisual texts (where applicable) | AUDIO      | Make sure the speaking person speaks slowly, has good pronunciation and vocalises clearly.                 | Does the speaker speak slowly, with clear pronunciation and vocalisation?              | Politis et al., 2014<br>CEDEC, 2020 |



| ASPECT                               | SUB-ASPECT | RECOMMENDATION                                                                                                        | QUESTION                                                                                 | SOURCE                              |
|--------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------|
| Audiovisual texts (where applicable) | AUDIO      | Voices with different accents can also be used.                                                                       | Are voices with different accents used in the audio?                                     | ISO/IEC 23859                       |
| Audiovisual texts (where applicable) | AUDIO      | Make sure the audio description has pauses when necessary and gives people enough time to understand the information. | Does the audio description have pauses to allow for good access to information by users? | Politis et al., 2014<br>CEDEC, 2020 |
| Audiovisual texts (where applicable) | AUDIO      | Do not hesitate to repeat important information several times.                                                        | Is important information repeated several times in the audio?                            | Politis et al., 2014<br>CEDEC, 2020 |
| Audiovisual texts (where applicable) | AUDIO      | Read the text so that emotions are clearly understood.                                                                | Is the text read in a way that conveys emotions clearly?                                 | Politis et al., 2014<br>CEDEC, 2020 |
| Audiovisual texts (where applicable) | AUDIO      | A textual transcription of the audio should be provided.                                                              | Is a textual transcription of the audio provided?                                        | Politis et al., 2014<br>CEDEC, 2020 |

### Visual Aids and Pictograms

| RECOMMENDATION                                                                                   | QUESTION                                                                                                                  | SOURCE                                                                                          |
|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Ensure that students are able to physically and effortlessly identify and select the pictograms. | Are pictograms physically and effortlessly identified so that students can select the relevant pictograms?                | CEAPAT-IMSERO, 2018<br>Fundación ONCE, 2013<br>ISO 9186-1:2014<br>CEAPAT, 2023<br>Berrios, 2020 |
| Make sure that the pictograms are implemented on a suitable support and material.                | Have the pictograms been implemented on a suitable support for the learner?<br><br>Are there physical or sensory barriers | CEAPAT-IMSERO, 2018<br>Fundación ONCE, 2013<br>ISO 9186-1:2014<br>CEAPAT, 2023<br>Berrios, 2020 |



| RECOMMENDATION                                                                                                                                                                                                                         | QUESTION                                                                                                                                                                                                                  | SOURCE                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                        | to accessing these resources?                                                                                                                                                                                             |                                                                                                  |
| Ensure that pictograms are appropriately sized for the students' physical or visual mode of access.                                                                                                                                    | Are pictograms of an appropriate size for the student's physical or visual mode of access?                                                                                                                                | CEAPAT-IMSERSO, 2018<br>Fundación ONCE, 2013<br>ISO 9186-1:2014<br>CEAPAT, 2023<br>Berrios, 2020 |
| Check that pictograms have high contrast for students with low vision (pictograms for cortical visual impairment: red, yellow, black, white).                                                                                          | Do pictograms have the appropriate colours for the visual characteristics of the learners?                                                                                                                                | CEAPAT-IMSERSO, 2018<br>Fundación ONCE, 2013<br>ISO 9186-1:2014<br>CEAPAT, 2023<br>Berrios, 2020 |
| Ensure that pictograms have a white background. In the case of users of Augmentative and Alternative Communication, pictograms may have a coloured background or border according to the grammatical category of the word represented. | Do pictograms have a white background? In the case of users of Augmentative and Alternative Communication, do pictograms have coloured backgrounds or borders, according to the grammatical category of the word at hand? | CEAPAT-IMSERSO, 2018<br>Fundación ONCE, 2013<br>ISO 9186-1:2014<br>CEAPAT, 2023<br>Berrios, 2020 |
| Pictograms should facilitate participation and should be used to ask or answer questions.                                                                                                                                              | Do students have access to pictograms to facilitate participation? Are pictograms used to allow for the asking or answering of questions?                                                                                 | CEAPAT-IMSERSO, 2018<br>Fundación ONCE, 2013<br>ISO 9186-1:2014<br>CEAPAT, 2023<br>Berrios, 2020 |
| Ensure that the pictographic system used is already known by the learner.                                                                                                                                                              | Does the learner know the pictographic system used?                                                                                                                                                                       | CEAPAT-IMSERSO, 2018<br>Fundación ONCE, 2013<br>ISO 9186-1:2014<br>CEAPAT, 2023<br>Berrios, 2020 |

| RECOMMENDATION                                                                                                                                                                                                                                                                                                     | QUESTION                                                                                                                                                                                                     | SOURCE                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Make sure that pictograms are appropriate to the student's culture.                                                                                                                                                                                                                                                | Are pictograms culturally appropriate?                                                                                                                                                                       | CEAPAT-IMSERSO, 2018<br>Fundación ONCE, 2013<br>ISO 9186-1:2014<br>CEAPAT, 2023<br>Berrios, 2020 |
| Verify that pictograms are appropriate for the student's age                                                                                                                                                                                                                                                       | Are pictograms appropriate for the student's age?                                                                                                                                                            | CEAPAT-IMSERSO, 2018<br>Fundación ONCE, 2013<br>ISO 9186-1:2014<br>CEAPAT, 2023<br>Berrios, 2020 |
| Pictograms should represent concepts easily understood by the student.                                                                                                                                                                                                                                             | Do pictograms represent concepts that the student understands or can understand?                                                                                                                             | CEAPAT-IMSERSO, 2018<br>Fundación ONCE, 2013<br>ISO 9186-1:2014<br>CEAPAT, 2023<br>Berrios, 2020 |
| Ensure that pictograms help to promote personal autonomy when performing tasks related to the educational resource (e.g., pictograms to give instructions, to navigate in the educational resource, to distribute information in space, to indicate the type of statement or to structure the type of activities). | Do pictograms promote personal autonomy to perform tasks related to the educational resource?                                                                                                                | CEAPAT-IMSERSO, 2018<br>Fundación ONCE, 2013<br>ISO 9186-1:2014<br>CEAPAT, 2023<br>Berrios, 2020 |
| Pictograms are often used with non-literate learners. Use pictograms in relation to the literacy level of the learners.                                                                                                                                                                                            | Are pictograms non-literate learner friendly?<br><br>Are different pictograms used to allow for learners' various literacy levels or for students who do not know the language of the material?              | CEAPAT-IMSERSO, 2018<br>Fundación ONCE, 2013<br>ISO 9186-1:2014<br>CEAPAT, 2023<br>Berrios, 2020 |
| Pictograms should always have a caption or transcription above or below the picture written in the student's native language. In the case of bilingual contexts, one language can be placed above and the other below.                                                                                             | Do pictograms have a text with a transcription of what they are representing in the student's native language? In bilingual contexts, is one language placed above and the other language below the picture? | CEAPAT-IMSERSO, 2018<br>Fundación ONCE, 2013<br>ISO 9186-1:2014<br>CEAPAT, 2023<br>Berrios, 2020 |



| RECOMMENDATION                                                                                                                                                                                                                                                                                                                                                 | QUESTION                                                                                                                                                                                                           | SOURCE                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Do not translate the text into pictograms word for word, as the resource can be cognitively overloaded and inaccessible. Use pictograms for the most relevant words in the text or use other visual aids (visual scenes).                                                                                                                                      | Is the word for word translation of the text into pictograms avoided? Are pictograms or other visual aids used for the most relevant words in the text?                                                            |                                                                                                  |
| Many pictograms are abstract, and their meaning cannot be interpreted unless they are taught beforehand. Make sure that these pictograms are explained beforehand.                                                                                                                                                                                             | Have abstract pictograms been explained for the student beforehand?                                                                                                                                                | CEAPAT-IMSERSO, 2018<br>Fundación ONCE, 2013<br>ISO 9186-1:2014<br>CEAPAT, 2023<br>Berrios, 2020 |
| Some pictograms are very simple, while others have a high number of details. Make sure that you use the most appropriate ones for the learner.                                                                                                                                                                                                                 | Is there a balance between what you want to represent and the amount of detail in the pictograms?                                                                                                                  | CEAPAT-IMSERSO, 2018<br>Fundación ONCE, 2013<br>ISO 9186-1:2014<br>CEAPAT, 2023<br>Berrios, 2020 |
| Open access pictograms can be either pictograms for augmentative and alternative communication or pictograms for graphic illustration. Try not to mix pictograms of one type with those of another. Additionally, avoid mixing pictograms from different sources, since pictographic systems form a unified set in terms of shape, colours, or interpretation. | Do pictograms come from a single source?<br><br>If they come from multiple sources, are they compatible so that there is no confusion in the design?                                                               | CEAPAT-IMSERSO, 2018<br>Fundación ONCE, 2013<br>ISO 9186-1:2014<br>CEAPAT, 2023<br>Berrios, 2020 |
| Use the same pictograms throughout the resource, in other words, do not change the type of pictogram used for the same concept throughout the resource.                                                                                                                                                                                                        | Is the same pictogram used consistently to represent the same concept?                                                                                                                                             | CEAPAT-IMSERSO, 2018<br>Fundación ONCE, 2013<br>ISO 9186-1:2014<br>CEAPAT, 2023<br>Berrios, 2020 |
| Consider using other types of graphic visual support for texts, such as visual scenes and scientific illustrations, provided it is not necessary to illustrate concept by concept. Illustrations are useful for complex scientific contents. Visual                                                                                                            | Does the OER use different types of visual aids depending on whether it refers to an isolated concept or to other types of communication units, such as complex or specialised concepts and complex communication? |                                                                                                  |

| RECOMMENDATION                                                                                                                                                                                                                                                                                             | QUESTION | SOURCE |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| <p>scenes provide a broad context for narrative or historical events, while pictograms are used in augmentative communication to isolate concepts.</p> <p>For these visual aids, physical and sensory accessibility must also be ensured. Open access photo banks are another useful resource for OER.</p> |          |        |





## 5.4 Technological accessibility

**Technological accessibility:** Technological accessibility relates to the availability of digital tools and technologies that can assist learners in accessing and interacting with IA-OERs. They include two subdimensions: Multimedia guidelines (image, video, audio and video games), and Guidelines based on WCAG (perceivable, operable, readable, robust and conformance)

### Multimedia

| ASPECT | RECOMMENDATION                                                                                                                          | QUESTION                                                                                                                   | SOURCE                                                                                               |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Image  | Use quality images with adequate size, resolution (number of pixels), brightness, exposure, etc in the OER.                             | Does the OER use quality images with adequate size, resolution (number of pixels), brightness, exposure, etc?              | Politis et al., 2014<br>UNESCO, 2019<br>Monsalve et al., 2018                                        |
| Image  | Provide an alternative text with sense to describe each image.                                                                          | Does the OER provide an alternative text with sense to describe each image?                                                | Achieve Inc, 2011<br>Politis et al., 2014.<br>UNESCO, 2019<br>Monsalve et al., 2018<br>WCAG2.2 - 1.1 |
| Image  | Use images that are easy to understand or are related to the text to help learners understand it or help emphasise contents in the OER. | Are the images in the OER easy to understand and related to the text to ensure understanding and/or to emphasise contents? | Politis et al., 2014<br>UNESCO, 2019<br>Monsalve et al., 2018                                        |
| Image  | Ensure that there is sufficient contrast between the background and foreground of the images in the OER.                                | Do images in the OER have sufficient contrast between the background and foreground?                                       | UNE-71362                                                                                            |
| Image  | Do not include text in the images of the OER.                                                                                           | Is the use of text in the images of the OER avoided?                                                                       | Information for all.<br>2010<br>UNESCO, 2019<br>Monsalve et al., 2018<br>WCAG2.2 - 1.4               |
| Image  | Do not include tables with text as images in the OER.                                                                                   | Is the use of tables with texts as images avoided in the OER?                                                              | Politis et al., 2014<br>UNESCO, 2019<br>Monsalve et al., 2018                                        |
| Video  | Avoid slow and fast motion.                                                                                                             | Have you avoided slow and fast motion?                                                                                     | Politis et al., 2014                                                                                 |

| ASPECT       | RECOMMENDATION                                                                                                                    | QUESTION                                                                                                                                   | SOURCE                                                                                     |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Video</b> | Ensure that the video and sound are clear and of very high quality.                                                               | Are the video and sound clear and of very high quality?                                                                                    | Politis et al., 2014                                                                       |
| <b>Video</b> | Ensure that there is sufficient contrast between the background and foreground in the video.                                      | Is there sufficient contrast between the background and foreground in the video?                                                           | UNE-71362                                                                                  |
| <b>Video</b> | Make sure sound and full-screen controls are easy to find and use in the OER.                                                     | Are sounds and full-screen controls easy to find and use in the OER?                                                                       | Politis et al., 2014                                                                       |
| <b>Video</b> | Create short informative videos to ensure that they are no longer than 20-30 minutes.                                             | Have you created short informative videos to ensure that are no longer than 20-30 minutes?                                                 | Politis et al., 2014                                                                       |
| <b>Video</b> | Provide synchronised alternatives for the video, such as subtitling, audio description, full transcription or sign language (SL). | Have you provided synchronised alternatives for the video such as subtitling, audio description, full transcription or sign language (SL)? | UNE-71362<br>Politis et al., 2014<br>UNESCO, 2019<br>Monsalve et al., 2018<br>WCAG2.2 -1.1 |
| <b>Video</b> | If the video includes sounds, follow the specific recommendations for audio.                                                      | Have you followed the specific recommendations for the audio in the video?                                                                 | Politis et al., 2014                                                                       |
| <b>Video</b> | Follow specific recommendations to add subtitles or captions.                                                                     | Have you followed specific recommendations to add subtitles or captions?                                                                   | Politis et al., 2014                                                                       |
| <b>Video</b> | Follow specific recommendations to add audio description.                                                                         | Have you followed specific recommendations to add audio description?                                                                       | Politis et al., 2014                                                                       |
| <b>Video</b> | Follow specific recommendations to add sign language.                                                                             | Have you followed specific recommendations to add sign language?                                                                           | Politis et al., 2014                                                                       |



| ASPECT       | RECOMMENDATION                                                                                        | QUESTION                                                                                     | SOURCE                                                                |
|--------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>Video</b> | Provide textual description and transcription of the video with sense.                                | Have you provided textual description and transcription of the video, with sense?            | Politis et al., 2014<br>Quality Matters Program, 2011<br>WCAG2.2 -1.1 |
| <b>Video</b> | Ensure that students have control over the management of the video reproduction and its alternatives. | Do students have control over the management of the video reproduction and its alternatives? | UNE-71362                                                             |
| <b>Audio</b> | Make sure the quality of the sound, its pitch, duration, intensity and timbre is appropriate.         | Are the quality of the sound, its pitch, duration, intensity and timbre appropriate?         | Politis et al., 2014                                                  |
| <b>Audio</b> | Make sure the person speaking has a clear pronunciation and vocalises clearly.                        | Does the person speaking have good pronunciation and vocalise clearly?                       | Politis et al., 2014                                                  |
| <b>Audio</b> | Make sure the people who speak in the video do it slowly and not too softly neither too loudly.       | Does the person who speaks in the video do it slowly and neither too softly nor loudly?      | Politis et al., 2014                                                  |
| <b>Audio</b> | Make sure the sound is at the right volume.                                                           | Is the sound at the right volume?                                                            | Politis et al., 2014                                                  |
| <b>Audio</b> | Use pauses at times when they are necessary.                                                          | Does the audio have pauses at times when they are necessary?                                 | Politis et al., 2014                                                  |
| <b>Audio</b> | Give learners enough time to understand the information.                                              | Do learners have enough time to understand the information?                                  | Politis et al., 2014                                                  |
| <b>Audio</b> | Do not hesitate to repeat important information several times.                                        | Does the audio repeat important information several times?                                   | Politis et al., 2014                                                  |
| <b>Audio</b> | The people who speak have to express emotions clearly.                                                | Does the person who speaks express emotions clearly?                                         | Politis et al., 2014                                                  |
| <b>Audio</b> | Provide textual transcription of the audio.                                                           | Is textual transcription of the audio provided?                                              | Politis et al., 2014<br>Quality Matters Program, 2011                 |



| ASPECT            | RECOMMENDATION                                                                                                            | QUESTION                                                                                                | SOURCE                                                                                                              |
|-------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|                   |                                                                                                                           |                                                                                                         | WCAG2.2 -1.1                                                                                                        |
| <b>Audio</b>      | Avoid background noise or lower it.                                                                                       | Is background noise avoided or lowered?                                                                 | Politis et al., 2014<br>UNESCO, 2019<br>Monsalve et al., 2018<br>WCAG2.2 -1.4                                       |
| <b>Audio</b>      | Provide mechanisms in the OER to pause or stop the audios that are easily found and used.                                 | Does the OER provide mechanisms to pause or stop the audio? Are these mechanisms easily found and used? | WCAG2.2 - 1.4                                                                                                       |
| <b>Audio</b>      | Provide separate volume controls or mutes for effects, speech and background/music.                                       | Does the OER provide separate volume controls or mutes for effects, speech and background/music?        | Neto et al., 2019                                                                                                   |
| <b>Audio</b>      | Ensure that students have control over the management of the audio reproduction and its textual alternative.              | Do the students have control over the management of the audio reproduction and its textual alternative? | Politis et al., 2014                                                                                                |
| <b>Audio</b>      | Provide contents in DAISY format for digital talking books.                                                               | Does the OER provide contents in DAISY format for digital talking books?                                | <a href="https://www.w3.org/TR/SMIL/smil-daisy-profile.html">https://www.w3.org/TR/SMIL/smil-daisy-profile.html</a> |
| <b>Videogames</b> | Use distinct sound / music design for all interactive objects and events.                                                 | Does the OER use distinct sound / music design for all interactive objects and events?                  | Neto et al., 2019                                                                                                   |
| <b>Videogames</b> | Use stereo, binaural or surround sound for game objects localisation.                                                     | Does the OER use stereo, binaural or surround sound for game objects localisation?                      | Neto et al., 2019                                                                                                   |
| <b>Videogames</b> | Ensure game controls are consistent, intuitive and naturally mapped (controlled through the most conventional game keys). | Are the game controls consistent, intuitive and naturally mapped?                                       | Neto et al., 2019                                                                                                   |



| ASPECT            | RECOMMENDATION                                                                                                                                                 | QUESTION                                                                                                                                                   | SOURCE            |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Videogames</b> | Provide a tutorial to introduce all general controls necessary to make progress in the game through a series of small tasks.                                   | Does the game provide a tutorial to introduce all general controls necessary to make progress in the game, through a series of small tasks?                | Neto et al., 2019 |
| <b>Videogames</b> | Ensure the game is easy to learn but harder to master, regardless of whether the user is visually impaired or not.                                             | Is the game easy to learn but harder to master, regardless of whether the user is visually impaired or not?                                                | Neto et al., 2019 |
| <b>Videogames</b> | Ensure the game world reacts to the players and remembers their passage through it, being perceived by the player, including visually impaired ones.           | Does the game world react to the player and remember their passage through it, being perceived by the player, including visually impaired ones?            | Neto et al., 2019 |
| <b>Videogames</b> | Ensure the game goals are clear for players, including those players who are visually impaired.                                                                | Are the game goals clear for players, including those players who are visually impaired?                                                                   | Neto et al., 2019 |
| <b>Videogames</b> | Ensure the players (including visually impaired people) have a sense of control and influence on the game world, since all actions provide immediate feedback. | Do the players (including visually impaired people) have a sense of control and influence on the game world, since all actions provide immediate feedback? | Neto et al., 2019 |
| <b>Videogames</b> | Ensure the game provides appropriate (audiovisual) feedback to users and reacts in a consistent, immediate, challenging and exciting way to players' actions.  | Does the game provide appropriate (audiovisual) feedback to users and reacts in a consistent, immediate, challenging and exciting way to players' actions? | Neto et al., 2019 |
| <b>Videogames</b> | Ensure the game does not put an unnecessary burden                                                                                                             | Does the game put an unnecessary burden                                                                                                                    | Neto et al., 2019 |



| ASPECT            | RECOMMENDATION                                                                                                                                                                                                                                         | QUESTION                                                                                                                                                                                                                                          | SOURCE            |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
|                   | on the player (regardless of whether he/she is visually impaired or not).                                                                                                                                                                              | sary burden on the player (regardless of whether he/she is visually impaired or not)?                                                                                                                                                             |                   |
| <b>Videogames</b> | Ensure the game story encourages immersion when the game has a story component.                                                                                                                                                                        | Does the game story encourage immersion when the game has a story component?                                                                                                                                                                      | Neto et al., 2019 |
| <b>Videogames</b> | Ensure the player, regardless of whether she/he is visually impaired or not, feels that her/his actions have meaning and directly affect the game world.                                                                                               | Does the player, regardless of whether she/he is visually impaired or not, feel that her/his actions have meaning and directly affect the game world?                                                                                             | Neto et al., 2019 |
| <b>Videogames</b> | Ensure that the player, regardless of whether she/he is visually impaired or not, builds a sense of identity throughout the game, knows clearly her/his goals and skills and develops a desire to learn new skills to achieve their goals in the game. | Does the player, regardless of whether she/he is visually impaired or not, build a sense of identity throughout the game, and know clearly her/his goals and skills, and develop a desire to learn new skills to achieve their goals in the game? | Neto et al., 2019 |
| <b>Videogames</b> | Ensure that the game challenges are ordered in a way that the players apply the knowledge obtained in solving the previous problems to solve the next ones.                                                                                            | Are the game challenges ordered in a way that the players apply the knowledge obtained in solving the previous problems to solve the next ones?                                                                                                   | Neto et al., 2019 |
| <b>Videogames</b> | Provide information “On Demand” and “Just in Time.”                                                                                                                                                                                                    | Does the game provide information “On Demand” and “Just in Time”?                                                                                                                                                                                 | Neto et al., 2019 |

## Guidelines based on WCAG2.2



Co-funded by  
the European Union

Funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or EACEA. Neither the European Union nor the granting authority can be held responsible for them.

| ASPECT             | RECOMMENDATION                                                                                                                                                                                                                                     | QUESTION                                                                                                                                                                                                                                     | SOURCE                                  |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| <b>Perceivable</b> | Ensure that the information is not provided exclusively through sensory information such as colour, depth, shape, size, visual location, orientation or sound.                                                                                     | Is the information not provided exclusively through sensory information such as colour, depth, shape, size, visual location, orientation or sound?                                                                                           | UNE-71362<br>Hasley et al., 2021        |
| <b>Perceivable</b> | Provide resources in alternative formats (e.g., .doc, .odt, .pdf).                                                                                                                                                                                 | Does the OER provide resources in alternative formats (e.g., .doc, .odt, .pdf)?                                                                                                                                                              | OER Africa, 2015                        |
| <b>Perceivable</b> | The OER should allow to be adapted for use in some non-online context where required (i.e., printed, CD, USB).                                                                                                                                     | Can the OER be adapted for use in some non-online context where required (i.e., printed, CD, USB)?                                                                                                                                           | UNESCO, 2019<br>Ehlers, 2012            |
| <b>Perceivable</b> | Provide means for students to adjust the perceptive characteristics of the information (type, colour, shape, size, etc.) to their needs/personal preferences and ensure that this configuration is maintained when the student uses the OER again. | Does the OER provide means for students to adjust perceptive characteristics of the information (type, colour, shape, size, etc.) to their needs/personal preferences? Are these preferences maintained when the student uses the OER again? | UNE-71362<br>DUA- Monsalve et al., 2018 |
| <b>Perceivable</b> | Facilitate that information, structure, and relationships conveyed through presentation are programmatically determined or are available in text.                                                                                                  | Are the information, structure, and relationships conveyed through presentation programmatically determined or available in text?                                                                                                            | WCAG2.2 -1.3                            |
| <b>Perceivable</b> | Ensure that a correct reading sequence is programmatically determined when the content sequence affects its meaning.                                                                                                                               | Does the content sequence affect its meaning? Is the correct reading sequence programmatically determined?                                                                                                                                   | WCAG2.2 -1.3                            |



| ASPECT             | RECOMMENDATION                                                                                                                                                                    | QUESTION                                                                                                                                                              | SOURCE                                                                         |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Perceivable</b> | Ensure that content does not restrict its view and operation to a single display orientation (such as portrait or landscape), unless a specific display orientation is essential. | Does the content restrict its view and operation to a single display orientation (such as portrait or landscape), unless a specific display orientation is essential? | WCAG2.2 -1.3<br>Monsalve et al., 2018                                          |
| <b>Perceivable</b> | Ensure that there is coherence between icons and text with the same purpose.                                                                                                      | Is there coherence between icons and text with the same purpose?                                                                                                      | Politis et al., 2014.<br>UNESCO, 2019<br>Monsalve et al., 2018<br>WCAG2.2 -1.3 |
| <b>Perceivable</b> | Ensure the application interface and screen layout are efficient, consistent and visually pleasing.                                                                               | Are the application interface and screen layout efficient, consistent and visually pleasing?                                                                          | Neto et al., 2019                                                              |
| <b>Perceivable</b> | Use alternative means (i.e., pattern) rather than colour to differentiate content and functionality, where needed.                                                                | Does the OER use alternative means (i.e., pattern) rather than colour to differentiate content and functionality where needed?                                        | UNESCO, 2019<br>WCAG2.2 -1.4                                                   |
| <b>Perceivable</b> | Ensure that links are distinguished by means different from colour (i.e., box, text label, underline).                                                                            | Are links distinguished by means different from colour (i.e., box, text label, underline)?                                                                            | UNESCO, 2019                                                                   |
| <b>Perceivable</b> | Provide high contrast between text/UI and background, including buttons and dialogue boxes                                                                                        | Does the OER provide high contrast between text/UI and background, including buttons and dialogue boxes?                                                              | Neto et al., 2019                                                              |
| <b>Perceivable</b> | Ensure sufficient contrast between text, images, videos and background.                                                                                                           | Is there sufficient contrast between text, images, videos and background?                                                                                             | UNESCO, 2019                                                                   |
| <b>Perceivable</b> | Ensure that texts can be resized without assistive technology up to 200 percent without loss of content or                                                                        | Except for captions and images of text, can the text be resized without assistive technology up to 200 percent                                                        | WCAG2.2 -1.4                                                                   |





| ASPECT             | RECOMMENDATION                                                                                                                                                                                                    | QUESTION                                                                                                                                                                                                                                  | SOURCE                                |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
|                    | functionality with the exception of captions and images of text.                                                                                                                                                  | without loss of content or functionality?                                                                                                                                                                                                 |                                       |
| <b>Perceivable</b> | When tables are provided, use them correctly. They have to be well structured, described and simple, avoiding blank, combined, divided or nested cells as far as possible. They should also include a header row. | If there are tables, are they used correctly: are they well-structured and described as far as possible? Are the tables simple, avoiding blank, combined, divided or nested cells as far as possible? Do the tables include a header row? | UNE-71362                             |
| <b>Perceivable</b> | If there are lists, use them correctly and ensure they are well labelled.                                                                                                                                         | If there are lists, are they used correctly and well labelled?                                                                                                                                                                            | UNE-71362                             |
| <b>Perceivable</b> | Ensure that the content is presented without loss of information or functionality, and without requiring scrolling.                                                                                               | Is the content presented without loss of information, functionality, and without requiring scrolling?                                                                                                                                     | WCAG2.2 -1.4                          |
| <b>Perceivable</b> | Provide means to adjust spaces between letters, words and lines.                                                                                                                                                  | Are there means to adjust spaces between letters, words and lines?                                                                                                                                                                        | UNESCO, 2019<br>Monsalve et al., 2018 |
| <b>Perceivable</b> | Use simple clear text formatting and an easily readable default font size.                                                                                                                                        | Is the text formatting simple? Does the text have an easily readable default font size?                                                                                                                                                   | Neto et al., 2019                     |
| <b>Operable</b>    | Ensure any fatigue or boredom is minimised by varying activities and pacing during the learning.                                                                                                                  | Is the fatigue or boredom minimised by varying activities and pacing during the learning?                                                                                                                                                 | Neto et al., 2019                     |



| ASPECT          | RECOMMENDATION                                                                                                                                                                                                                                                                                                                                                                                                                                 | QUESTION                                                                                                                                                                                                                                                                                                                                                                                                                         | SOURCE                                                     |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <b>Operable</b> | The functionality of every digital educational material should be easy to handle through standard input devices (keyboard, mouse) and also through non-standard ones using additional assistive technology (buttons, emulators, etc). Users should be allowed to press the keys, the mouse or other input devices at the speed that suits their needs. Keyboard shortcuts or hotkeys are provided for major links and important form controls. | Is the functionality of every digital educational material easy to handle through standard input devices (keyboard, mouse) and also through non-standard ones using additional assistive technology (buttons, emulators, etc)?Are users allowed to press the keys, the mouse or other input devices at the speed that suits their needs?Are there any keyboard shortcuts or hotkeys for major links and important form controls? | Neto et al., 2019<br>Monsalve et al., 2018<br>WCAG2.2 -2.1 |
| <b>Operable</b> | Provide users enough time to read and use content. Allow for time adjustments.                                                                                                                                                                                                                                                                                                                                                                 | Do users have enough time to read and use content?Can users adjust time?                                                                                                                                                                                                                                                                                                                                                         | WCAG2.2 - 2.2<br>UNE-71362<br>Monsalve et al., 2018        |
| <b>Operable</b> | Ensure that for any moving, blinking, auto-updating or scrolling information 1) that starts automatically, 2) it lasts more than five seconds, or 3) it is presented in parallel to other content, there is a mechanism for the user to pause, stop, or hide it, unless it is part of an activity where it is essential.                                                                                                                       | Does any moving, blinking, auto-updating or scrolling information that (1) starts automatically, (2) lasts more than five seconds, or (3) is presented in parallel with other content, have a mechanism for the user to pause, stop, or hide it unless it is part of an activity where it is essential?                                                                                                                          | WCAG2.2 2.2                                                |
| <b>Operable</b> | Provide controls to pause, stop or move in multimedia contents                                                                                                                                                                                                                                                                                                                                                                                 | Are there controls to pause, stop or move in multimedia contents?                                                                                                                                                                                                                                                                                                                                                                | UNE-71362<br>Monsalve et al., 2018                         |



| ASPECT          | RECOMMENDATION                                                                                                                                                                                                                    | QUESTION                                                                                                                                                                                                                                | SOURCE                    |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <b>Operable</b> | Ensure learning interruption is supported, so that students can easily turn the application on and off and be able to save the application in different states.                                                                   | Can learning interruption be supported, so that students can easily turn the application on and off and be able to save the application in different states?                                                                            | Neto et al., 2019         |
| <b>Operable</b> | Ensure that in case of interruptions or re-authentication, the system continues in the same place avoiding asking the user to introduce data again.                                                                               | In case of interruptions or re-authentication, can the system continue in the same place avoiding asking the user to introduce data again?                                                                                              | WCAG2.2 -2.2              |
| <b>Operable</b> | Do not design content in a way that is known to cause seizures or physical reactions. Avoid flashes.                                                                                                                              | Is the content designed in a way that is known to cause seizures or physical reactions, avoiding flashes?                                                                                                                               | WCAG2.2 -2.3              |
| <b>Operable</b> | Ensure the navigation is consistent, logical and minimal for students (including those with visual impairment).                                                                                                                   | Is the navigation consistent, logical and minimal for students (including the visually impaired ones)?                                                                                                                                  | Neto et al., 2019         |
| <b>Operable</b> | Provide visible focus from keyboard, mouse or alternative input device, ensuring that the area of the focus indicator is as least as large as the area of the component focused and that it has a contrast ratio of at least 3:1. | Is there a visible focus from keyboard, mouse or alternative input device that ensures that the area of the focus indicator is as least as large as the area of the component focused and that it has a contrast ratio of at least 3:1? | UNE-71362<br>WCAG2.2 -2.4 |
| <b>Operable</b> | Ensure that each section or screen has a title or heading.                                                                                                                                                                        | Does each section or screen have a title or heading?                                                                                                                                                                                    | UNE-71362<br>WCAG2.2 -2.4 |
| <b>Operable</b> | Ensure that the tab or focus order makes sense to preserve                                                                                                                                                                        | Does the tab or focus order                                                                                                                                                                                                             | WCAG2.2 -2.4              |



| ASPECT          | RECOMMENDATION                                                                                                                                                                                                                                            | QUESTION                                                                                                                                                                                                                                     | SOURCE                                           |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
|                 | meaning and operability.                                                                                                                                                                                                                                  | make sense to preserve meaning and operability?                                                                                                                                                                                              |                                                  |
| <b>Operable</b> | Provide at least two mechanisms to locate information: e.g., search engine, maps, or index of contents.                                                                                                                                                   | Are at least two mechanisms to locate information: e.g., search engine, maps, or index of contents provided?                                                                                                                                 | UNE-71362                                        |
| <b>Operable</b> | Ensure that form fields have appropriate text labels.                                                                                                                                                                                                     | Do form fields have appropriate text labels?                                                                                                                                                                                                 | UNESCO, 2019<br>WCAG2.2 -2.4                     |
| <b>Operable</b> | Add labels and extra instructions when content requires user input (e.g., format of a date) when need it.                                                                                                                                                 | Have labels and extra instructions been added if they are necessary, when content requires user input (e.g., format of a date)?                                                                                                              | UNE-71362                                        |
| <b>Operable</b> | Ensure that the layout is appropriate for the content presentation.                                                                                                                                                                                       | Is the layout appropriate for the content presentation?                                                                                                                                                                                      | Mhlanga, 2010                                    |
| <b>Operable</b> | Ensure that headings and labels describe topic or purpose.                                                                                                                                                                                                | Do headings and labels describe topic or purpose?                                                                                                                                                                                            | WCAG2.2 -2.4                                     |
| <b>Operable</b> | Ensure that users know where they are, where they have been and where to go next.                                                                                                                                                                         | Do users know where they are, where they have been and where to go next?                                                                                                                                                                     | Baya'a et al., 2009<br>UNE-71362<br>Ehlers, 2012 |
| <b>Operable</b> | If the image is a link, identify the link destination or purpose clearly.                                                                                                                                                                                 | If the image is a link, is the link destination or purpose clearly identified?                                                                                                                                                               | WCAG2.2 -2.4                                     |
| <b>Operable</b> | Ensure that functionalities that can be operated by device motion or user motion can also be operated by user interface components and that responding to motion can be disabled to prevent accidental actuation, except when motion is essential for the | Can functionalities that can be operated by device motion or user motion also be operated by user interface components?<br><br>Can responding to motion be disabled to prevent accidental actuation, except when motion is essential for the | WCAG2.2 -2.5<br>EDIA-CEDEC, 2020<br>UNE-71362    |

| ASPECT          | RECOMMENDATION                                                                                                                                                | QUESTION                                                                                                                                                     | SOURCE                                     |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
|                 | function.                                                                                                                                                     | tion is essential for the function?                                                                                                                          |                                            |
| <b>Operable</b> | Ensure that all functionality that uses a dragging movement for operation can be achieved by a single pointer without dragging, unless dragging is essential. | Can all functionality that uses a dragging movement for operation be achieved by a single pointer without dragging, unless dragging is essential?            | WCAG2.2 -2.5                               |
| <b>Operable</b> | Ensure interactive elements / virtual controls are large and well-spaced, particularly on small or touch screens.                                             | Are interactive elements / virtual controls large and well-spaced, particularly on small or touch screens?                                                   | Neto et al., 2019<br>Monsalve et al., 2018 |
| <b>Operable</b> | Ensure that buttons and elements in the screen can change their size to facilitate the interaction with the user.                                             | Can buttons and elements in the screen change their size to facilitate the interaction with the user?                                                        | Monsalve et al., 2018                      |
| <b>Operable</b> | Ensure that forms are easy to fill out and provide tools to help fill them out and to prevent and/or correct errors that may be made when filling them out.   | Are forms easy to fill out?<br><br>Do forms provide tools to help fill them out and to prevent and/or correct errors that may be made when filling them out? | JISC Tiger Project, 2011,<br>UNE-71362     |
| <b>Readable</b> | Design the OER to facilitate readability and minimise distractions.                                                                                           | Is the OER designed to facilitate readability and minimise distractions?                                                                                     | Quality Matters Program, 2011              |
| <b>Readable</b> | Adapt the text complexity to the users 'reading capacity.                                                                                                     | Can the text complexity be adapted to the users 'reading capacity?                                                                                           | Monsalve et al., 2018 - W3C                |



| ASPECT          | RECOMMENDATION                                                                                                                                                                                                                                                                                 | QUESTION                                                                                                                                                                                                                                                                                                      | SOURCE                                                               |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>Readable</b> | Ensure that the human language of each passage or phrase in the content can be programmatically determined except for proper names, technical terms and words of indeterminate language.                                                                                                       | Can the human language of each passage or phrase in the content be programmatically determined (except for proper names, technical terms, words of indeterminate language)?                                                                                                                                   | Monsalve et al., 2018 - W3C                                          |
| <b>Readable</b> | Provide a mechanism for identifying specific definitions of words or phrases used in an unusual or restricted way, including idioms and jargon.                                                                                                                                                | Is there a mechanism for identifying specific definitions of words or phrases used in an unusual or restricted way, including idioms and jargon?                                                                                                                                                              | WCAG2.2 -3.1                                                         |
| <b>Readable</b> | Avoid abbreviations, but if they must be included, provide a mechanism for identifying their expanded form or meaning.                                                                                                                                                                         | Are abbreviations avoided in the OER? If abbreviations must be included, is there a mechanism that helps identify their expanded form or meaning?                                                                                                                                                             | Monsalve et al., 2018 - W3C<br>UNESCO, 2019<br>Monsalve et al., 2018 |
| <b>Readable</b> | When texts require a reading ability that is more advanced than the lower secondary education level after removal of proper names and titles, provide supplemental content, or a version that does not require reading ability that is more advanced than the lower secondary education level. | When texts require a reading ability that is more advanced than the lower secondary education level after removal of proper names and titles, does the OER provide supplemental content, or a version that does not require a reading ability that is more advanced than the lower secondary education level? | WCAG2.2 2.2                                                          |
| <b>Readable</b> | Ensure that changes of context are not initiated if a component receives the focus, only by user request.                                                                                                                                                                                      | Are changes of context not initiated if a component receives the focus, only by user request?                                                                                                                                                                                                                 | WCAG2.2 -3.2                                                         |

| ASPECT          | RECOMMENDATION                                                                                                                                                                                             | QUESTION                                                                                                                                                                                       | SOURCE                                           |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| <b>Readable</b> | Ensure that navigational or help mechanisms that are repeated on the OER occur in the same relative order each time they are repeated, unless a change is initiated by the user.                           | Do the navigational or help mechanisms that are repeated in the OER occur in the same relative order each time they are repeated, unless a change is initiated by the user?                    | WCAG2.2 -3.2<br>Hasley et al., 2021              |
| <b>Readable</b> | Ensure that consistency in the identification and appearance (form, size, colour, location, etc.) of the elements that have the same functionality (links, icons, buttons, etc.) is maintained throughout. | Is consistency in the identification and appearance (form, size, colour, location, etc.) of the elements that have the same functionality (links, icons, buttons, etc.) maintained throughout? | UNE-71362<br>Hasley et al., 2021<br>WCAG2.2 -3.4 |
| <b>Readable</b> | Provide more assistance to avoid and correct mistakes.                                                                                                                                                     | Does the OER provide assistance to avoid and correct mistakes?                                                                                                                                 | Hasley et al., 2021                              |
| <b>Readable</b> | If an input error is automatically detected, describe the error to the user, and if suggestions for correction are known, also provide these suggestions to the user.                                      | If an input error is automatically detected, does the OER describe the error to the user? If suggestions for correction are known, does the OER provide these suggestions to the user?         | WCAG2.2 -3.3                                     |
| <b>Readable</b> | Provide labels or instructions when content requires user input.                                                                                                                                           | Does the OER provide labels or instructions when content requires user input?                                                                                                                  | WCAG2.2 2.2                                      |
| <b>Readable</b> | Provide a mechanism for reviewing, confirming, and correcting information before the user finalises the submission of data.                                                                                | Is there a mechanism in place for users to review, confirm, and correct information before the user finalises the submission of data?                                                          | UNE-71362                                        |
| <b>Readable</b> | Provide context-sensitive help.                                                                                                                                                                            | Does the OER provide context-sensitive help?                                                                                                                                                   | WCAG2.2 -3.3                                     |



| ASPECT          | RECOMMENDATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | QUESTION                                                                                                                                                                                                                                                                                                                                                                                                                                     | SOURCE                            |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>Readable</b> | <p>Ensure that a cognitive function test (such as remembering a password or solving a puzzle) is not required in an authentication process, facilitating at least one of the following alternatives:</p> <p>(1) Another authentication method that does not rely on a cognitive function test, (2) Assistance to complete the cognitive function, (3) the cognitive function test is to recognise objects, or (4) the cognitive function test is to identify non-text content.</p> | <p>If there is a cognitive function test (such as remembering a password or solving a puzzle), does the OER facilitate one of the following alternatives?</p> <p>(1) Another authentication method that does not rely on a cognitive function test, (2) assistance to complete the cognitive function test, (3) the cognitive function test is to recognise objects, or (4) the cognitive function test is to identify non-text content.</p> | WCAG2.2 -3.3                      |
| <b>Robust</b>   | <p>Ensure the OER is available in different standard formats such as XML, Daisy, ePub, Tagged PDF, Audio/Video Cassette, DVD/DVD-ROM, or Blu-ray Disk Production Standards.</p>                                                                                                                                                                                                                                                                                                    | <p>Can the OER be read by a range of assistive technology?</p> <p>Is the OER available in different standard formats such as XML, Daisy, ePub, Tagged PDF, Audio/Video Cassette, DVD/DVD-ROM, or Blu-ray Disk Production Standards?</p>                                                                                                                                                                                                      | Achieve Inc, 2011<br>UNESCO, 2019 |
| <b>Robust</b>   | <p>Ensure that the OER complies with NCAM Guidelines for accessible Movies, Web and Multimedia.</p>                                                                                                                                                                                                                                                                                                                                                                                | <p>Does the OER comply with NCAM Guidelines for accessible Movies, Web and Multimedia?</p>                                                                                                                                                                                                                                                                                                                                                   | Achieve Inc, 2011                 |
| <b>Robust</b>   | <p>Ensure that the OER is available in an Open Learning Management System (LMS).</p>                                                                                                                                                                                                                                                                                                                                                                                               | <p>Is the OER available in an Open Learning Management System (LMS)?</p>                                                                                                                                                                                                                                                                                                                                                                     | Achieve Inc, 2011                 |





| ASPECT        | RECOMMENDATION                                                                                                                                                                                                                                                                                                | QUESTION                                                                                                                                                                                                                                                                                      | SOURCE                                                                   |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>Robust</b> | Ensure that all content and functionality provided with different controls are available using assistive technologies such as screen readers and provide guidance on how to obtain accommodation to them.                                                                                                     | Are all content and functionality with different controls available using assistive technologies such as screen readers? Does the OER provide guidance on how to obtain accommodation to assistive technologies?                                                                              | WCAG2.2 1.3.2<br>QLT 2022<br>Quality Matters Program, 2011<br>LORI, 2009 |
| <b>Robust</b> | Let users operate touch screens with one finger and reduced gestures.                                                                                                                                                                                                                                         | Can users operate touch screens with one finger and reduced gestures?                                                                                                                                                                                                                         | Monsalve et al., 2018                                                    |
| <b>Robust</b> | Reduce accidental activation of controls.                                                                                                                                                                                                                                                                     | Can the accidental activation of controls be reduced?                                                                                                                                                                                                                                         | Monsalve et al., 2018                                                    |
| <b>Robust</b> | Do not rely solely on device motion to control the content, except if the user needs this kind of interaction.                                                                                                                                                                                                | If the user needs device motion to control the content, is it available?                                                                                                                                                                                                                      | Monsalve et al., 2018                                                    |
| <b>Robust</b> | Ensure that name, role, states, properties and values of all user interface components (buttons, menus, form fields, links, components generated by scripts, etc), can be programmatically set; and the notification of changes to these items is available to user agents, including assistive technologies. | Can name, role, states, properties and values of all user interface components (buttons, menus, form fields, links, components generated by scripts, etc), be programmatically set? Is the notification of changes to these items available to user agents, including assistive technologies? | WCAG2.2 -4.1                                                             |
| <b>Robust</b> | Ensure that the elements of the learning environment (i.e., content, activities, etc.) change their state and position because they prevent the confusion and unnecessary effort of creating a new mental                                                                                                     | Do the elements of the learning environment (i.e., content, activities, etc.) change their state and position only when necessary to prevent the confusion and unnecessary effort                                                                                                             | UNE-71362                                                                |



| ASPECT             | RECOMMENDATION                                                                                                                                                                                                                                                      | QUESTION                                                                                                                                                                                                                                                                                   | SOURCE                                                                   |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
|                    | image each time.                                                                                                                                                                                                                                                    | of creating a new mental image each time?                                                                                                                                                                                                                                                  |                                                                          |
| <b>Robust</b>      | Show students their progress while advancing through the contents (the parts that have been successfully completed and those that have not been completed yet). Provide clear, obvious and available information that does not interfere with the learning session. | Is the students' progress shown while advancing through the contents (the parts that have been successfully completed and those that have not been completed yet)?<br><br>Does the OER provide clear, obvious and available information that does not interfere with the learning session? | UNE-71362<br>Neto et al., 2019                                           |
| <b>Conformance</b> | Ensure that if the OER presents a sequence of steps that needs to be completed in order to accomplish an activity, all steps in the process conform at the specified level.                                                                                         | If the OER presents a sequence of steps that needs to be completed in order to accomplish an activity, do all steps in the process conform at the specified level?                                                                                                                         | WCAG2.2 -5.2                                                             |
| <b>Conformance</b> | Provide officially supported assistive technologies, which are already fully accessible. Ensure that any third-party tools used are accessible.                                                                                                                     | Are officially supported assistive technologies provided, which are already fully accessible? Are any third-party tools used accessible?                                                                                                                                                   | WCAG2.2 1.3.2<br>QLT 2022<br>Quality Matters Program, 2011<br>LORI, 2009 |
| <b>Conformance</b> | Ensure that if technologies are used in a way that is not accessibility supported, or if they are used in a non-conforming way, they do not block the ability of users to access the rest of the contents.                                                          | If technologies are used in a way that is not accessibility supported, or if they are used in a non-conforming way, do they block the ability of users to access the rest of the contents?                                                                                                 | WCAG2.2 -5.2                                                             |



## 6 Bibliographic references used to create the quality framework

Achieve Inc., 2011 Rubrics for Evaluating Open Education Resource (OER) Objects.

<https://www.achieve.org/files/AchieveOERRubrics.pdf>

AENOR (2005). Norma UNE 153020. Audiodescripción para personas con discapacidad visual.

Requisitos para la audiodescripción y elaboración de audioguías.

Alberta University, 2014. Quality etoolkit. <https://caqc.alberta.ca/review-process/standards/program-standards/>

Arrufat, M. A., Abasolo, A. y Martínez, S. (2021). Análisis de los códigos visogestuales en el entorno digital. Revista de Estudios de Lenguas de Signos REVLES, 3: 158-183.

Ávila, C., Baldiris, S., Fabregat, R. y Graf, S. (2016). Cocreation and evaluation of inclusive and accessible open educational resources: A mapping toward the IMS caliper. IEEE

Bakken, B., & Bridges, B. (2011). National Standards for Quality Online Courses (version 2). Vienna, VA: International Association for K--12 Online Learning iNACOL. Retrieved January 5, 2013.

<https://www.aurora-institute.org/wp-content/uploads/national-standards-for-quality-online-courses-v2.pdf>

Baya'a, N., Shehade, H.M., & Baya'a, A.R. (2009). A rubric for evaluating web-based learning environments. British Journal of Educational Technology, 40 (4), 761-763. <https://bera-journals.onlinelibrary.wiley.com/doi/pdfdirect/10.1111/j.1467-8535.2008.00864.x>

Berrios, O. 2020. Qué dicen las normas sobre pictogramas: ¿cómo comprobar que se entienden?

<https://www.plenainclusion.org/noticias/que-dicen-las-normas-sobre-pictogramas-como-comprobar-que-se-entienden/#:~:text=Este%20es%20el%20%C3%B3digo%20de,ISO%209186%2D1>

Bethel, E. (2020). Open Textbooks: Quality and Relevance for Postsecondary Study in The Bahamas. The International Review of Research in Open and Distributed Learning, 21(2), 61–80.

<https://www.erudit.org/en/journals/irrodl/2020-v21-n2-irrodl05261/1069028ar.pdf>

Binns, F., & Otto, A. (2006). Quality assurance in open distance education - Towards a culture of quality : A case study from the Kyambogo University, Uganda. In B.N. Koul, & A.S. Kanwar (Eds.), Perspectives on distance education : Towards a culture of quality, (pp.31-44). Vancouver, BC : Commonwealth of Learning. [https://www.researchgate.net/publication/277101345\\_Towards\\_a\\_Culture\\_of\\_Quality](https://www.researchgate.net/publication/277101345_Towards_a_Culture_of_Quality)

California State University. (Hrsg). MERLOT Peer Review Report Form -V 17.6. [https://info.MERLOT.org/MERLOT/help/assets/docs/Peer\\_Review\\_form\\_CC\\_0519.pdf](https://info.MERLOT.org/MERLOT/help/assets/docs/Peer_Review_form_CC_0519.pdf)

Camilleri, A.F., & Tannhäuser, A-C. (Eds.) (2012). Open learning recognition: Taking open educational resources a step further. Brussels, Belgium : EFQUEL - European Foundation for Quality in e-Learning. [https://www.researchgate.net/publication/260423351\\_Open\\_learning\\_recognition\\_Taking\\_open\\_educational\\_resources\\_a\\_step\\_further](https://www.researchgate.net/publication/260423351_Open_learning_recognition_Taking_open_educational_resources_a_step_further)



- CEAPAT, 2023. 10 fichas sobre la accesibilidad cognitiva por derecho. Ficha 5. Señalización en el entorno. [https://ceapat.imserso.es/publicaciones/informe-publicacion/-/asset\\_publisher/jbbrNfFLTBpz/content/10-fichas-sobre-la-accesibilidad-cognitiva-por-derecho/20123](https://ceapat.imserso.es/publicaciones/informe-publicacion/-/asset_publisher/jbbrNfFLTBpz/content/10-fichas-sobre-la-accesibilidad-cognitiva-por-derecho/20123)
- CEAPAT-IMSERSO, 2018. Proyecto Tu opinión CuentAA. <https://ceapat.imserso.es/documents/20123/363420/informetuopinioncuentaceapat.pdf>
- CEDEC, (2020). 12 recomendaciones para elaborar materiales accesibles e inclusivos. Cedec. <https://cedec.intef.es/12-recomendaciones-para-elaborar-materiales-accesibles-e-inclusivos/>
- Cenacchi, M. A., San Marín, P. S., & Monjelat, N. G. (2021). Los principios de accesibilidad-DHD aplicados al diseño y desarrollo de un juego musical. *Lúdicamente*, 10(20), 1–19. [https://ri.conicet.gov.ar/bitstream/handle/11336/185820/CONICET\\_Digital\\_Nro.77e6153c-2177-4430-a5b6-f0e4fa5677ee\\_A.pdf?sequence=2](https://ri.conicet.gov.ar/bitstream/handle/11336/185820/CONICET_Digital_Nro.77e6153c-2177-4430-a5b6-f0e4fa5677ee_A.pdf?sequence=2)
- Chimbo, B., Ingavélez-Guerra, P., & Otón-Tortosa, S. (2021, October). Accessibility Metadata: A Proposal for the OER's Evaluation Considering Quality Models. In 2021 IEEE Fifth Ecuador Technical Chapters Meeting (ETCM) (pp. 1-6). [https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9590799&casa\\_token=da-oCRrr6kcAAAAA:MyKLn1vpatahtp0uKV4Mj1rYEXk8km1gHtQr-clLKZMPwwW4tO5JpX30oY5Y4WtNVYsZleM6Q&tag=1](https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9590799&casa_token=da-oCRrr6kcAAAAA:MyKLn1vpatahtp0uKV4Mj1rYEXk8km1gHtQr-clLKZMPwwW4tO5JpX30oY5Y4WtNVYsZleM6Q&tag=1)
- CNLSE, 2017. Guía de buenas prácticas para la incorporación de la lengua de signos española en televisión. Madrid: Real Patronato sobre Discapacidad. <https://www.sis.net/documentos/ficha/529550.pdf>
- Creative Commons. About Creative Commons Licenses <https://creativecommons.org/share-your-work/cclicenses/>
- Daniela, L. (Ed.). (2022). Inclusive Digital Education. Springer Nature. [https://www.researchgate.net/profile/Gema-De-Pablo/publication/364747503\\_Breaking\\_the\\_Digital\\_Gender\\_Gap\\_with\\_Inclusive\\_Digital\\_Education/links/63b30b23097c7832ca84cbf1/Breaking-the-Digital-Gender-Gap-with-Inclusive-Digital-Education.pdf](https://www.researchgate.net/profile/Gema-De-Pablo/publication/364747503_Breaking_the_Digital_Gender_Gap_with_Inclusive_Digital_Education/links/63b30b23097c7832ca84cbf1/Breaking-the-Digital-Gender-Gap-with-Inclusive-Digital-Education.pdf)
- de la Rosa Gómez, A., Meza Cano, J. M., & Miranda Díaz, G. A. (2019). Validation of a Rubric to Evaluate Open Educational Resources for Learning. *Behavioral Sciences*, 9(12). <https://www.mdpi.com/2076-328X/9/12/126>
- Descrivedendo. (2021). 10 punti per realizzare una descrizione efficace. <https://www.descrivedendo.it/home-2/le-linee-guida/>
- Ehlers, U-D. (2004). Quality in e-learning from a learner's perspective. *European Journal of Open, Distance and e-Learning*. <https://old.eurodl.org/?p=archives&sp=full&year=2004&halfyear=2..&article=101>
- Elias, M., Oelen, A., Tavakoli, M., Kismihok, G., & Auer, S. (2020). Quality Evaluation of Open Educational Resources. In C. Alario-Hoyos, M. J. Rodríguez-Triana, M. Scheffel, I. Arnedillo-Sánchez,

- & S. M. Dennerlein (Eds.), Lecture Notes in Computer Science. Addressing Global Challenges and Quality Education (Vol. 12315, pp. 410–415). Springer International Publishing. [https://www.researchgate.net/profile/Mirette-Elias/publication/342509702\\_Quality\\_Evaluation\\_of\\_Open\\_Educational\\_Resources/links/5ef83b0c299bf18816ede134/Quality-Evaluation-of-Open-Educational-Resources.pdf](https://www.researchgate.net/profile/Mirette-Elias/publication/342509702_Quality_Evaluation_of_Open_Educational_Resources/links/5ef83b0c299bf18816ede134/Quality-Evaluation-of-Open-Educational-Resources.pdf)
- Equality and Human Rights Commission, (2019). Reasonable adjustments for disabled pupils. [https://www.equalityhumanrights.com/sites/default/files/guidance-reasonable-adjustments-for-disabled-pupils-in-england\\_0.docx](https://www.equalityhumanrights.com/sites/default/files/guidance-reasonable-adjustments-for-disabled-pupils-in-england_0.docx)
- Fitzgerald & Byers (2002). A rubric for Selecting Inquiry-Based Activities. Teaching Strategies, 22-25. [https://www.researchgate.net/publication/267156453\\_A\\_Rubric\\_for\\_Selecting\\_Inquiry-Based\\_Activities?tp=eyJjb250ZXh0Ijp7Im-ZpcnN0UGFnZSI6InB1YmxpY2F0aW9uliwicGFnZSI6InB1YmxpY2F0aW9uIn19#read](https://www.researchgate.net/publication/267156453_A_Rubric_for_Selecting_Inquiry-Based_Activities?tp=eyJjb250ZXh0Ijp7Im-ZpcnN0UGFnZSI6InB1YmxpY2F0aW9uliwicGFnZSI6InB1YmxpY2F0aW9uIn19#read)
- Fundación ONCE, 2013. Pautas de diseño de pictogramas para todas las personas. Proyecto de biblioteca digital de gráficos orientativos en accesibilidad cognitiva urbana. Dirección de Accesibilidad Universal de la Fundación Once. [https://accesibilidadcognitivaurbana.fundaciononce.es/docs/Manual\\_pictogramas.pdf](https://accesibilidadcognitivaurbana.fundaciononce.es/docs/Manual_pictogramas.pdf)
- Gordillo Méndez, A., Barra Arias, E., & Quemada Vives, J. (2018). Estimación de calidad de objetos de aprendizaje en repositorios de recursos educativos abiertos basada en las interacciones de los estudiantes. Educación XX1, 21(1), 285–302 <https://redined.educacion.gob.es/xmlui/bitstream/handle/11162/160449/20196-39962-1-SM.pdf?sequence=1>
- Guía para la creación de materiales educativos digitales y accesibles del proyecto CREA [https://emtic.educarex.es/images/innovated/pdf/Guia\\_DUA\\_CREA.pdf](https://emtic.educarex.es/images/innovated/pdf/Guia_DUA_CREA.pdf)
- Harris, R., Miske, S., & Attig, G. (2004). Embracing Diversity: Toolkit for Creating Inclusive Learning-Friendly Environments. UNESCO Bangkok. <https://files.eric.ed.gov/fulltext/ED496229.pdf>
- Hashey, A., Perez, L., & Stahl, S. Open Education Resources: Ensuring Inclusive Learning in Uncertain Times. <https://aem.cast.org/binaries/content/assets/common/publications/aem/oers-inclusive-learning-uncertain-times.pdf>
- Haughey, M. y Muirhead, B. (2005). Evaluating Learning Objects for Schools. e-Journal of Instructional Science and Technology 8 (1). <https://files.eric.ed.gov/fulltext/EJ850358.pdf>
- Huertas, E., Biscan, I., Ejsing, C., Kerber, L., Kozłowska, L., Ortega, S. M., ... & Seppmann, G. (2018). Considerations for quality assurance of e-learning provision. Report from the ENQA Working Group VIII on quality assurance and e-learning (Occasional Papers 26). <https://www.engq.eu/wp-content/uploads/Considerations-for-QA-of-e-learning-provision.pdf>
- IEEE Standard for Learning Object Metadata, in IEEE Std 1484.12.1-2020 , vol., no., pp.1-50, 16 Nov. 2020, <http://doi.org/10.1109/IEEESTD.2020.9262118>.



- Inclusion Europe. Information for all. European standards for making information easy to read and understand. 2010. [https://www.inclusion-europe.eu/wp-content/uploads/2017/06/EN\\_Information\\_for\\_all.pdf](https://www.inclusion-europe.eu/wp-content/uploads/2017/06/EN_Information_for_all.pdf)
- Iniesto, F.; McAndrew, P.; Minocha, S. y Coughlan T. (2019). Auditing the Accessibility of MOOCs: A Four-Component Approach. En: Scheffel, M., Broisin, J., Pammer-Schindler, V., Ioannou, A., Schneider, J. (eds), Transforming Learning with Meaningful Technologies. EC-TEL 2019. Lecture Notes in Computer Science, vol 11722. Springer, Cham. [https://doi.org/10.1007/978-3-030-29736-7\\_59](https://doi.org/10.1007/978-3-030-29736-7_59)  
<https://oro.open.ac.uk/62256/1/EC-TEL%20paper%20Audit.pdf>
- JISC Tiger Project, 2011. TIGER contributes to OER infokit (quality considerations). Project Tiger Blog. <https://tigeroer.wordpress.com/2011/06/10/tiger-contributes-to-oer-infokit-quality-considerations/>
- Jung, I., & Lee, T. (2014). QA in e-ASEM OER project. Unpublished research. <https://www.surveymonkey.com/s/X5THFTF>
- Jung, I., Sasaki, T., & Latchem, C. (2016). A framework for assessing fitness for purpose in open educational resources. <https://link.springer.com/content/pdf/10.1186/s41239-016-0002-5.pdf>
- Kabani, M., Heijnen-Maathuis, E., & Lignell, M. (2019). Mainstreaming inclusive education: Sharing good practices. Save The Children Sweden. <https://resourcecentre.savethechildren.net/document/mainstreaming-inclusive-education-sharing-good-practices/>
- Kawachi, P. (2014). Guidelines on Quality Assurance for Open Educational Resources. [https://ton.oer4pacific.org/id/eprint/28/1/TIPSFramework\\_Version%202%5B1%5D%20Copy.pdf](https://ton.oer4pacific.org/id/eprint/28/1/TIPSFramework_Version%202%5B1%5D%20Copy.pdf)
- Kawachi, P. (2014). Quality assurance guidelines for open educational resources: TIPS framework. [https://ton.oer4pacific.org/id/eprint/28/1/TIPSFramework\\_Version%202%5B1%5D%20Copy.pdf](https://ton.oer4pacific.org/id/eprint/28/1/TIPSFramework_Version%202%5B1%5D%20Copy.pdf)
- Kimmons, R. (2015). OER Quality and Adaptation in K-12: Comparing Teacher Evaluations of Copyright-Restricted, Open, and Open/Adapted Textbooks. International Review of Research in Open and Distributed Learning, 16(5), 39–57. <https://doi.org/10.19173/irrodl.v16i5.2341>
- Kurilovas, E. y Dagiene, V. (2009). Learning Objects and Virtual Learning Environments Technical Evaluation Criteria. Electronic Journal of e-Learning, 7(2), 127-136. <https://files.eric.ed.gov/fulltext/EJ867110.pdf>
- La Aleman de Garza, L., Sancho Vinuesa, T., & Gómez Zermeño, M. G. (2014). Indicators of pedagogical quality for the design of a Massive Open Online Course for teacher training. RUSC. Universities and Knowledge Society Journal, 12(1), 104. <https://doi.org/10.7238/rusc.v12i1.2260>
- Latchem, C. (2012). Quality assurance toolkit for open and distance non--formal education. Vancouver, BC : Commonwealth of Learning. Retrieved December 20, 2012. <https://oasis.col.org/server/api/core/bitstreams/6a1fa7c5-0b94-4122-b479-0431ba93d9b1/content>
- Leicester University (2010). Quality criteria in OERs. OTTER / Institute of Learning Innovation website. Retrieved February 10, 2015, from <http://www2.le.ac.uk/departments/beyond-distance-research->



[alliance/projects/otter/about-oers/Quality%20criteria%20work%20in%20progress.pdf/view](https://alliance/projects/otter/about-oers/Quality%20criteria%20work%20in%20progress.pdf/view)

LOM -ES V1.0 (Norma UNE-71361:2010). [www.lom-es.es](http://www.lom-es.es)

Lübben, Müskens, Zawacki-Richter. Quality of OER: Test Theoretical Development and Validation of an Assessment Tool. <https://library.oapen.org/bitstream/handle/20.500.12657/61863/978-3-658-38703-7.pdf?sequence=1#page=144>

Marín, V. I., Orellana, M. L., & Peré, N. (2019). Open educational resources for research training: quality assurance through a collaborative evaluation. *Research in Learning Technology*, 27(0). <https://journal.alt.ac.uk/index.php/rlt/article/download/2271/2582>

Martín, A. S., Mendoza, J. L. C., & Martín, A. P. (2017). Uso de recursos educativos abiertos en educación superior. *Universidad & ciencia*, 6, 124-137. <https://revistas.unica.cu/index.php/uciencia/article/download/706/1126>

Mayrberger, K.; Zawacki-Richter, O. y Müsken W. (2018). Qualitätsentwicklung von OER Vorschlag zur Erstellung eines Qualitätssicherungsinstruments für OER am Beispiel der Hamburg Open Online University. <https://doi.org/10.25592/978.3.924330.67.5>

Mhlanga, E. (2010). Quality criteria for maintaining quality in open schools. In L. Cameron (Ed.), *Quality assurance toolkit for open schools*, (pp.29-53). Vancouver, BC : Commonwealth of Learning. Retrieved April 4, 2014, from <http://www.col.org/PublicationDocuments/pubQAOSToolkit.pdf>

Monsalve Ríos, C. L.; Medina Monje, C. y Díaz Cardona, W. (2018). Recomendaciones para la construcción de Recursos Educativos Abiertos inclusivos y accesibles. Universidad de Antioquía (Colombia). [https://bibliotecadigital.udea.edu.co/bitstream/10495/11661/6/MonsalveClara\\_2018\\_ConstruccionRecursosEducativos.pdf](https://bibliotecadigital.udea.edu.co/bitstream/10495/11661/6/MonsalveClara_2018_ConstruccionRecursosEducativos.pdf)

Murillo, F.J. (2020) Educación y desigualdades sociales: cómo avanzar hacia una educación inclusiva. Cátedra UNESCO en Educación para la Justicia Social de la Universidad Autónoma de Madrid (UAM). <https://upnatv.unavarra.es/unes/francisco-javier-murillo>

Navarrete, R. y Luján-Mora, S. (2013). Accessibility considerations in learning objects and open educational. *Proceedings of the 6th International Conference of Education, Research and Innovation (IC-ERI 2013)*, pp. 521-530. <https://sergiolujanmora.es/verpdf/158>

Navarrete, R., Luján-Mora, S. (2018). Bridging the accessibility gap in Open Educational Resources. *Univ Access Inf Soc* 17, 755–774. <https://link.springer.com/content/pdf/10.1007/s10209-017-0529-9.pdf>

Nesbit, J. C., Belfer, K., & Leacock, T. (2003). Learning object review instrument (LORI). *E-learning research and assessment network*, 33-68. [https://www.academia.edu/7927907/Learning\\_Object\\_Review\\_Instrument\\_LORI](https://www.academia.edu/7927907/Learning_Object_Review_Instrument_LORI)

Nesbit, J., Belfer, K., & Leacock, T. (2007). Learning object review instrument (LORI). [https://www.academia.edu/7927907/Learning\\_Object\\_Review\\_Instrument\\_LORI](https://www.academia.edu/7927907/Learning_Object_Review_Instrument_LORI)



- OER Africa (2015). Faculty Guide for evaluating Open Education Resources. <https://www.oerafrica.org/content/evaluating-open-content>
- Open Textbook rubric. Prince Georges Community College. 2017. [https://pgcc.libguides.com/ld.php?content\\_id=57241857](https://pgcc.libguides.com/ld.php?content_id=57241857)
- Ossiannilsson, E., Altınay, Z., & Altınay, F. (2017). Towards fostering quality in open online education through OER and MOOC practices. Open Education: from OERs to MOOCs, 189-204. [https://link.springer.com/chapter/10.1007/978-3-662-52925-6\\_10](https://link.springer.com/chapter/10.1007/978-3-662-52925-6_10)
- Pereira Baz, M.A. & Valdera López, C. (2017) Guide for creators of Open Educational Resources. EDIA Project. CEDEC. Instituto Nacional de Tecnologías Educativas y de Formación del Profesorado (INTEF). [http://descargas.educalab.es/cedec/guia\\_rea\\_cedec\\_v2017/guia\\_rea\\_cedec\\_v2017\\_english/edia\\_project.html](http://descargas.educalab.es/cedec/guia_rea_cedec_v2017/guia_rea_cedec_v2017_english/edia_project.html)
- Pérez-Martínez, R., & González-Laredo, M. (2018). Abierta UGR, un diagnóstico para la calidad: Cómo mejorar las iniciativas MOOC con estándares y referentes. In Asociación Red Campus Virtuales (Ed.), VIII Jornadas Internacionales de Campus Virtuales (JICV' 18) (pp. 1–4). <http://riull.ull.es/xmlui/handle/915/8072>
- Pinto, M., Gómez-Camarero, C., Fernández-Ramos, A., & Doucet, A. V. (2017). Evaluareed: Development of an online educational resources assessment instrument. Investigación bibliotecológica, 31(72), 227-248. <https://www.scielo.org.mx/pdf/ib/v31n72/2448-8321-ib-31-72-00227.pdf>
- Politis, Y., Deveril, D., Navarro, S. B., Avila, C., De Lera, E., Monjo, T., & Goodman, L. (2014). Introducing the Inclusive Learning Handbook: An OER for teachers and policy makers. In EDULEARN14 Proceedings (pp. 5463-5469). IATED. [https://www.researchgate.net/profile/Yurgos-Politis/publication/315760950\\_Introducing\\_the\\_inclusive\\_learning\\_handbook\\_an\\_OER\\_for\\_teachers\\_and\\_policy\\_makers/links/599ef3bfaca272dff134fb87/Introducing-the-inclusive-learning-handbook-an-OER-for-teachers-and-policy-makers.pdf](https://www.researchgate.net/profile/Yurgos-Politis/publication/315760950_Introducing_the_inclusive_learning_handbook_an_OER_for_teachers_and_policy_makers/links/599ef3bfaca272dff134fb87/Introducing-the-inclusive-learning-handbook-an-OER-for-teachers-and-policy-makers.pdf)
- Portugal, C.; Guimarães, M.; Moura, M.; Magro Junior, J. C. Exploring Inclusive Design and Digital Humanities: Enabling Bilingual Digital Narratives for Deaf Children. Design and Technology Education: An International Journal, [S. l.], v. 28, n. 2, p. 122–135, 2023. <https://openjournals.ljmu.ac.uk/DATE/article/view/1208>
- Proyecto EDIA-CEDEC (2020). Lista de comprobación de calidad de un recurso Educativo. <https://cedec.intef.es/lista-de-comprobacion-de-calidad-de-un-recurso-educativo-abierto/>
- Proyecto REA DUA Andalucía. <https://www.juntadeandalucia.es/educacion/portals/web/transformacion-digital-educativa/rea>
- QLT. CSU Quality Learning & Teaching Rubric. California State University. 2022. <https://ocs.calstate.edu/sites/default/files/2023-06/QLT-Rubric-3rd-Edition-2022.pdf>





- Quality Matters Rubric Standards 2011-2013 (2013). <https://es.slideshare.net/slideshow/qm-standards-20112013/39417595>
- Reinken, C., Greiff, P., Draxler-Weber, N., y Hoppe, U. (2021, September). Quality of OER from the Perspective of Lecturers—Online Survey of Quality Criteria for Quality Assurance. In International Conference on Business Informatics Research (pp. 36-50). Cham: Springer International Publishing. [https://doi.org/10.1007/978-3-030-87205-2\\_3](https://doi.org/10.1007/978-3-030-87205-2_3)
- Remael, A., Reviers, N., & Vercauteren, G. (Eds.). (2015). Pictures painted in words: ADLAB audio description guidelines. EUT Edizioni Università di Trieste.
- Revista Iberoamericana de Tecnologías del Aprendizaje, 11(3), 167-176. [https://ieeexplore.ieee.org/iel7/6245520/6461616/07516709.pdf?casa\\_token=Ld2PTbi8VnwAAAAA:3u5flc2AktDKu5s6bMsL0LYatb\\_I5V4tGqEZMHnCmLc8UbUzHmj1nCcdiuPHYgOwf1uISpeSfA](https://ieeexplore.ieee.org/iel7/6245520/6461616/07516709.pdf?casa_token=Ld2PTbi8VnwAAAAA:3u5flc2AktDKu5s6bMsL0LYatb_I5V4tGqEZMHnCmLc8UbUzHmj1nCcdiuPHYgOwf1uISpeSfA)
- RNIB, & VocalEyes. (2003). Talking images guide: Museums, galleries and heritage sites : improving access for blind and partially sighted people. RNIB : Vocaleyes.
- Rodríguez-Fuentes, A., Aláin-Botaccio, L., & García-García, F. (2022). Presentation and evaluation of a digital tool for sign language (Presentación y evaluación de una herramienta digital para la lengua de signos). Culture and Education, 34(3), 658-688. <https://doi.org/10.1080/11356405.2022.2058793>
- Roig Vila, Rosabel; Mengual-Andres, Santiago; Suarez Guerrero, Cristobal (2014). Evaluación de la calidad pedagógica de los MOOC. [https://rua.ua.es/dspace/bitstream/10045/37206/1/2014\\_Roig-Vila\\_etal\\_Profesorado.pdf](https://rua.ua.es/dspace/bitstream/10045/37206/1/2014_Roig-Vila_etal_Profesorado.pdf)
- Romero Peláez, A. E., & Morocho Yunga, J. C. (2016). Características inherentes para OER's accesibles. Opción, 32(8), 674-688.. ISSN: 1012-1587. <https://www.redalyc.org/pdf/310/31048481039.pdf>
- Romero-Peláez, A., Segarra-Faggioni, V., Piedra, N. y Tovar, E. (2019). A Proposal of Quality Assessment of OER Based on Emergent Technology. IEEE Global Engineering Education Conference (EDUCON), Dubai, United Arab Emirates, pp. 1114-1119. [https://ieeexplore.ieee.org/iel7/8718248/8725024/08725067.pdf?casa\\_token=el5\\_jlKxiiAAAAA:qt8tNZZ-MRPp01BQOmZPApMmYrrH\\_iB-wKl\\_LzI-NVhJPAlHeoz27jromegzyEJCcmG41rVaWLg](https://ieeexplore.ieee.org/iel7/8718248/8725024/08725067.pdf?casa_token=el5_jlKxiiAAAAA:qt8tNZZ-MRPp01BQOmZPApMmYrrH_iB-wKl_LzI-NVhJPAlHeoz27jromegzyEJCcmG41rVaWLg)
- Rubio Pulido, 2018 CREA con Diseño Universal para el Aprendizaje. Consejería de Educación, Ciencia y Formación Profesional. Junta de Extremadura. <https://emtic.educarex.es/224-emic/atencion-a-la-diversidad/3079-crea-con-diseno-universal-para-el-aprendizaje>
- Save the children (2019). Guía de buenas prácticas en educación inclusiva. [https://www.savethechildren.es/sites/default/files/imce/docs/guia\\_de\\_buenas\\_practicas\\_en\\_educacion\\_inclusiva\\_vok.pdf](https://www.savethechildren.es/sites/default/files/imce/docs/guia_de_buenas_practicas_en_educacion_inclusiva_vok.pdf)
- Schuer, R., y Baas, M. (2023). Reuse of OER, a Process Model Approach. In Distributed Learning Ecosystems: Concepts, Resources, and Repositories (pp. 117-137). Wiesbaden: Springer Fachmedien Wiesbaden. <https://library.oapen.org/bitstream/handle/20.500.12657/61863/978-3-658-38703->



[7.pdf?sequence=1#page=123](#)

Sloan (2013). A quality scorecard for the administration of online education programs. Newburyport, MA : The Sloan Consortium. Retrieved October 8, 2013, from [http://sloanconsortium.org/quality\\_scoreboard\\_online\\_program](http://sloanconsortium.org/quality_scoreboard_online_program)

Soler Vilageliu, Olga; Marta Bosch-Baliarda & Pilar Orero. (2015) Hbb4All: Diseño de experimentos con usuarios para evaluar la recepción de LSE en TV. Congreso CNLSE de la Lengua de Signos Española 2015.

Sreedher, R. (Ed.) (2009). Quality assurance of multimedia learning materials. New Delhi : Commonwealth Educational Media Centre for Asia. Retrieved March 3, 2013.  
<https://www.cemca.org/ckfinder/userfiles/files/QA MLM%201-5.pdf>

The UDL Guidelines. <https://udlguidelines.cast.org>

Tlili, A., Huang, R., Chang, T. W., Nascimbeni, F. y Burgos, D. (2019). Open educational resources and practices in China: A systematic literature review. Sustainability, 11(18), 4867.  
<https://www.mdpi.com/2071-1050/11/18/4867/pdf>

UNE-71362. (2020). Calidad de los materiales educativos digitales

UNESCO (2015). <https://unesdoc.unesco.org/ark:/48223/pf0000245656>

UNESCO (2016). Learning for All: Guidelines on the Inclusion of Learners with Disabilities in Open and Distance Learning.  
<https://unesdoc.unesco.org/ark:/48223/pf0000244355>  
<https://unesdoc.unesco.org/ark:/48223/pf0000244355>

Unesco (2019). Guidelines on the development of open educational resources policies. <https://unesdoc.unesco.org/ark:/48223/pf0000371129>

UNESCO 2017. A Guide for ensuring inclusion and equity in education.  
<https://unesdoc.unesco.org/ark:/48223/pf0000248254>

UNESCO 2020. Towards inclusion in education: status, trends and challenges: the UNESCO Salamanca Statement 25 years on. <https://doi.org/10.54675/ASIM9654>

Vahdati, S., Lange, C., & Auer, S. (2015). OpenCourseWare Observatory - Does the Quality of Open-CourseWare Live up to its Promise? In J. Baron, G. Lynch, N. Maziarz, P. Blikstein, A. Merceron, & G. Siemens (Eds.), Proceedings of the Fifth International Conference on Learning Analytics And Knowledge (pp. 73–82). ACM. <https://dl.acm.org/doi/pdf/10.1145/2723576.2723605>

Valerio Neto, L.; Fontoura Junior, Paulo H. F.; Bordini, Rogrio A.; Otsuka, Joice L. y Beder, Delano M. (2019). Details on the Design and Evaluation Process of an Educational Game Considering Issues for Visually Impaired People Inclusion. Educational Technology & Society 22(3). 4-18.  
[https://www.jstor.org/stable/pdf/26896706.pdf?casa\\_to-ken=sHPOM8IN20YAAAAA:q4jkYERM7JuEEA6\\_7YIEwqq9ff2uWFY4yEJzXMIjIRMDbTc3EPUSYPe-ffSVWbE3PKYv8GuGgMA20OrEAs-5ZZ5JXCRE\\_DnY0DSrD4vVpY-cmOMxVBZM](https://www.jstor.org/stable/pdf/26896706.pdf?casa_to-ken=sHPOM8IN20YAAAAA:q4jkYERM7JuEEA6_7YIEwqq9ff2uWFY4yEJzXMIjIRMDbTc3EPUSYPe-ffSVWbE3PKYv8GuGgMA20OrEAs-5ZZ5JXCRE_DnY0DSrD4vVpY-cmOMxVBZM)



Co-funded by  
the European Union

Funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or EACEA. Neither the European Union nor the granting authority can be held responsible for them.

- Vinoba Vinayagamoorthy, Maxine Glancy, Christoph Ziegler, and Richard Schäffer. (2019). Personalising the TV Experience using Augmented Reality: An Exploratory Study on Delivering Synchronised Sign Language Interpretation. In Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19). Association for Computing Machinery, New York, NY, USA, Paper 532, 1–12. <https://doi.org/10.1145/3290605.3300762>
- Vittaz, N.; Vilanova, G. y Varas, J. (2023). Recursos Educativos Abiertos Inclusivos (REAI), prácticas educativas abiertas (PEA) en el nivel superior en entornos virtuales. ICT-UNPA, 15, Vol. 3, 97-126. <https://publicaciones.unpa.edu.ar/index.php/ICTUNPA/article/view/985>
- Wakin, S. (2020). Evaluating and Selecting OER. <https://lagccnsdoer.commons.gc.cuny.edu/oer-evaluation/>
- Williams, K., Kear, K., & Rosewell, J. (2012). Quality assessment for e-learning : A benchmarking approach (2nd edn.). Heerlen, Netherlands : European Association of Distance Teaching Universities (EADTU). Retrieved May 26, 2014, from <http://oro.open.ac.uk/34632/2/3D5D7C.pdf>
- Zawacki-Richter, O. y Mayrberger, K. (2017). Qualität von OER: Internationale Bestandsaufnahme von Instrumenten zur Qualitätssicherung von Open Educational Resources (OER)-Schritte zu einem deutschen Modell am Beispiel der Hamburg Open Online University. Universität Hamburg, Universitätskolleg. [https://www.researchgate.net/publication/322676518\\_Qualitat\\_von\\_OER\\_Internationale\\_Bestandsaufnahme\\_von\\_Instrumenten\\_zur\\_Qualitatssicherung\\_von\\_Open\\_Educational\\_Resources\\_OER\\_-\\_Schritte\\_zu\\_einem\\_deutschen\\_Modell\\_am\\_Beiispiel\\_der\\_Hamburg\\_Open\\_Online\\_Uni](https://www.researchgate.net/publication/322676518_Qualitat_von_OER_Internationale_Bestandsaufnahme_von_Instrumenten_zur_Qualitatssicherung_von_Open_Educational_Resources_OER_-_Schritte_zu_einem_deutschen_Modell_am_Beiispiel_der_Hamburg_Open_Online_Uni)
- Zawacki-Richter, O., Müskens, W. y Marín, V. I. (2022/2023). Quality assurance of open educational resources. In Handbook of Open, Distance and Digital Education (pp. 1-19). Singapore: Springer Singapore. [https://link.springer.com/content/pdf/10.1007/978-981-19-0351-9\\_43-1.pdf](https://link.springer.com/content/pdf/10.1007/978-981-19-0351-9_43-1.pdf)
- Zhang, X., Tilili, A., Nascimbeni, F., Burgos, D., Huang, R., Chang, T. W., ... & Khribi, M. K. (2020). Accessibility within open educational resources and practices for disabled learners: A systematic literature review. Smart Learning Environments, 7, 1-19. <https://link.springer.com/content/pdf/10.1186/s40561-019-0113-2.pdf>



# European Quality Development System for Inclusive Education and Teacher Training

